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Design and Development of a 2-DOF Autogenous TIG Seam-Welding Machine

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Abstract

This project presents the development of a compact, automated seam-welding machine designed to bridge the gap between large industrial welding systems and the needs of small workshops, university laboratories, and hobbyist environments. While industrial automated welding solutions offer exceptional performance, they remain financially and technically inaccessible to many small-scale users. Our design provides a practical, reliable, and cost-effective alternative built entirely from readily available components, ensuring ease of fabrication, maintenance, and long-term operation.

The mechanical structure emphasizes simplicity and rigidity through a lightweight square-beam chassis and a dedicated X-Z welding mechanism optimized for thin-sheet longitudinal seam welding. Gas Tungsten Arc Welding (GTAW) without filler metal was selected for its precision and low defect rate, aligning with the small-scale, high-accuracy nature of the target applications. A pneumatic clamping system ensures workpiece stability under thermal expansion and vibration.

Automation is achieved through a PLC-based control system integrated with an HMI for operator interaction and an Industrial Internet of Things (IIoT) layer for monitoring, data logging, and future scalability. The final system demonstrates how accessible automation can replicate essential industrial welding processes in a smaller, affordable, and educationally valuable format, offering a viable path for modernizing small workshops and academic settings.

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Chapter 1: Introduction

1.1 Introduction

This chapter discusses the history and the current state of automatic welding, and who this project is made for.

1.2 Brief history of automatic welding

The history of automatic welding reflects a steady transition from rigid mechanization to adaptive, intelligence-driven robotic systems. Although commonly linked to industrial robots, automatic welding predates the first robotic manipulators by more than four decades. Its development traces a clear engineering progression beginning with early hard-automation concepts and leading to today's sensor-rich, autonomous welding platforms.

Automatic welding began in 1920 when P. O. Nobel of General Electric introduced a voltage-regulated bare-wire system capable of maintaining arc length and feed rate without manual adjustment. This innovation enabled consistent repair of heavy industrial components such as motor shafts and crane wheels. The introduction of Submerged Arc Welding in the 1930s brought a major leap in productivity by shielding the arc beneath granular flux, achieving very high deposition rates essential to World War II shipbuilding and pipe-mill fabrication (Figure 1.1). These early systems delivered excellent repeatability but were limited to fixed geometries and required extensive retooling for any change in part design.



Figure 1.1 Welding the hull of the T-34 tank with an automatic welding, Russia, 1939.

A decisive shift occurred in 1961 with the introduction of Unimate, (Figure 1.2) the first programmable industrial robot used for welding. Unlike track-bound mechanisms, this manipulator allowed the welding torch to follow variable paths defined by stored programs. The move from hydraulic to electromechanical actuators in the 1970s produced the first fully electric, microprocessor-controlled robots, such as the ASEA IRB 6 (Figure 1.3), which delivered the positional accuracy needed for reliable arc-welding automation. Purpose-built welding robots soon followed, including Yaskawa's Motoman-L10 (Figure 1.4), supporting the rapid expansion of automated GMAW in automotive manufacturing.



Figure 1.2 First Unimate Robot Ever Installed on an Assembly Line, 1961



Figure 1.3 The ASEA IRB 6 manufactured by ASEA, was the first microcomputer controlled electric industrial robot, using Intel's first chipset. The robot allowed movement in 5 axes



Figure 1.4 The Motoman L10 was the first robot Yaskawa introduced on the market.

1.3 The current state of automatic welding

Modern automatic welding stands at the intersection of sensing, computation, and intelligent automation, evolving from early voltage-regulated arc systems into today's adaptive and connected platforms. Vision-based seam tracking (Figure 1.5), laser profiling, and thermal-distortion compensation now allow machines to adjust to joint variations in real time, while AI-driven systems predict bead shape, penetration, and defects before they occur. The industry's growth reflects this shift: valued at \$15.5 billion in 2024 and projected to reach \$22.3 billion by 2030, adoption is accelerating across automotive, aerospace, shipbuilding, heavy machinery, and even medical manufacturing. Labor shortages, safety demands, and the need for consistent, high-volume precision are driving investment in robotic cells, smart sensors, and fully integrated monitoring systems. Collaborative robots have further opened the door for small and medium workshops, offering precision with safe interaction and modest space requirements. Despite challenges—upfront costs, programming skill demands, and system maintenance—modern welding is moving steadily toward fully autonomous, self-optimizing systems designed for industrial connectivity. This trajectory firmly establishes automated welding as a backbone of contemporary manufacturing and a decisive force in shaping future industrial competitiveness.

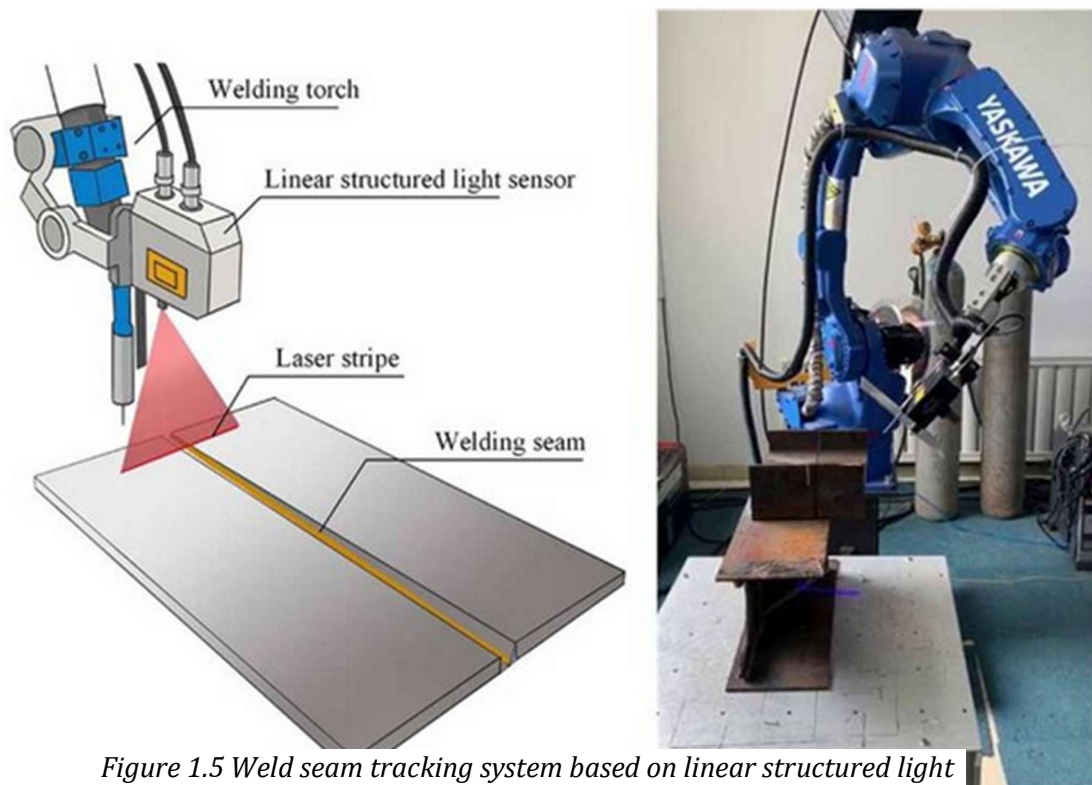


Figure 1.5 Weld seam tracking system based on linear structured light

1.4 Who is this project for?

This project is positioned to serve the segment that lies between large industrial plants and small to medium workshops, university labs, and individual hobbyists. These users often lack the financial and technical capacity to acquire or support a full robotic welding system. The proposed machine offers an accessible alternative by combining low cost, straightforward maintenance, and a flexible seam-welding design built entirely from off-the-shelf components. It provides a controlled, repeatable welding process comparable to that of an industrial environment, while remaining simple enough for students, small workshops, and enthusiasts to adopt, customize, and sustain with minimal investment.



Figure 1.6 Texas Tech University Welding Lab

Chapter 2: Welding Process

2.1 Introduction

Welding is a process of joining two or more pieces of the same or dissimilar materials to achieve complete coalescence. This is the only method of developing monolithic structures and it is often accomplished by the use of heat and/or pressure. Although in its present form it has been used since about the beginning of the 20th century, it is fast replacing other joining processes like riveting and bolting. At times, it may be used as an alternative to casting.

Presently, welding is used extensively for fabrication of vastly different components, including critical structures like boilers and pressure vessels, ships, off-shore structures, bridges, storage tanks and spheres, pipelines, railway coaches, anchor chains, missile and rocket parts, nuclear reactors, fertiliser and chemical plants, structurals, earth-moving equipment, plate and box girders, automobile bodies, press frames, and water turbines.

Welding is also used in heavy plate fabrication industries, pipe and tube fabrication, jointing drill bits to their shanks, automobile axles to brake drums, lead wire connections to transistors and diodes, sealing of containers of explosives like nitroglycerine, welding of cluster gears, and the like.

2.1.1 Classification of Welding and Allied Processes

Although almost all materials (metals, plastics, ceramics, and composites) can be welded but not by the same process. To achieve this universality many Welding and Allied Processes have been developed. Most of the industrially important processes amongst them classified depending upon the nature of heat source and its movement resulting in spot, seam or zonal welds; or on the extent of heat generation viz., low heat and high heat, are shown in Figure 2.1.

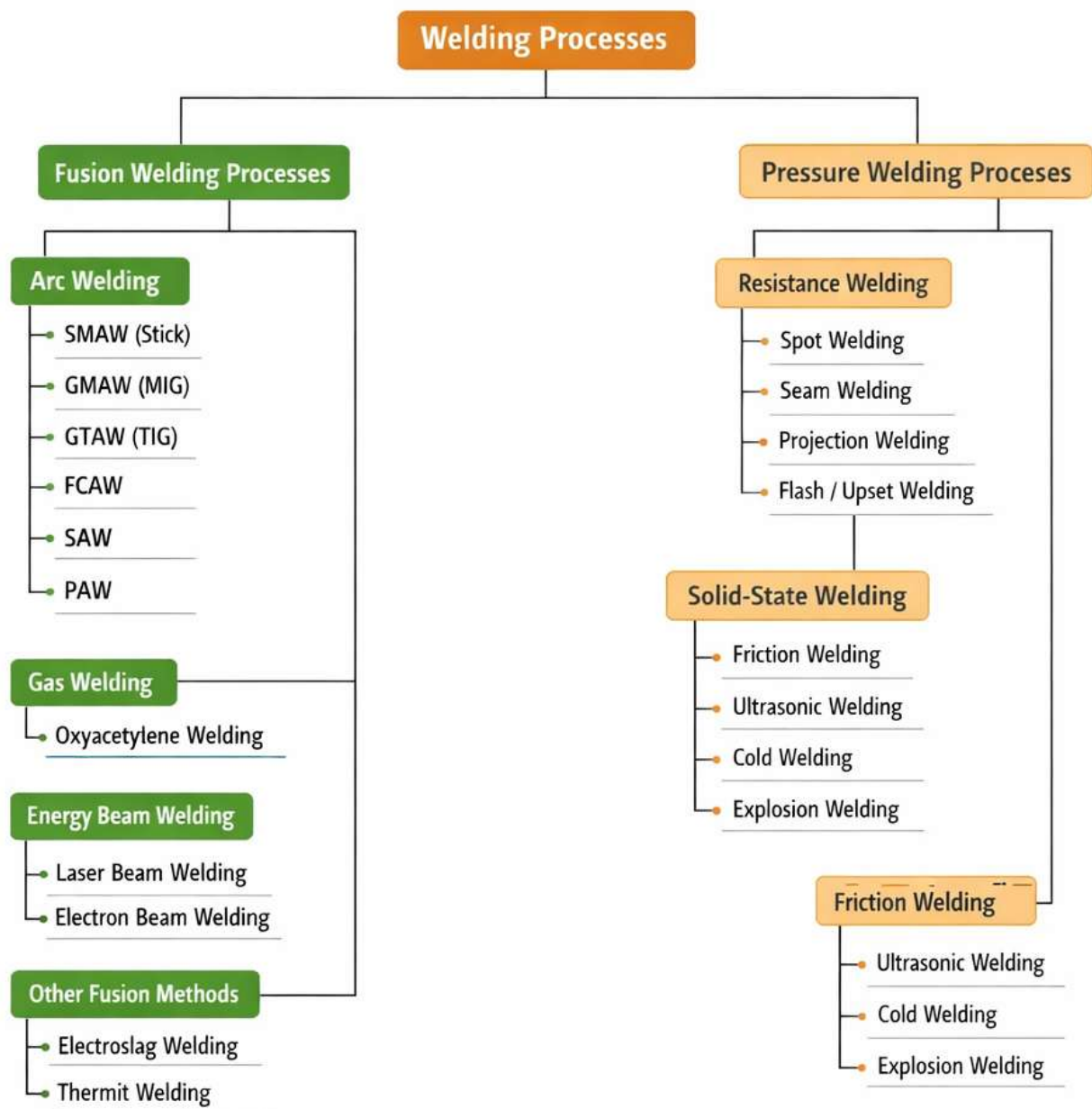


Figure 2.1 Various welding processes

2.2 Types of welding processes

2.2.1 Cast-Weld Processes

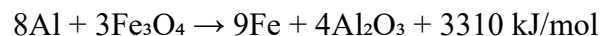
These processes involve a large amount of molten metal, resulting in properties close to that of castings. For achieving the desired joint properties, such welds are usually given normalization treatment.

Two processes in this class are Thermit Welding and Electroslag Welding.

2.2.1.1 Thermit welding

Thermit is a mixture of aluminum powder and metal oxide which, when ignited, results in a non-explosive exothermic reaction. The heat so generated melts and reduces the metal oxide to metallic form at a high temperature. This molten metal is used for joining metal parts by pouring it between them, resulting in what may be termed as a cast-weld joint.

One of the most used thermit mixtures is aluminum powder and ferric oxide which, on ignition, results in the following reaction:



The molten metal obtained has a high temperature of about 2450°C. This metal is poured into the sand mould surrounding the parts to be welded, as shown in Figure 2.2. The mould is broken soon after the solidification of the metal is complete, and the excess metal is removed by chisel and hammer to give the finished joint.

Thermit welding is extensively used for joining rails at site, cable conductors, reinforcing bars for R.C.C. structures, and for heavy repairs such as those of broken necks of rollers, and ship Lanterns.

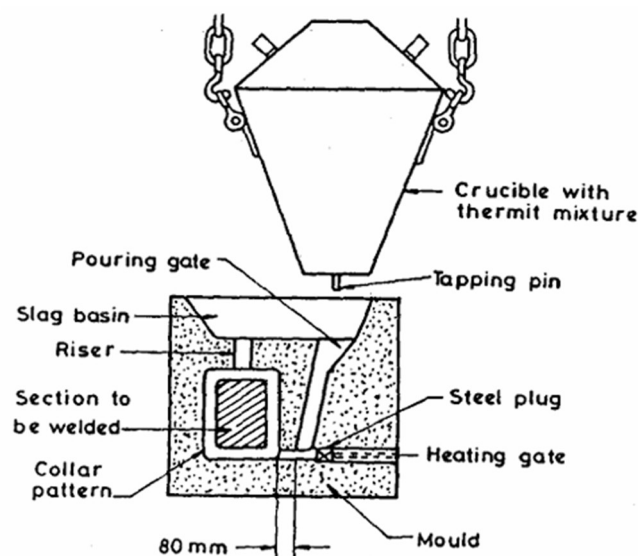


Figure 2.2 Thermit welding

2.2.1.2 Electroslag welding

Electroslag welding (ESW) is a fusion welding process for joining thick workpieces in a single run. This is not an arc welding process, though most of the set-up is similar to the usual arc welding processes, and arcing is required to initiate the process and may also occur subsequently when the process stability is disturbed.

Apart from the conventional ESW process in which the usual electrode wire with contact tube is employed, there is a popular variant of this process called the Consumable Guide ESW Process. Figure 2.3 shows the process diagram for the latter.

An essential feature of the ESW process is that welding is done with the weld joint in the vertical position.

Due to high heat input the weld pool in ESW is usually quite voluminous resulting in a weld with properties resembling that of a casting which makes post weld heat treatment (PWHT) essential to achieve the desired metallurgical structure to attain the required mechanical strength. Typically, ESW is extensively used in the construction of pressure vessels, press frames, water turbines and heavy plate fabrication industries.

2.2.2 Arc and Flame Welding Processes

The welding processes included in this class are those which make use of an electric arc or a flame obtained by burning an oxy-fuel gas mixture. The size of the weld pool evolved depends upon the energy input per unit time, and the extent of spread of arc or flame, however the volume of the molten metal in the weld pool, at any given time, is much smaller than that obtained in electroslag welding. These processes are either used to produce welds along seams (e.g., SMAW, GMAW, Oxy-fuel gas welding, etc.) or just at spots (e.g., stud welding, GTAW spot welding, etc.). Brief descriptions of industrially important processes in this class follow:

Seam Welding Processes

These processes are mainly used for welding workpieces along straight or curved seams of desired lengths and include all well-known arc and flame welding processes.

2.2.2.1 Shielded Metal Arc Welding

This process employs coated or covered electrodes for producing an arc to act as a heat source; the covering on burning provides the necessary shield to protect the molten metal from the ill effects of oxygen and nitrogen from the surrounding atmosphere. This process is more popularly known as Stick electrode welding or manual metal arc welding and is the single most used welding process in the world. Both ac and dc power sources can be used equally effectively. Figure 2.4 shows the basic circuit diagram for the process.

The weld pool produced depends upon the size of the covered electrode and the welding current used and may vary from very small to fairly large size. Larger sized pools are used only for downhand welding. This process is an all-position welding process and is used for all types of jobs. All metals for which covered electrodes are available can be welded by this process. Because this is a very versatile process, it is still extensively used in the fabrication of ships, bridges, pressure vessels, and structures; however, it is used in its manual mode only.

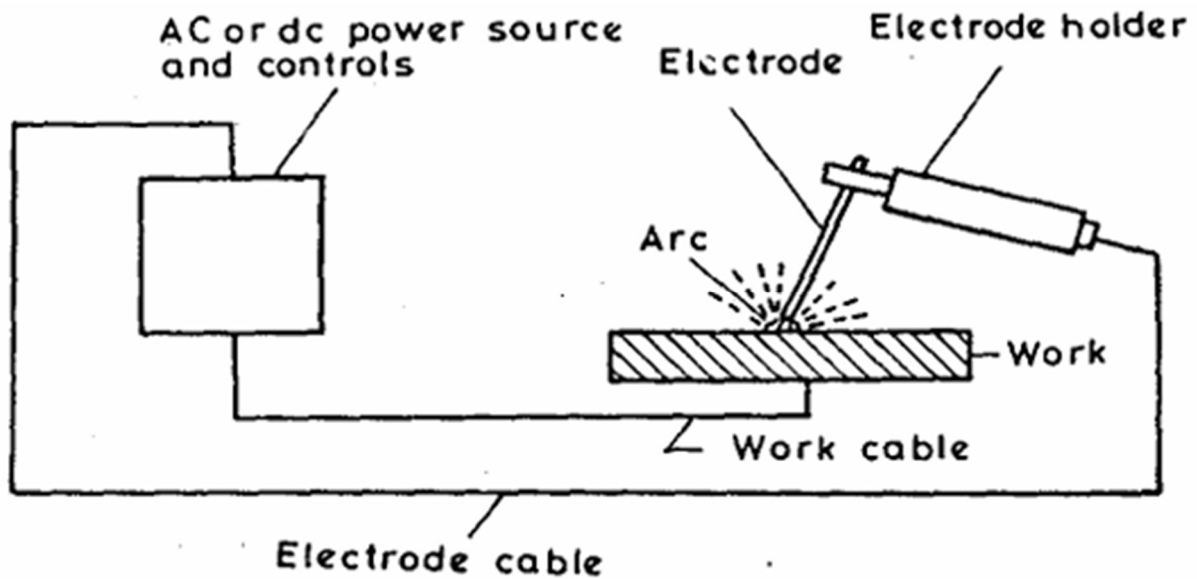


Figure 2.4 Shielded Metal Arc Welding

2.2.2.2 Submerged Arc Welding

Submerged arc welding (SAW) is a process in which continuous copper coated spooled wire is used in conjunction with loose granulated flux poured ahead of the arc so as to provide protective media to ward off the atmospheric gases from reacting with the molten metal pool. The electrode wire diameter may range between 2 and 10 mm. Both ac and dc power sources are used though dc with electrode positive (deep) is the preferred choice. SAW is mainly used in the downhand welding position in both automatic and semi-automatic modes. The former is a more popular mode and a set-up for the same is shown in Figure 2.5

The weld joint produced by submerged arc welding is of very high quality and consequently this process finds extensive use in joining thick plates in long, linear seams as are encountered in ships, pressure vessels, bridges, structural work, welded pipes, and nuclear reactors.

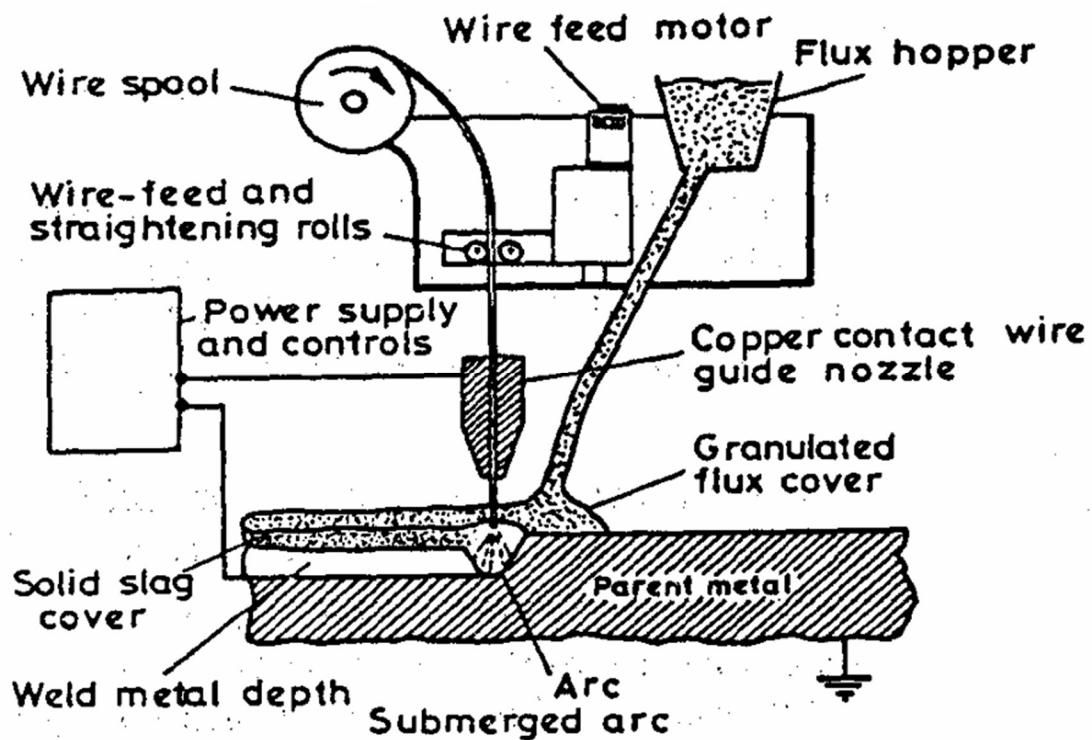


Figure 2.5 Submerged Arc Welding

2.2.2.3 Gas Tungsten Arc Welding

Gas tungsten arc welding (GTAW) or tungsten inert gas (TIG) welding employs a non-consumable tungsten electrode with an envelope of inert shielding gas (argon, helium, etc.) to protect both the electrode and the weld pool from the detrimental effects of surrounding atmospheric gases.

Both AC and DC power sources are used for GTAW. The tungsten electrode employed varies in diameter from 0.5 to 6.5 mm, and the current carrying capacity varies accordingly between 5 A and 650 A. The welding torch used for carrying current higher than 100 A is normally water-cooled. The process is used mainly in its manual mode. Figure 2.6 shows a schematic representation of the basic elements of an AC GTAW unit.

GTAW is an all-position welding process and gives the highest quality welds amongst the commonly employed arc welding processes and is, therefore, extensively used for welding most of the industrially useful metals and alloys, usually in thin grades. The aircraft industry, rocket and missile fabricators, and chemical and nuclear plant fabricators are the typical user industries of this process.

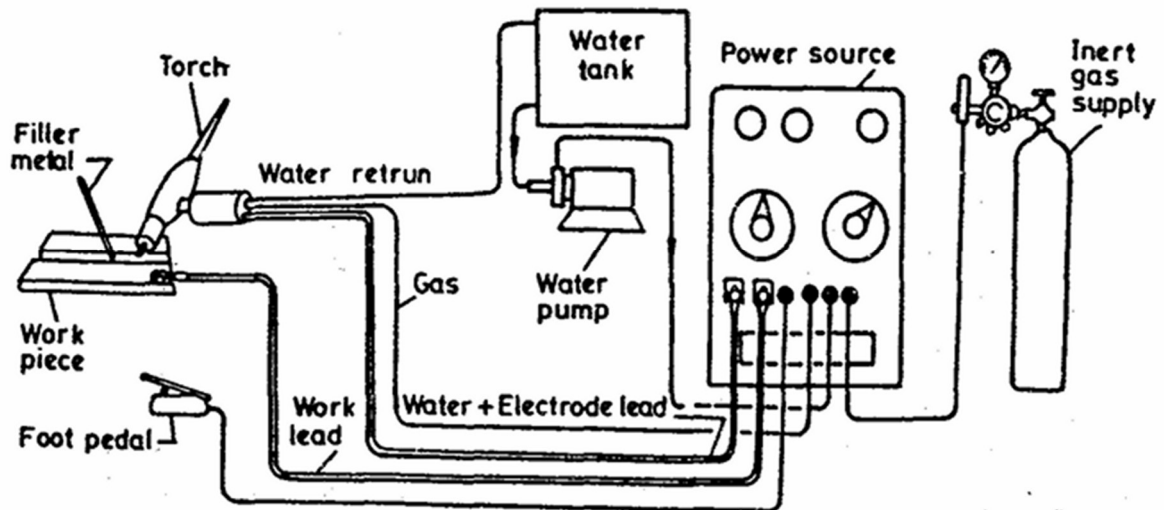


Figure 2.6 Gas Tungsten Arc Welding

2.2.2.4 Gas Metal Arc Welding

In gas metal arc welding (GMAW) process, a consumable wire of 0.8 to 2.4 mm diameter and wound in spool form is fed at a preset speed through a welding torch, wherein it is provided the electrical connection and the shielding gas. The arc, which is struck by direct contact between the wire electrode and the workpiece, is maintained at a constant length by the interaction of electrical parameters. The power source used is invariably of the rectified DC type. Both constant voltage and constant current power sources are in use.

Depending upon the work material, the shielding gas may be argon, helium, nitrogen, carbon dioxide, hydrogen, and their mixtures. When inert shielding gas is used, the process is more popularly known as MIG (metal inert gas) welding, and when CO₂ is used as the shielding gas, it is referred to as CO₂ welding or MAG (metal active gas) welding. GMAW is an all-position semi-automatic welding process, though its automatic versions are also available. A set-up for semi-automatic GMAW process is shown in Figure 2.7.

GMAW is a very versatile process and can be used for welding all metals for which compatible filler wires have been developed. However, its typical applications include medium-gauge fabrication such as structurals, earth-moving equipment, plate and box girders, and automobile bodies. This process has great potential for use with robotic welding systems.

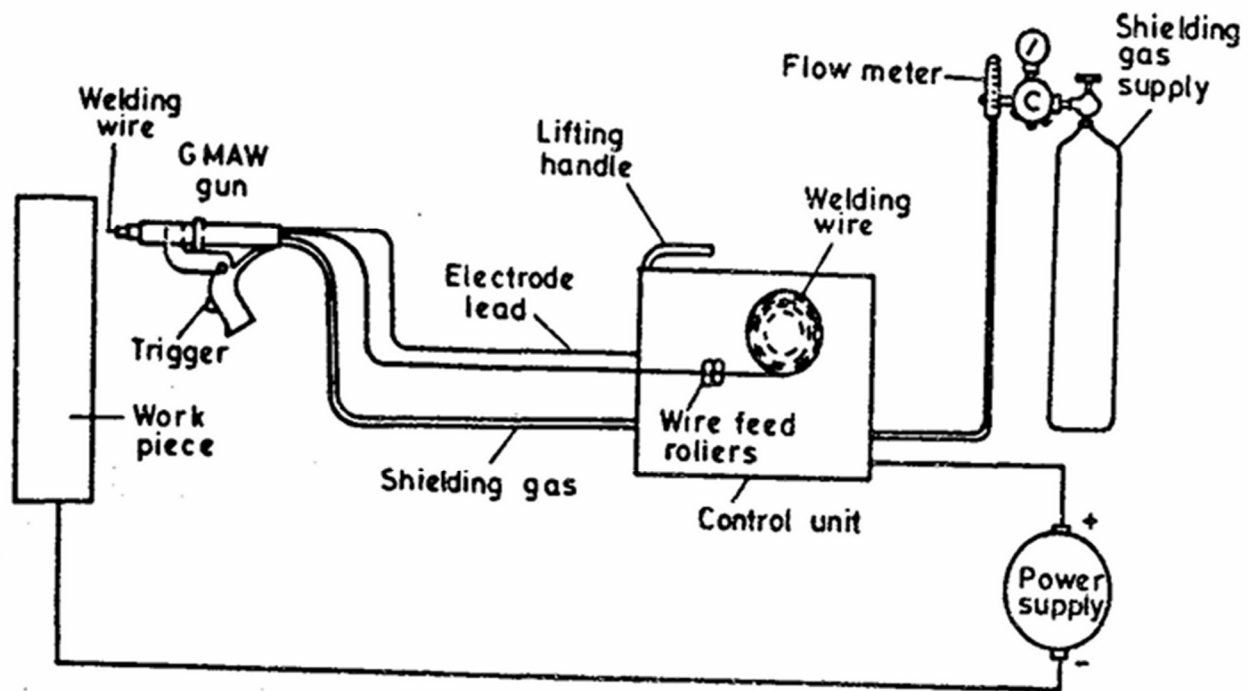


Figure 2.7 Gas Metal Arc Welding

2.2.2.5 Oxy-Acetylene Welding

In this process, acetylene gas is mixed with oxygen in the gas welding torch and is then burnt at the torch tip to give a flame with a temperature of about 3300°C, which can melt most of the ferrous and non-ferrous metals in common use. Figure 2.8 shows a standard set-up for oxy-acetylene welding.

Three types of flames are used in oxy-acetylene welding. The nature of the flame depends upon the ratio of the two gases. The neutral flame is most often used for welding most materials, such as low carbon steels, cast steel, and cast iron. The oxidising flame has a higher proportion of oxygen than acetylene and is used for welding manganese steel, brass, and bronze, whereas the carburising flame has a higher proportion of acetylene and is used for welding aluminium, nickel, and similar metals.

The heat transfer to the work in this process is very poor (about 30%) and may lead to a wide heat-affected zone (HAZ) around the weld. The welding speed is also accordingly low.

Typical applications of oxy-acetylene welding include welding of root runs in pipes and other multi-run welds, light fabrications such as ventilation and air-conditioning ducts, and motor vehicle repairs. A large percentage of general repair work is also carried out using this process.

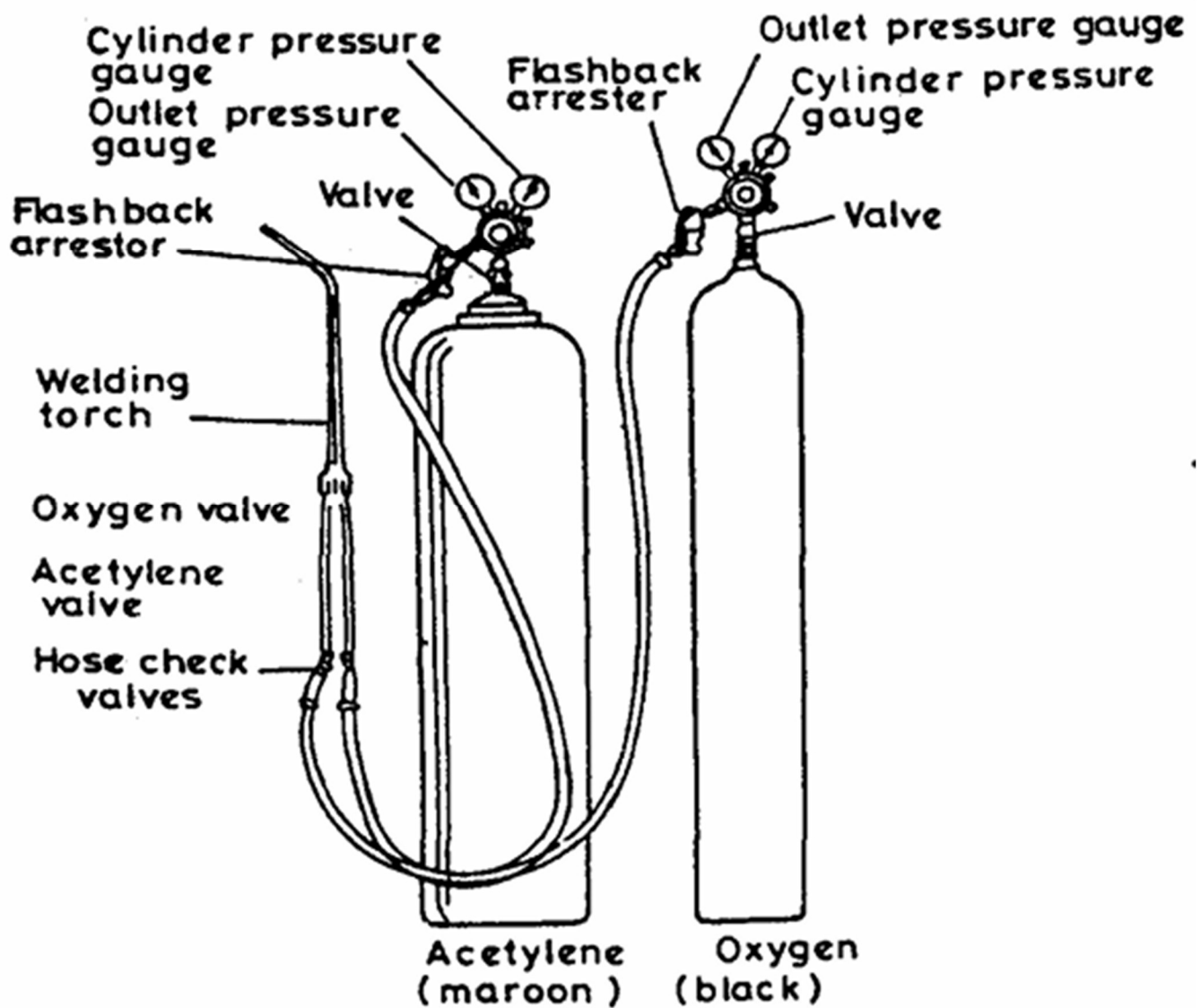


Figure 2.8 A standard set-up for Oxy-acetylene welding

2.2.3 Spot Welding Processes

In this class of processes, the materials are joined at a spot the size of which depends upon the design specifications and is controlled by the electrode size and the magnitude of the welding current.

Resistance Spot Welding

In the resistance spot welding process, overlapping sheets are welded by the flow of current between two cylindrical electrodes. The main equipment for spot welding is the spot-welding machine, which consists of a step-down transformer, a time control unit, and a pair of copper alloy electrodes.

As voltage plays no direct role in resistance welding, it is kept low, typically between 5 to 25 volts, while the current is usually very high (100–50,000 A). However, the current flows only for a short duration of time (0.06 to 3 seconds). Application of pressure to achieve forge between the two sheets is an essential aspect of the process. Figure 2.9 shows a general-purpose spot-welding machine.

Spot welding is mainly used for lap welding of thin sheets, particularly in the fabrication of automobile and refrigerator bodies, as well as high-quality work in aircraft engines.

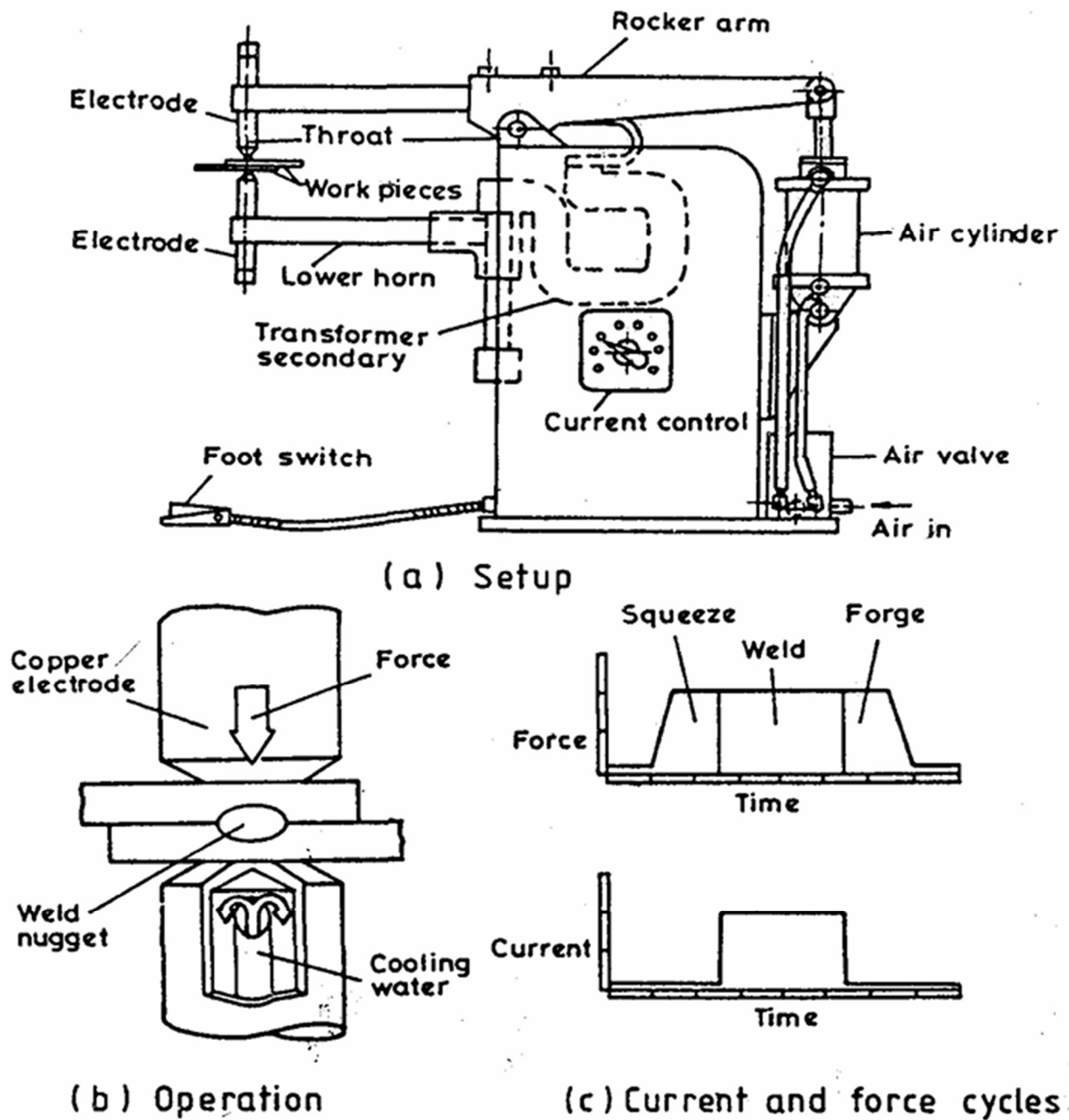


Figure 2.9 Essential features of a rocker arm type spot welder; operation details with current and force cycles

2.3 Arc welding parameters

The general parameters of welding involve current, length of arc, angle, manipulation, and speed.

Current: the amperage dictates the size and penetration of a weld.

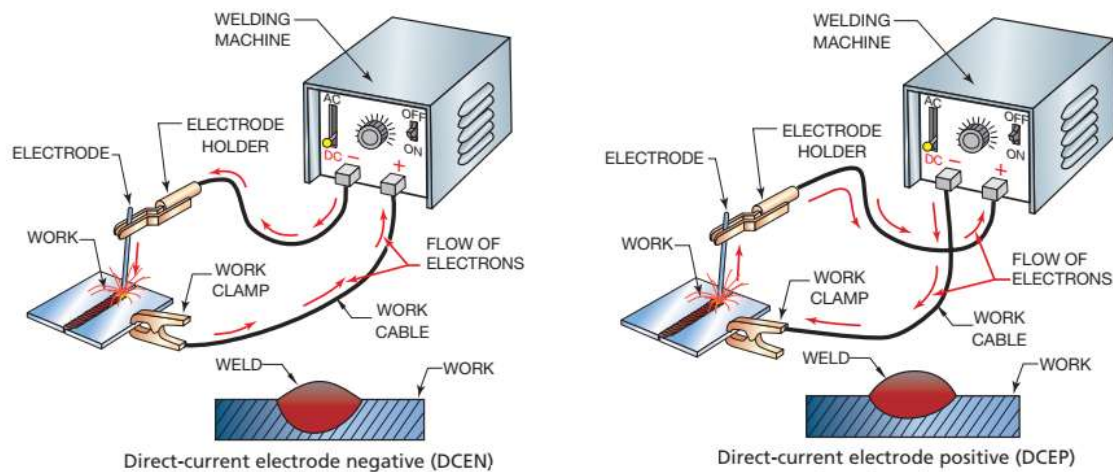


Figure 2.10 Direct Current welding

Length of Arc: the distance of the work plates and arc of a welding electrode.

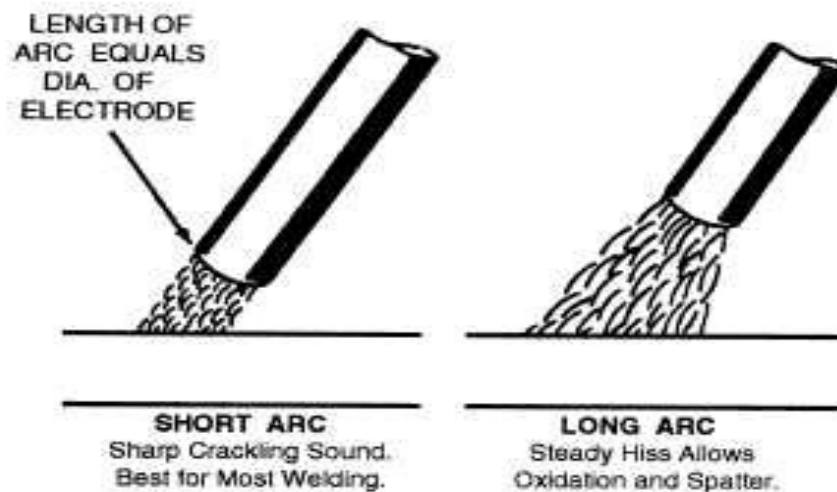


Figure 2.11 Arc length

Angle: includes both work and travel angle. The work angle is the relationship between the joint and the torch. Travel angle is the relationship between the torch and the line of travel.

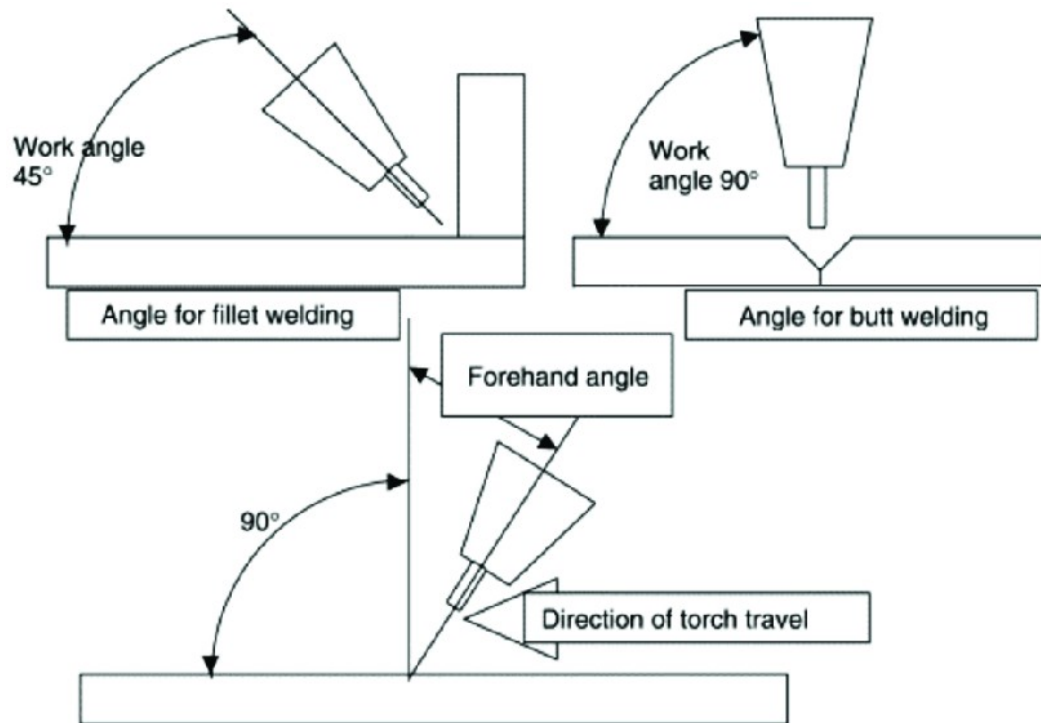


Figure 2.12 Various welding angles

Manipulation: the movement of the welder as they guide electrodes along the joint.

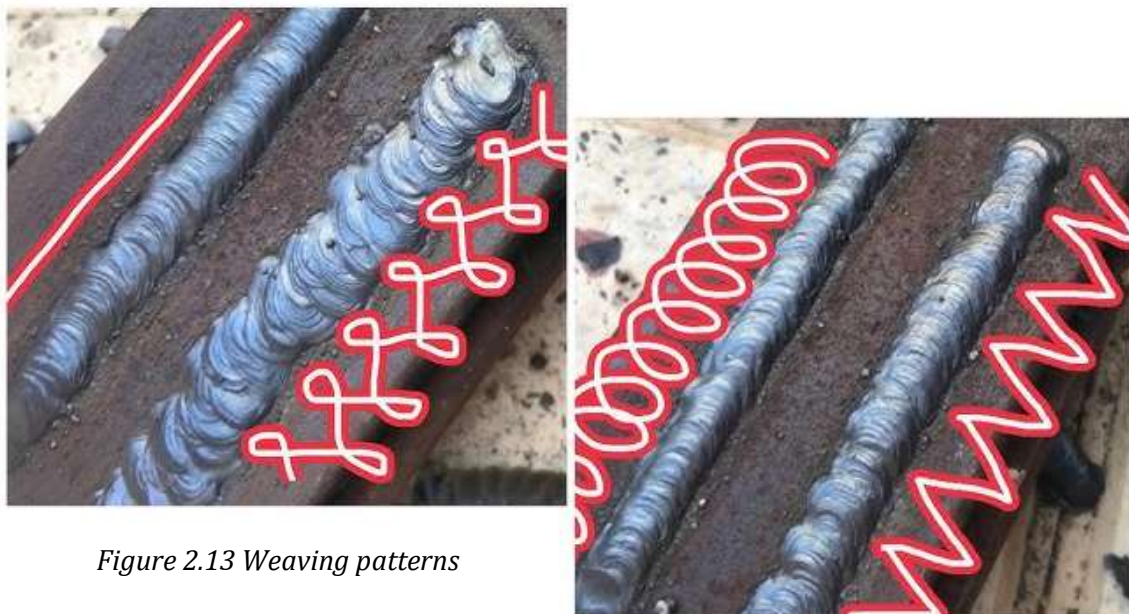


Figure 2.13 Weaving patterns

Speed: the quickness of movement in respect of the welding penetration.

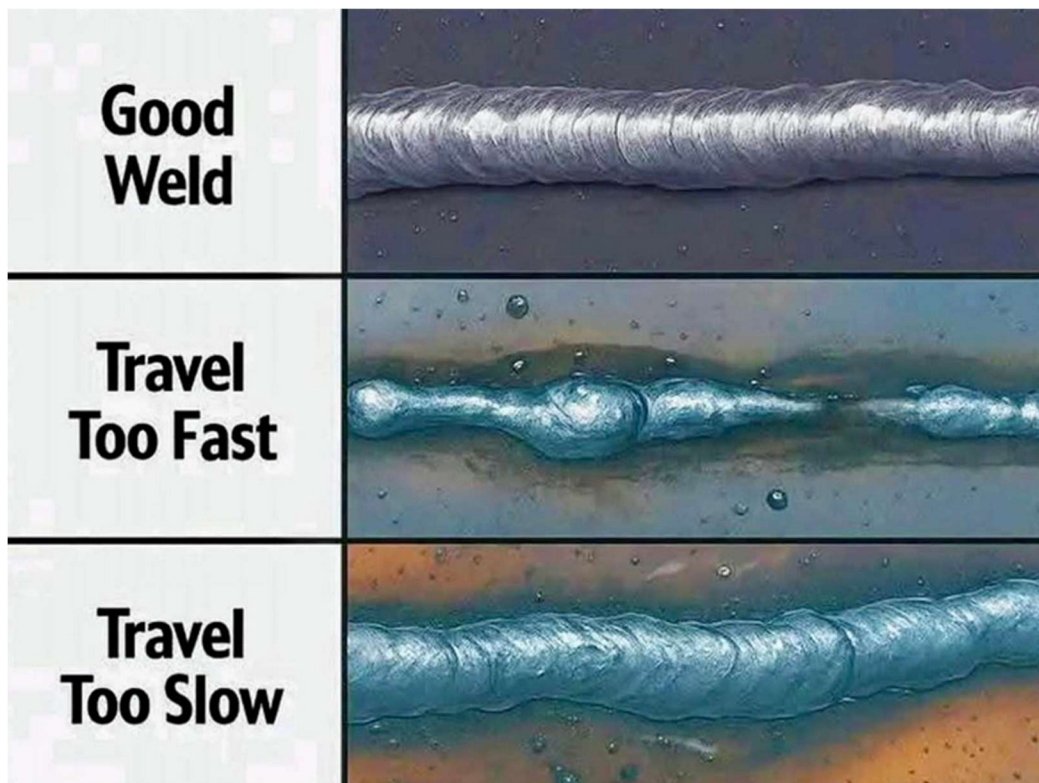


Figure 2.14 Speed effect on produced weld

Arc Voltage: the voltage between the electrode tip and the weld. The arc length is the distance from the electrode tip to the surface of the weld pool. Arc length is important for strong welding joints.

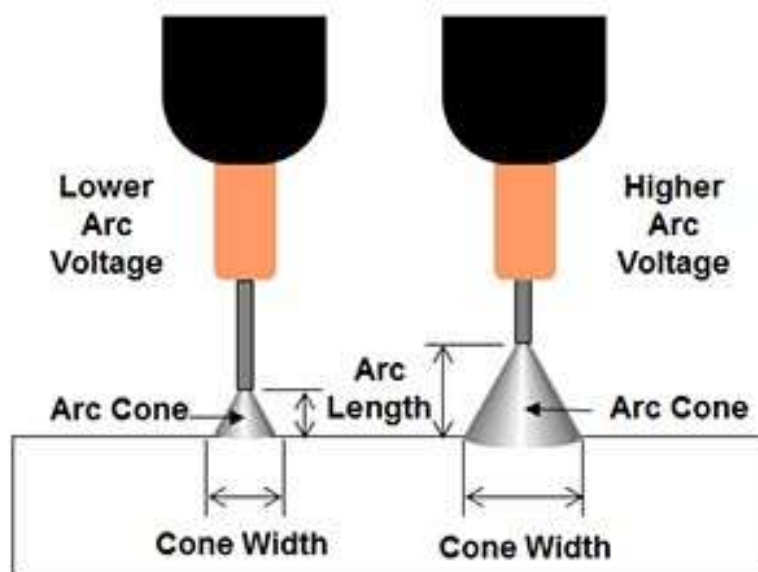


Figure 2.15 Arc voltage effect on cone width

Shielding Gas Composition: protects the arc and the molten weld pool from atmospheric oxygen and nitrogen. Proper protection can ensure a strong weld.

Metal Type	100% Argon	100% Helium	100% CO2
Carbon Steel	Good in TIG, not preferred in MIG	Not Preferred in TIG or MIG	High penetration in MIG, not suitable for TIG
Stainless Steel	Good in TIG, not preferred in MIG	Not Preferred in TIG or MIG	Not suitable with any method
Aluminum	Good in TIG and MIG	Good for DC TIG	Not suitable with any method
Copper Alloys	Good in TIG and MIG	Good in TIG	Not suitable with any method
Magnesium	Good in TIG and MIG	Good for DC TIG	Not suitable with any method
Nickel alloys	Good in TIG	Not Preferred in TIG or MIG	Not suitable with any method

Table 2.1 Common shielding gases used in arc welding

Inspection of parameters before, during, and after the welding process is essential. This inspection is required of the welder because they can see the progression of the weld as it happens, not just after the weld has been completed. Further, some non-destructive testing methods may not identify the strength of the weld, leaving the inspection up to the welder themselves.

2.4 Selected welding process

Speed, accuracy, and filler-metal requirements are the defining criteria when selecting a welding process for any automated system. High travel speeds increase throughput but reduce precision and raise the risk of porosity and bead instability, while slower speeds improve accuracy and reduce discontinuities—an essential factor when working with thin materials. The role of filler metal is also decisive: although it can strengthen a joint, many precision applications benefit from fusion welding without filler, producing a cleaner and more uniform bead.

Among the common processes used in automatic welding, primarily MIG and TIG (Figure 2.16), each offers distinct advantages. MIG welding delivers high deposition rates, faster operation, and excellent suitability for thicker sections, making it ideal for large-scale or high-volume production. However, its higher heat input, reliance on consumable wire, and lower arc precision make it less favorable for delicate components. TIG welding, by contrast, provides unmatched control, superior arc stability, minimal spatter, and the ability to weld thin sheets with exceptional accuracy, especially when performed without filler metal.

Given that our application involves a thin metal sheet of limited length and requires strictly longitudinal seam welding, TIG welding is the most appropriate choice. Its clean fusion process, low defect rate, and precise heat management align perfectly with the demands of the automated setup and ensure consistent, high-quality welds tailored to small-scale workshop environments.

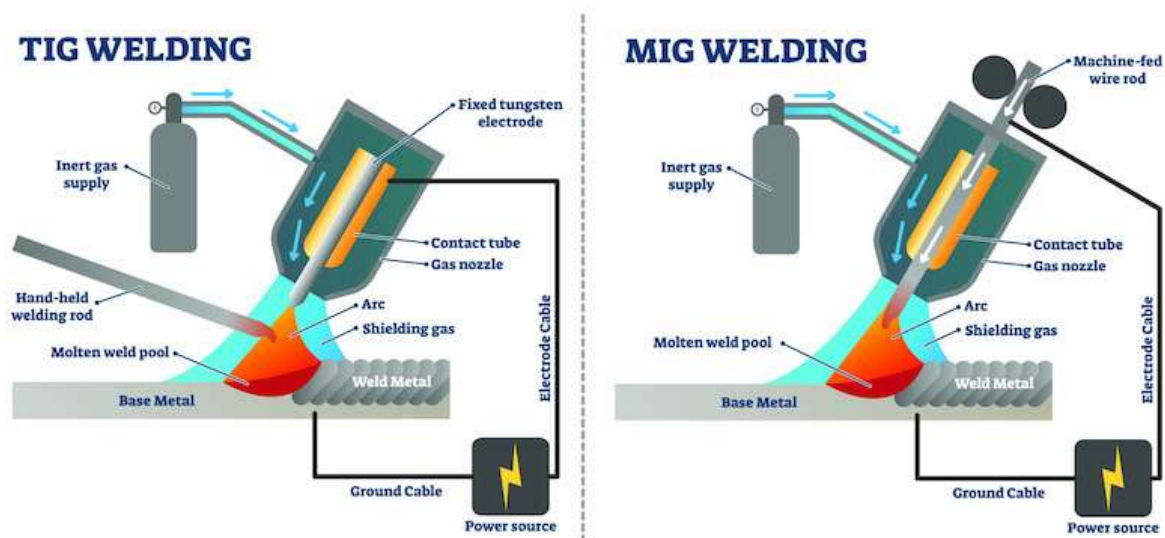


Figure 2.16 Schematics Comparing TIG and MIG welding processes

2.5 GTAW Specifications

2.5.1 Current Types in GTAW

Gas Tungsten Arc Welding can be performed using either alternating current or direct current (Figure 2.17), with AC generally used when a balanced thermal effect or surface cleaning is required. In most applications, however, direct current is used in one of two configurations: electrode negative and electrode positive. In the electrode negative mode, most of the heat is directed into the workpiece, providing deeper penetration and stable arc behavior suitable for standard welding tasks where controlled heat input is essential. In the electrode positive mode, more heat is concentrated at the electrode, resulting in reduced penetration but enhanced cleaning action on the material surface, which makes it useful for thin sections or situations where oxide removal is important. Each configuration serves specific purposes, and the choice depends on material characteristics, thickness, and the desired weld outcome.

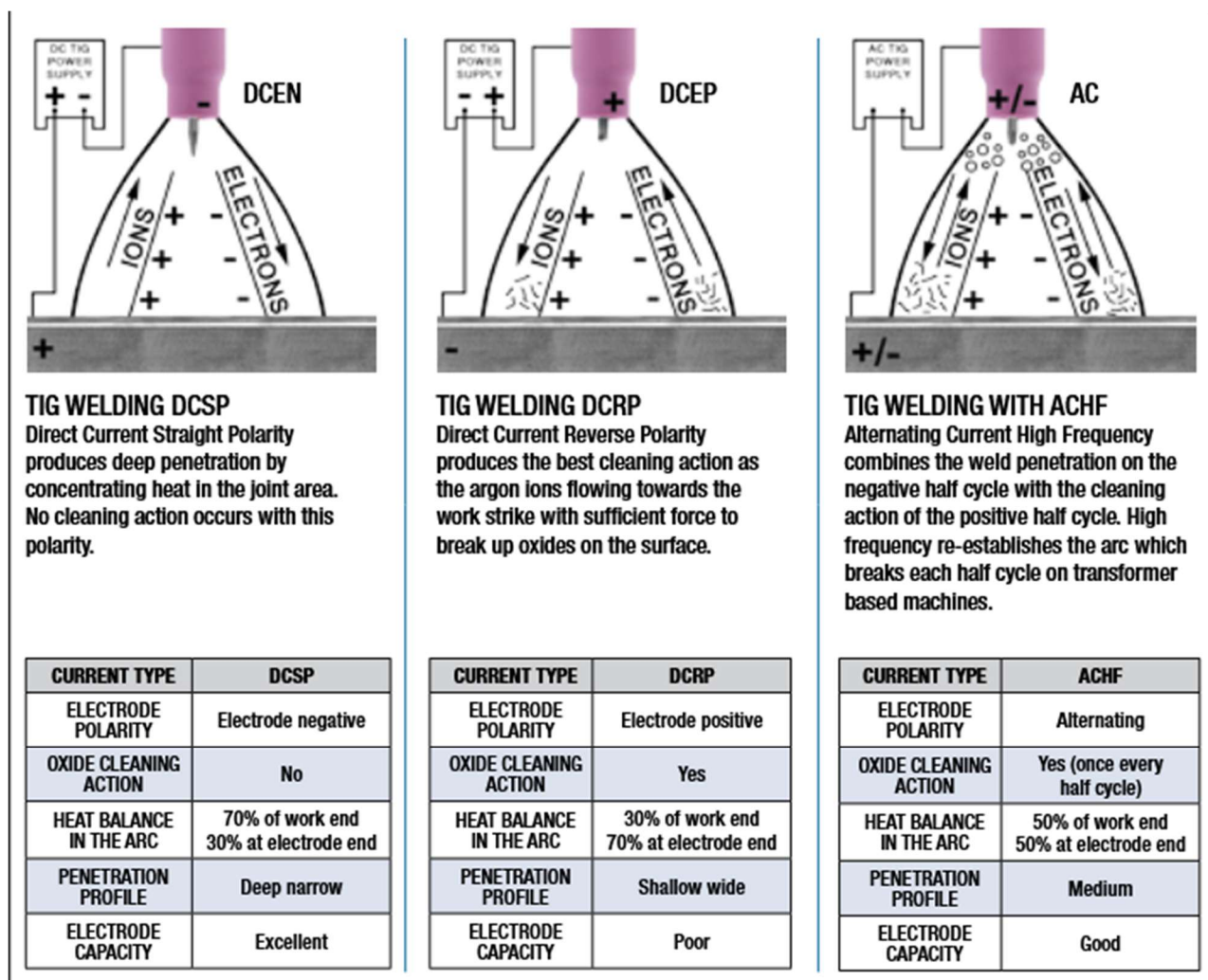


Figure 2.17 Discussing GTAW currents and their differences

2.5.2 Tungsten Electrode Types and Geometry

Tungsten electrodes serve as the terminal point for arc generation, and their performance is often enhanced by alloying with rare-earth oxides to improve thermionic emission. While thoriated electrodes have been a historical standard for DCEN applications due to their high current capacity, the radioactive nature of thorium oxide presents significant inhalation risks during grinding. Lanthanated electrodes have emerged as a superior, non-radioactive alternative, offering better arc-start reliability, lower operating temperatures, and versatility across both AC and DCEN configurations. The geometric preparation of the electrode tip is equally critical; acute angles (e.g., 30°) provide a constricted, high-density arc for deep penetration, while blunter angles (up to 120°) reduce peak current density to mitigate distortion in thin-section welding (figure 2.18).

Name	Color	Type	Current	Application
WP	Green	<u>Pure</u>	AC	Stable arc with alternating current, used to weld light alloys. Beware of the welding puddle contamination risk.
WT	Yellow 1%	<u>Thorium</u>	DC	Easy arc ignition, high current capacity and low welding puddle contamination risk. The use of AC current is difficult. Radioactive risks.
	Red 2%			
	Purple 3%			
WZ	White 0.7 to 0.9%	<u>Zirconium</u>	AC	Identical WP, with higher current capacity and less welding puddle contamination risk.
	Brown 0.15 to 0.50%			
WL	Black 0.8% to 1.2%	<u>Lanthanum</u>	AC/DC	Similar to WT, there is no radioactive risk, but it is less efficient.
	Gold 1.3% to 1.7%			
	Blue 1.8% to 2.2%			
WC	Grey 1.8% to 2.2%	<u>Cerium</u>	AC/DC	Similar to WT and WL but it has less current capacity.
WCL 1/1%	Pink	<u>Cerium-Lanthan</u>	DC	Simplified arc ignition and a long life span. It offers an excellent compromise.

Figure 2.18 The different types of electrodes used in orbital TIG welding

2.5.3 Shielding Gas in GTAW

Shielding gases act as a thermodynamic medium that determines arc ionization potential, heat transfer, and protection of the molten pool from atmospheric contamination. Argon is the industry standard due to its high density (figure 2.19), which provides efficient coverage at low flow rates, and its low ionization potential, which facilitates stable arc initiation. Helium, while requiring higher flow rates, offers superior thermal conductivity and a higher ionization potential, resulting in hotter arcs and significantly deeper penetration. Maintaining laminar flow is paramount; excessively high flow rates induce turbulence that pulls in ambient air, leading to porosity and oxidation, while insufficient flow fails to provide adequate volumetric coverage.



Figure 2.19 Argon gas tank

2.5.4 Filler Metal and Autogenous Welding

GTAW can be executed heterogeneously with filler metal or autogenously, where fusion is achieved solely through the base materials. Autogenous welding is highly efficient for thin-plate geometries and reduces the risk of inclusion defects, but it leaves the joint susceptible to the innate metallurgical flaws and solidification kinetics of the parent material. The introduction of filler metal allows for deliberate chemical buffering, such as adding grain refiners or stabilizers to offset brittle phase formation and improve impact toughness. Although filler metal improves mechanical properties in dissimilar joints, multi-pass operations can increase global residual thermal stresses due to repeated thermal cycling.

2.6 Welding Testing and Inspection

2.6.1 Destructive Testing

Destructive testing involves forcing a welded specimen under catastrophic mechanical loads to quantify properties such as ductility and ultimate tensile strength. Transverse tensile testing identifies the failure plane, with optimal joints fracturing in the base metal rather than the fusion zone. Bend testing evaluates macroscopic ductility and resistance to crack propagation within the fusion matrix and heat-affected zone (HAZ). Additionally, macrostructure examination uses chemical etchants to visually expose the internal topography, revealing penetration depth, HAZ extent, and localized lack of fusion.

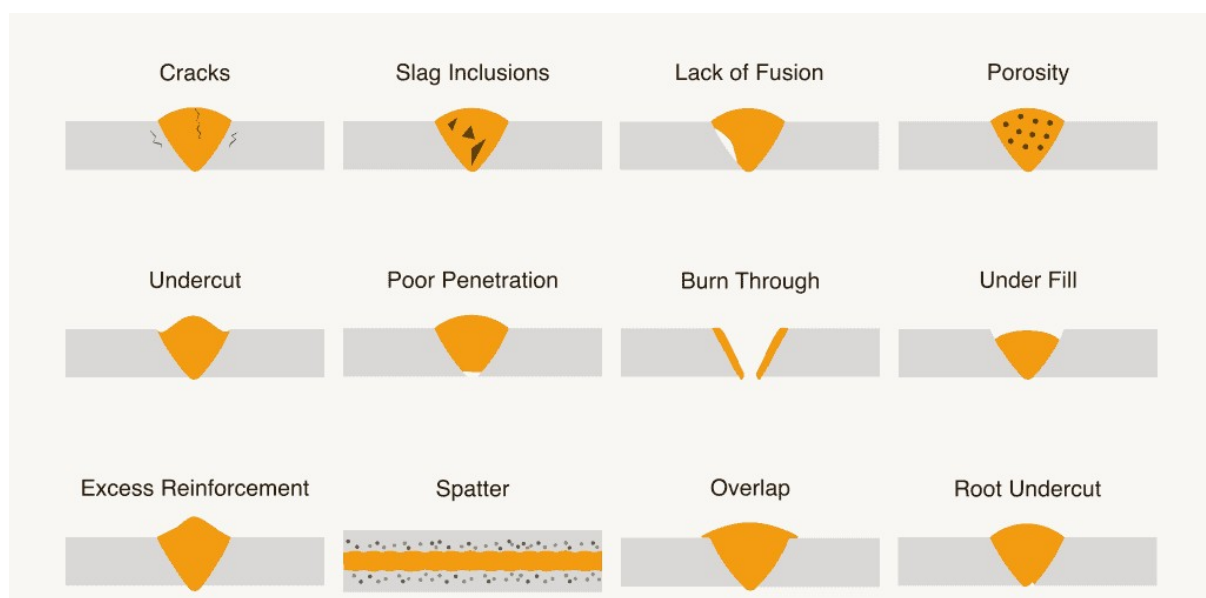


Figure 2.20 Common welding defects

2.6.2 Non-Destructive Testing

Non-destructive testing (NDT) validates the integrity of production components without impairing their future serviceability (figure 2.21). Visual testing (VT) and liquid penetrant testing (PT) are used to identify surface-breaking flaws such as undercut, porosity, and microscopic cracks. Volumetric interrogation is achieved through Ultrasonic Testing (UT), which uses acoustic waves to map internal planar flaws, or Radiographic Testing (RT), which uses ionizing radiation to detect internal voids. Modern applications may also utilize Eddy Current Testing (ECT) for high-speed detection of near-surface micro-cracks or Acoustic Emission (AE) to "listen" for real-time cracking during the cooling phase.



Figure 2.21 Nondestructive testing methods

2.7 Safety in TIG Welding

TIG welding presents severe occupational hazards including electrocution, radiation burns, and toxicological exposure. Electric shock is a primary risk, with secondary voltage shocks from the arc circuit potentially causing fatal cardiac arrest, particularly when using alternating current (AC). The plasma arc emits intense ultraviolet (UV) and infrared (IR) radiation, which can cause photokeratitis and permanent retinal damage (figure 2.21). Furthermore, the process generates a toxic plume of metal oxides and ozone, requiring localized mechanical exhaust systems and HEPA filtration to prevent respiratory diseases. Personal Protective Equipment (PPE) such as leather clothing, auto-darkening helmets, and specialized respirators act as the final barrier against these thermodynamic and chemical threats (figure 2.22).



Figure 2.22 Warning sign of "welding flash"



Figure 2.23 Personal Protective Equipment

2.8 Workpiece used in this project

By prioritizing application-specific parameters, this project achieves an optimal balance between system simplicity and operational precision. The scope encompasses commonly utilized metal sheets, with a primary focus on thickness up to 3 mm (figure 2.24), including both standard carbon and stainless-steel alloys. This targeted material selection facilitates the optimization of the autogenous welding process, significantly mitigating the risk of structural defects.

Furthermore, defining these workpiece constraints streamlines the mechanical requirements, ensuring a controlled weld path and consistent penetration. This focused approach also enables the integration of a specialized clamping assembly designed to manage thermal expansion effectively. By securing the workpiece according to its specific gauge, the system actively prevents thermal distortion during the critical heating and cooling phases of the welding cycle (figure 2.25).



Figure 2.24 Plain carbon steel sheet

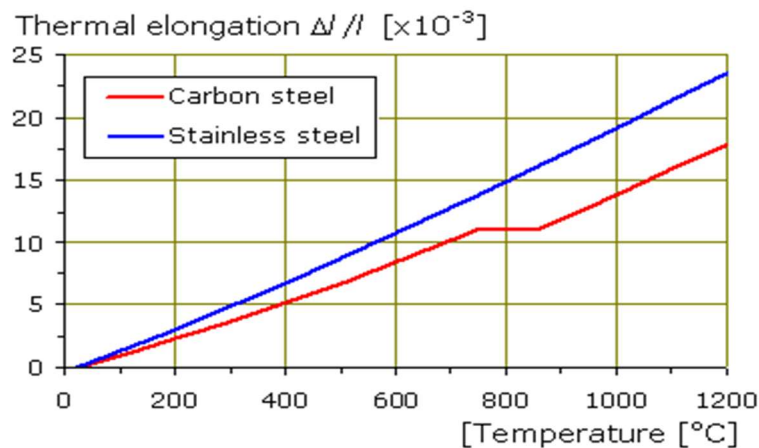


Figure 2.25 Thermal elongation and temperature relation

2.9 Selected welding components

2.9.1 GTAW inverter set

ShenYuan TIG/MMA-200A-21:

- Rated Input Voltage: 1~230 V / 1~110 V
- No-load Voltage: 56 V
- Current Range: 10–200 A
- Rated Duty Cycle: 60%
- Input Capacity: 4.5 KVA
- Insulation Class: F
- Weight: 7 kg
- Dimensions (cm): 40 × 22 × 29



Figure 2.26 ShenYuan TIG/MMA-200A-21 inverter system

2.9.2 Tungsten electrode

2% Thoriated Tungsten Electrodes (WT20)

- **Composition:** Contains approximately 2% Thorium Dioxide (ThO_2) dispersed throughout the tungsten.
- **Performance:** Offers high resistance to weld pool contamination and maintains a sharpened point under high-temperature conditions.
- **Identification:** Features a distinctive red tip for easy classification in the workshop.
- **Finish:** Supplied with a ground finish to ensure uniform electrical conductivity and consistent arc performance.



Figure 2.27 2% Thoriated Tungsten Electrodes (WT20)



Chapter 3: Mechanical System

3.1 Introduction

Our objective is to develop a simple, medium-scale, and reliable automated seam welding machine for university laboratories and small workshops. The design is guided by fundamental operational requirements such as weld length, welding speed, workpiece thickness, and the level of accuracy needed. Structural rigidity, ease of fabrication, and straightforward maintenance remain core priorities to ensure stable performance for end user.

The mechanical structure, motion system, and welding mechanism are engineered to deliver smooth torch travel, repeatable positioning, and dependable operation throughout repeated trials and student use. By relying on accessible manufacturing techniques and commonly available components, the system remains practical to build, operate, and service within the capabilities of academic settings and small-scale workshops. Overall, the design realizes a balanced solution that provides educational value, reliable functionality, and cost-effective performance for its intended users.



Figure 3.1 Industrial automatic seam welding machine

Large-scale automated welding machines (Figure 3.1) typically rely on complex industrial robotic systems intended for high-volume production. Such solutions exceed the needs of university laboratories and small workshops, both in scale and cost. For our application, we adopted a straightforward and dependable gantry-style chassis (Figure 3.2). This configuration offers a stable platform, predictable motion, and an overall structure that is easy to fabricate and maintain.

The system adopts a rigid, uncomplicated structure tailored to its low operating speeds and the modest size of its components, ensuring dependable performance without unnecessary bulk. The X-Z axis arrangement follows the proven layout of a basic two-axis milling setup (Figure 3.3), providing stable linear motion that preserves alignment and stand-off distance along the entire weld path. The welding mechanism follows the same practical design philosophy: it is intentionally limited to a single, well-defined function—longitudinal seam welding of thin metal sheets—to maximize precision, repeatability, and ease of control. By avoiding multi-purpose complexity, the design reduces mass, lowers fabrication and maintenance demands, and minimizes mechanical failure points. The result is a straightforward, serviceable mechanism that performs its intended task with consistent reliability and efficiency.



Figure 3.2 Gantry with robot arm



Figure 3.3 Two-Axis Mill

3.2 Welding mechanism

The goal is to make the system as easy to operate, assemble and maintain as possible. The decision was made to utilize metal sheets, as they are cheap, reliable and easy to transport and widely available in different materials. And they can be produced and processed very efficiently in comparison to other metal thicknesses.

3.2.1 Parts:

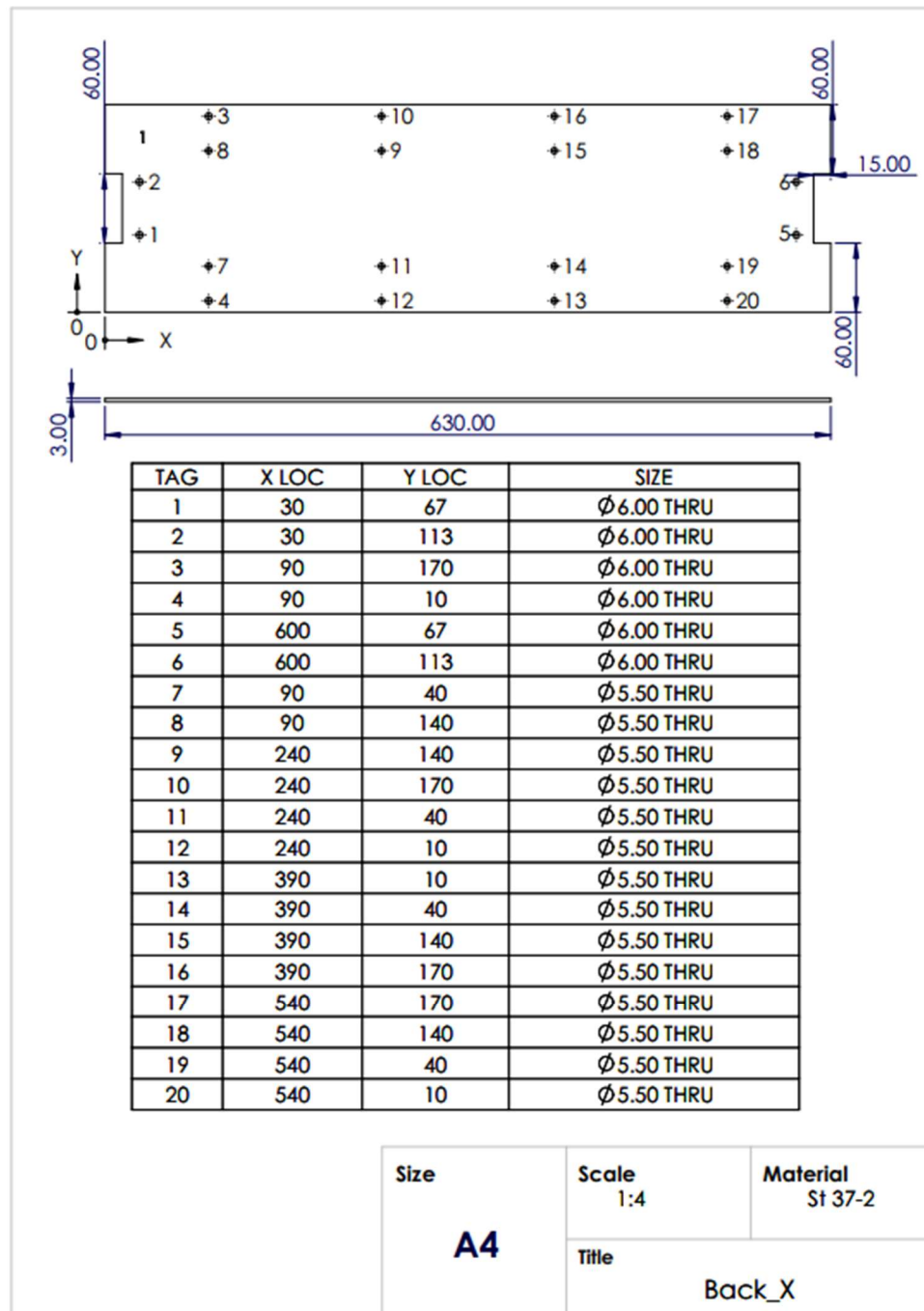


Figure 3.4 Back of X Axis, it holds Driver and driven side together, holds linear motion guides and ball screw (thickness=3mm)

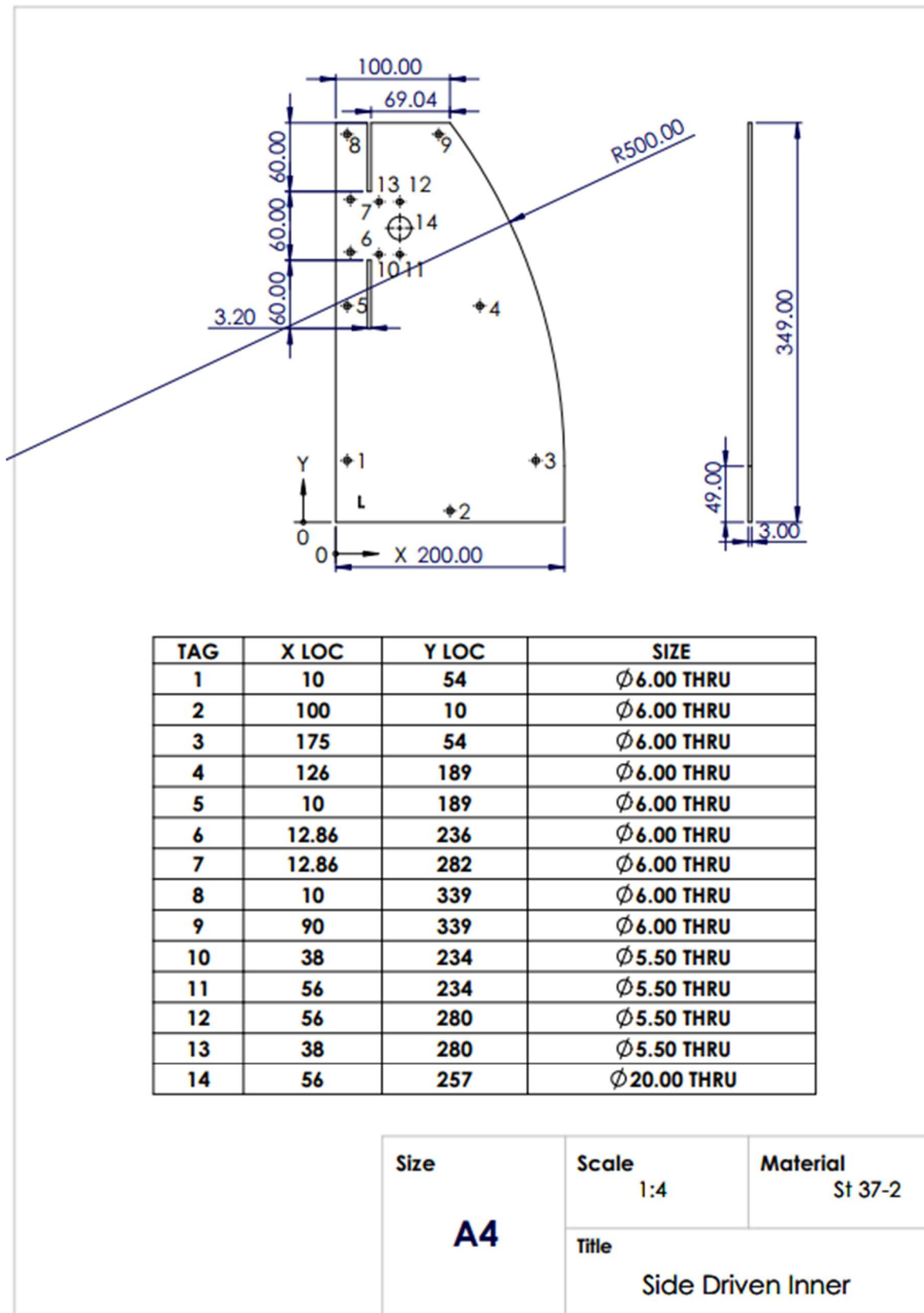


Figure 3.5 Inner part of the driven side, it holds the back of X-Axis in place in the driven side (thickness=3mm)

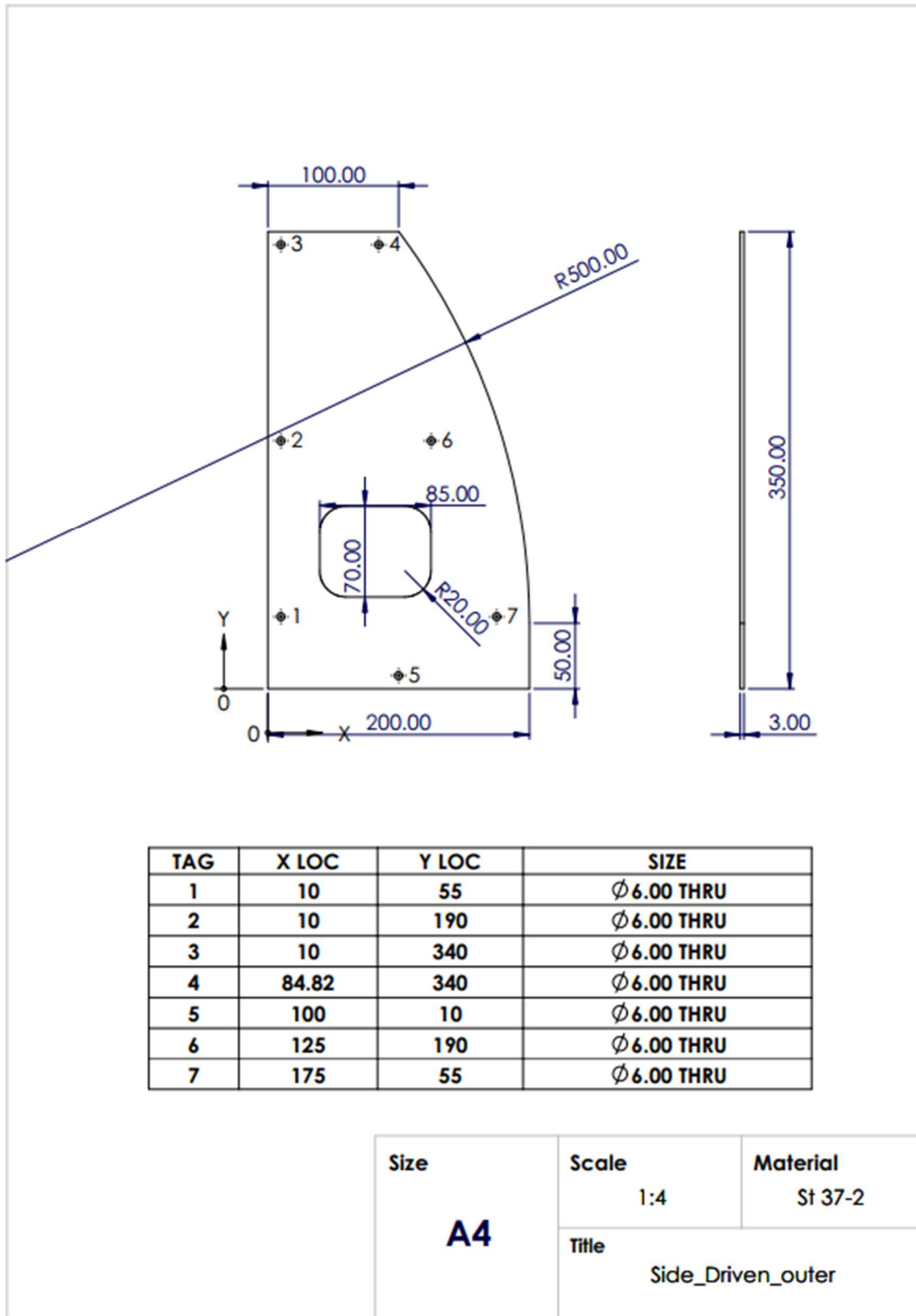


Figure 3.6 outer part of driven side, connected to the inner part by spacers and bolts gives the side greater inertia than if it were a thicker block (thickness=3mm)

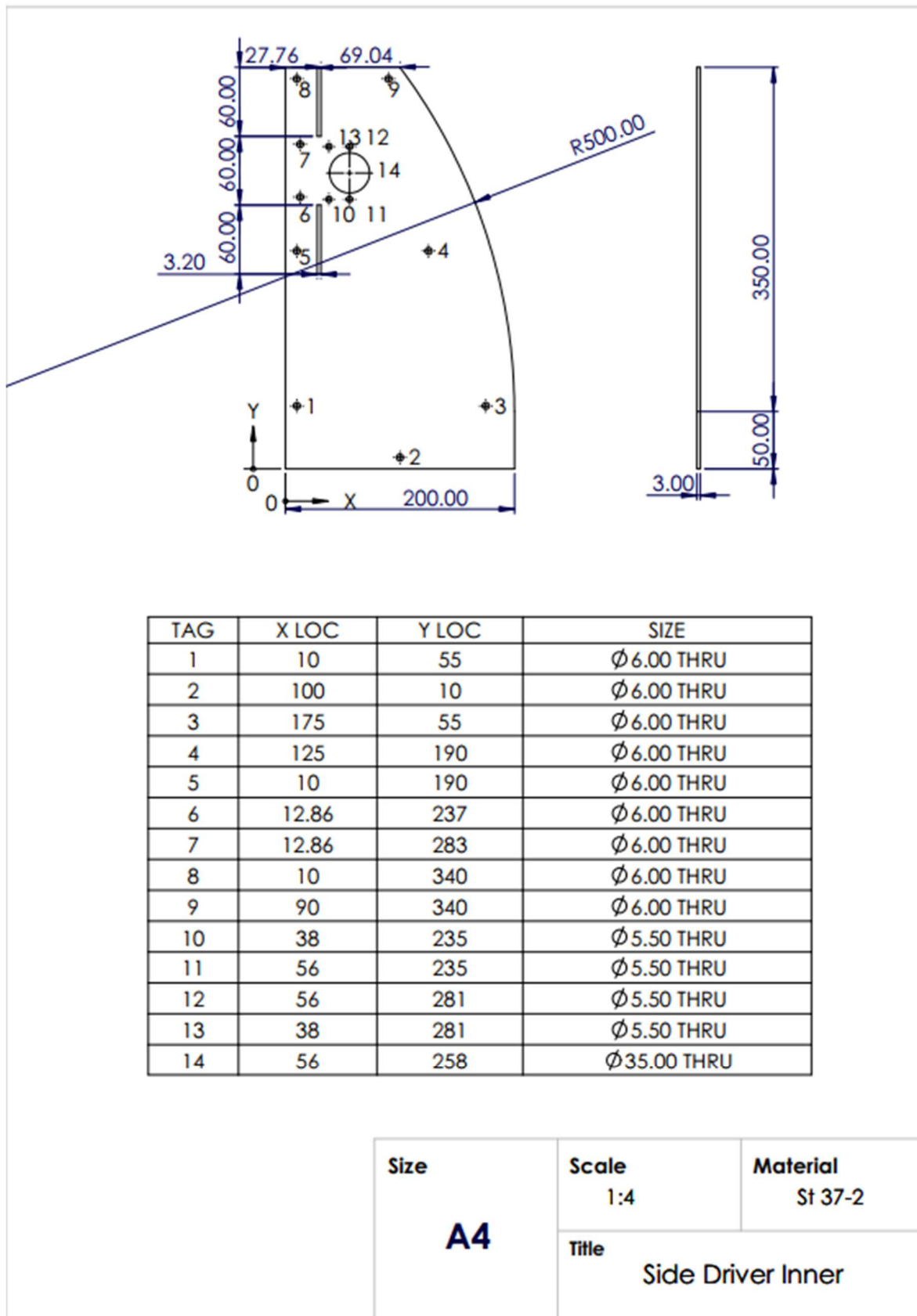


Figure 3.7 Inner part of drive side, where coupling will go through to mate with the ball screw (thickness=3mm)

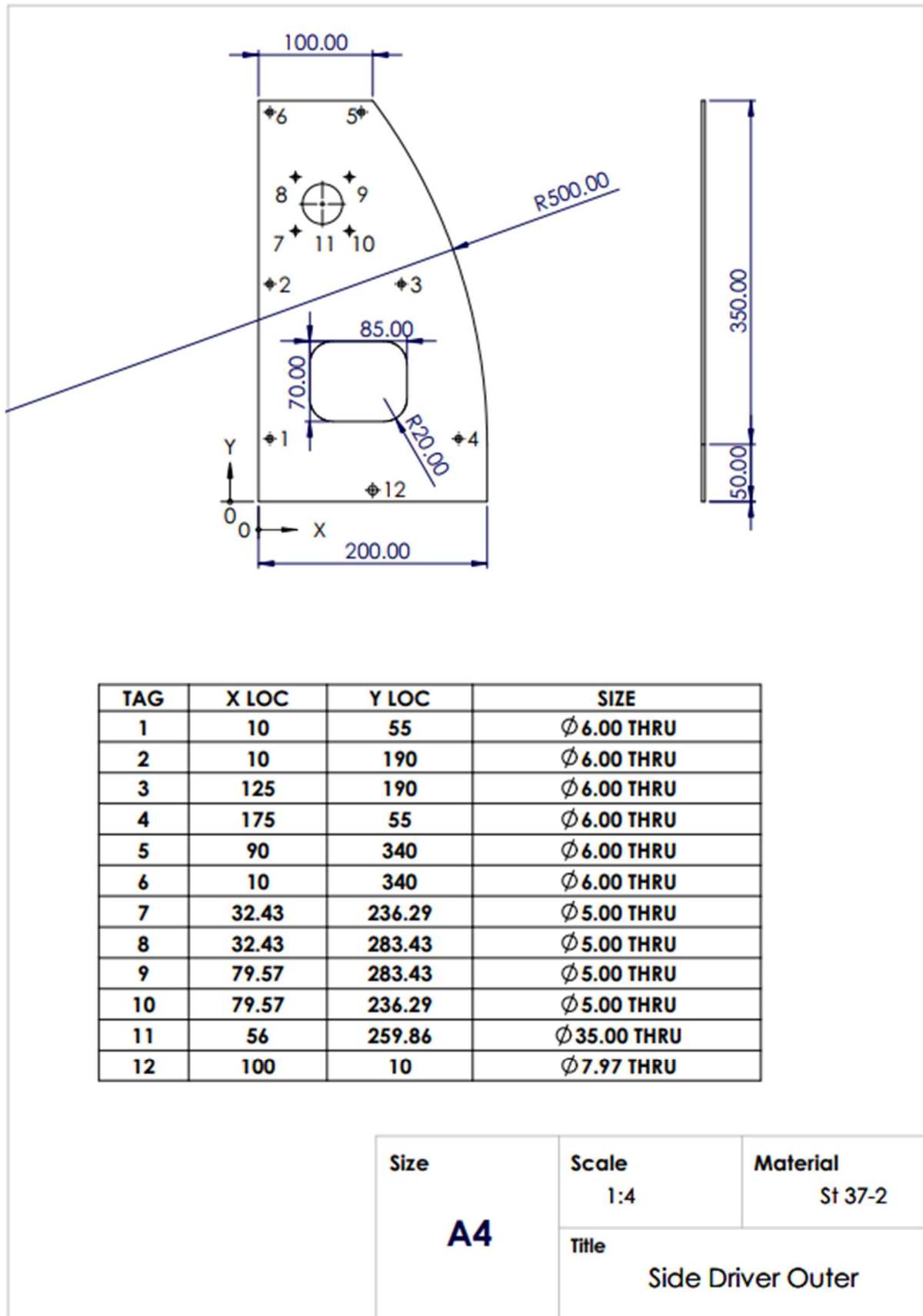
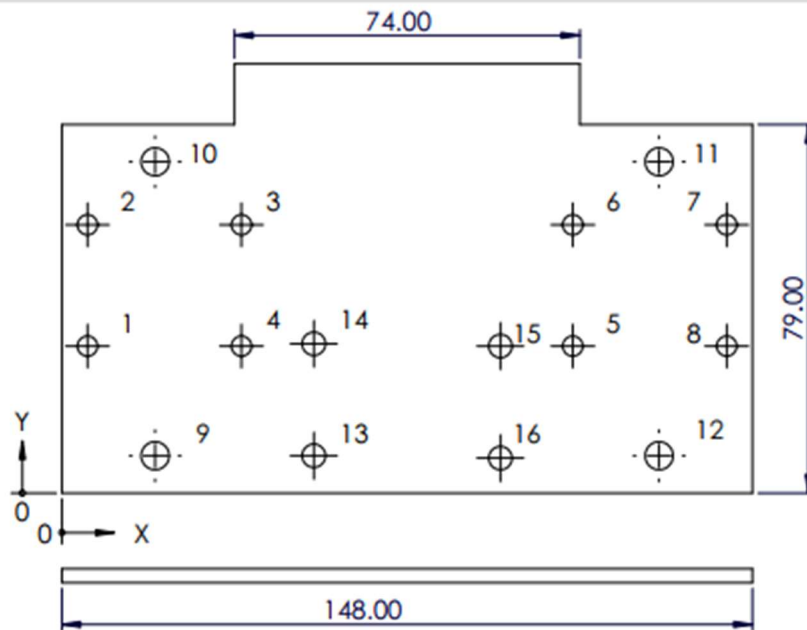


Figure 3.8 Outer part of driver side, where the motor will be held. The same connection in the driven side is used here to give equally effective vibration resistance. (thickness=3mm)



TAG	X LOC	Y LOC	SIZE
1	5.50	31.50	Ø 4.30 THRU
2	5.50	57.50	Ø 4.30 THRU
3	38.50	57.50	Ø 4.30 THRU
4	38.50	31.50	Ø 4.30 THRU
5	109.50	31.50	Ø 4.30 THRU
6	109.50	57.50	Ø 4.30 THRU
7	142.50	57.50	Ø 4.30 THRU
8	142.50	31.50	Ø 4.30 THRU
9	20	8	Ø 6.00 THRU
10	20	71	Ø 6.00 THRU
11	128	71	Ø 6.00 THRU
12	128	8	Ø 6.00 THRU
13	54	8	Ø 5.00 THRU
14	54	32	Ø 5.00 THRU
15	94	31.50	Ø 5.00 THRU
16	94	7.50	Ø 5.00 THRU

Size	Scale	Material
A4	1:1	St 37-2
	Title Torch Holder Base	

Figure 3.9 The base of torch holder, connected to the nut housing of leadscrew and to linear motion guides of Z-axis (thickness=3mm)

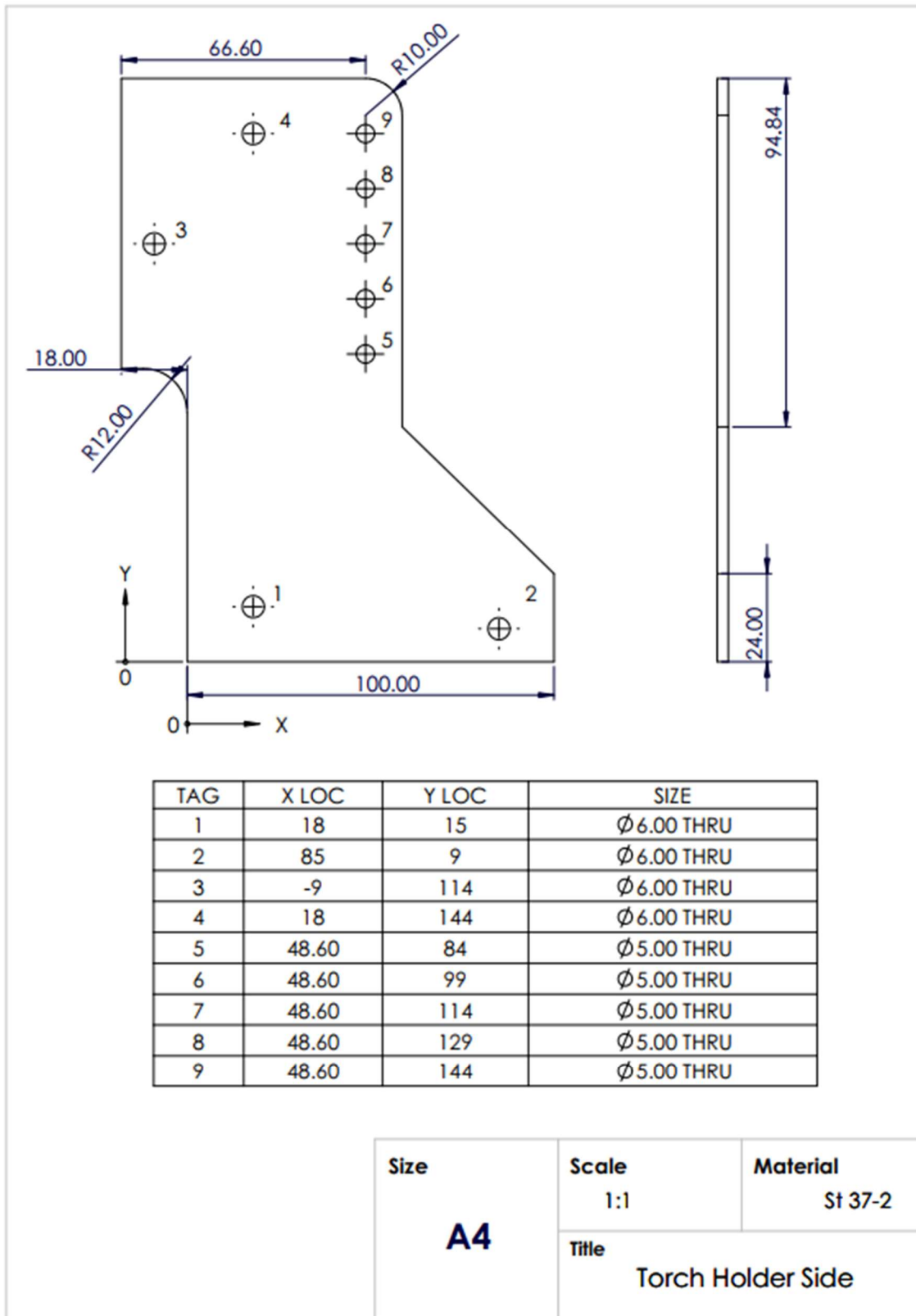


Figure 3.10 Sides of the torch holder, connected perpendicular to the holder base (thickness=3mm)

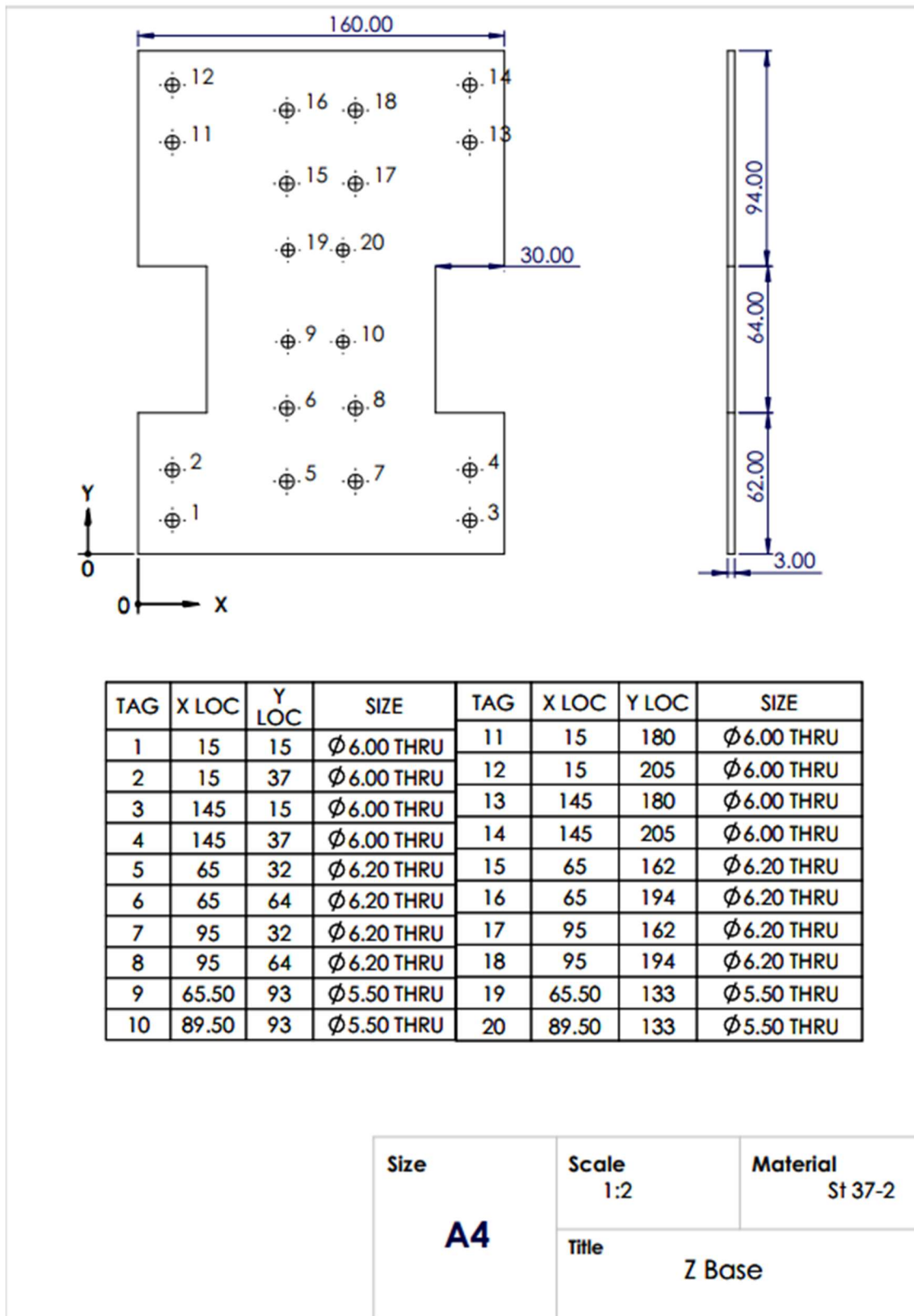
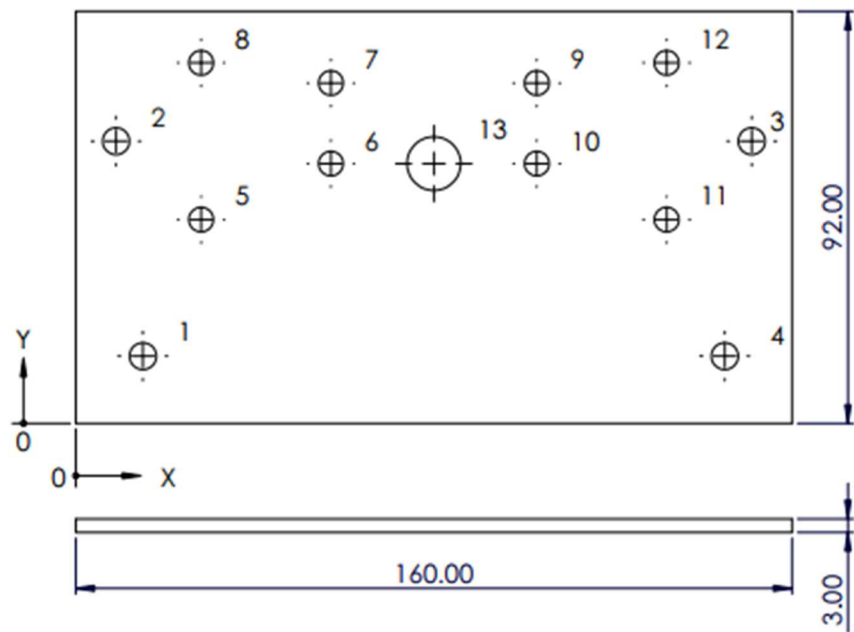


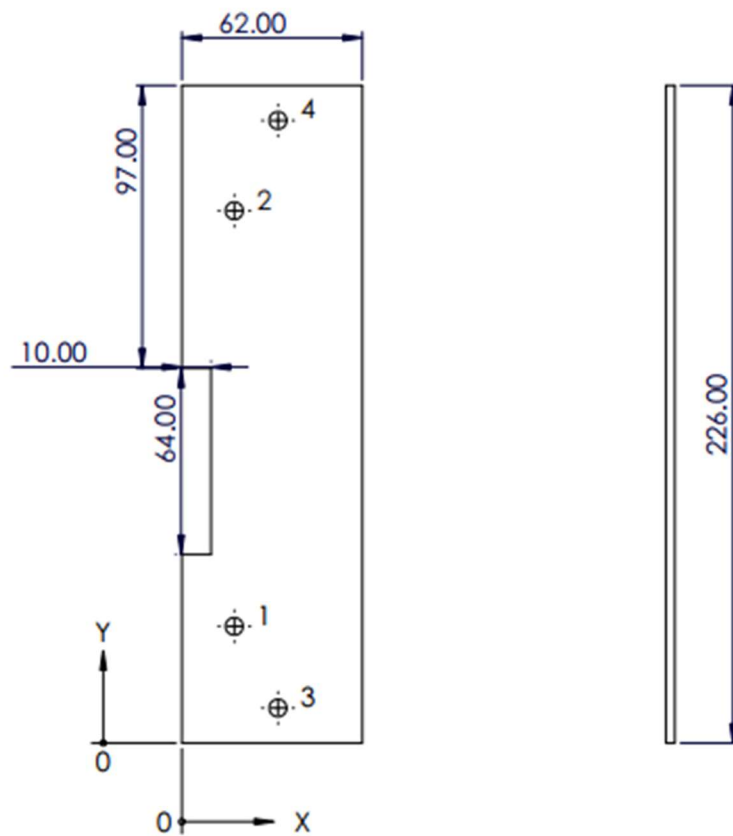
Figure 3.11 The base of Z-axis, connected to the nut housing and linear bearings of X-axis. It holds all the components of Z-axis motion (thickness=3mm)



TAG	X LOC	Y LOC	SIZE
1	15	15	Ø 6.00 THRU
2	9	63	Ø 6.00 THRU
3	151	63	Ø 6.00 THRU
4	145	15	Ø 6.00 THRU
5	28	45.50	Ø 5.50 THRU
6	57	58	Ø 5.50 THRU
7	57	76	Ø 5.50 THRU
8	28	80.50	Ø 5.50 THRU
9	103	76	Ø 5.50 THRU
10	103	58	Ø 5.50 THRU
11	132	45.50	Ø 5.50 THRU
12	132	80.50	Ø 5.50 THRU
13	80	58	Ø 12.00 THRU

Size	Scale	Material
A4	1:1	St 37-2
	Title	Z Bottom

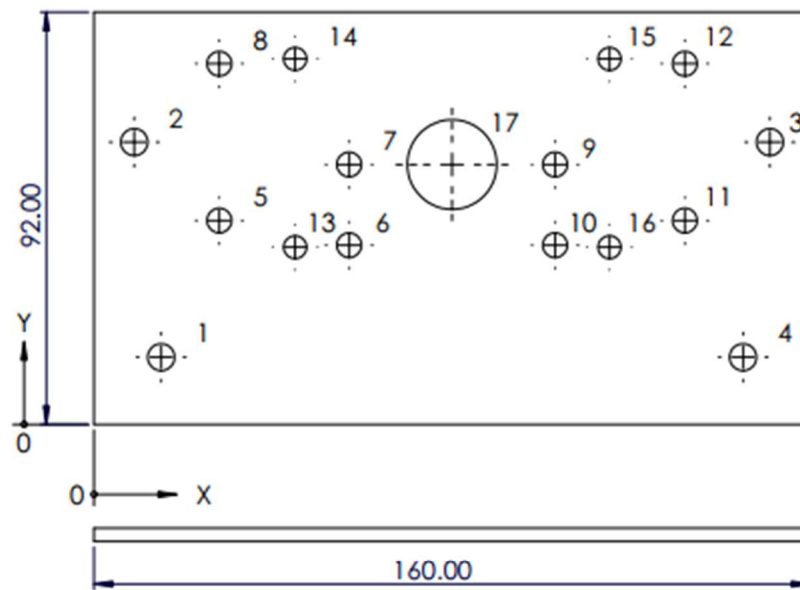
Figure 3.12 Bottom part of Z-axis assembly, holds BF bearing of leadscrew and shaft holders of motion guides (thickness=3mm)



TAG	X LOC	Y LOC	SIZE
1	18	40	Ø 6.00 THRU
2	18	183	Ø 6.00 THRU
3	33	12	Ø 6.00 THRU
4	33	214	Ø 6.00 THRU

Size A4	Scale 1:2	Material St 37-2
	Title Z Side	

Figure 3.13 Sides of Z-axis assembly, it closes it together (thickness=3mm)



TAG	X LOC	Y LOC	SIZE
1	15	15	Ø 6.00 THRU
2	9	63	Ø 6.00 THRU
3	151	63	Ø 6.00 THRU
4	145	15	Ø 6.00 THRU
5	28	45.50	Ø 5.50 THRU
6	57	40	Ø 5.50 THRU
7	57	58	Ø 5.50 THRU
8	28	80.50	Ø 5.50 THRU
9	103	58	Ø 5.50 THRU
10	103	40	Ø 5.50 THRU
11	132	45.50	Ø 5.50 THRU
12	132	80.50	Ø 5.50 THRU
13	44.93	39.55	Ø 5.11 THRU
14	44.93	81.57	Ø 5.11 THRU
15	115.17	81.57	Ø 5.11 THRU
16	115.17	39.55	Ø 5.11 THRU
17	80	58	Ø 20.00 THRU

Size

A4

Scale

1:1

Material

St 37-2

Title

Z Top

Figure 3.14 Top part of Z-axis assembly, holds BK bearing and shaft holders, also holding vertical hybrid servo motor (thickness=3mm)

3.2.2 Assembly:

After sketching each part individually, the project is assembled while maintaining robustness and convenience. All parts are assembled on SOLIDWORKS

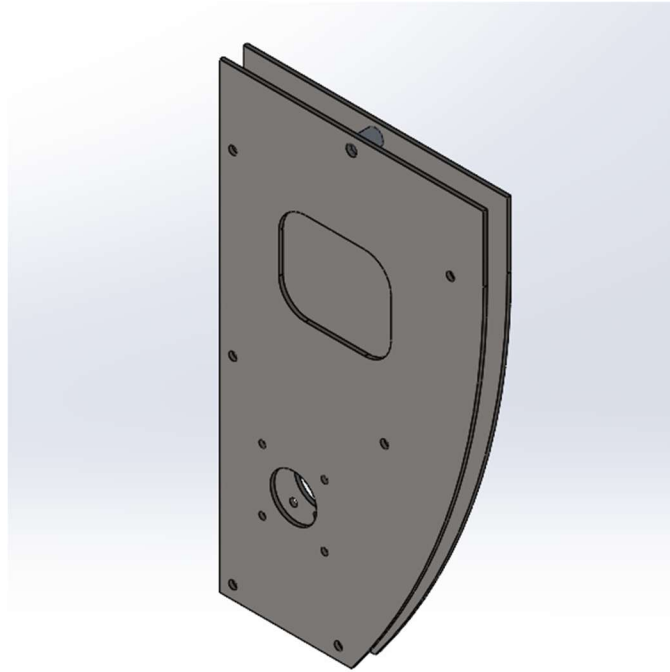


Figure 3.15 Driver side, assembled

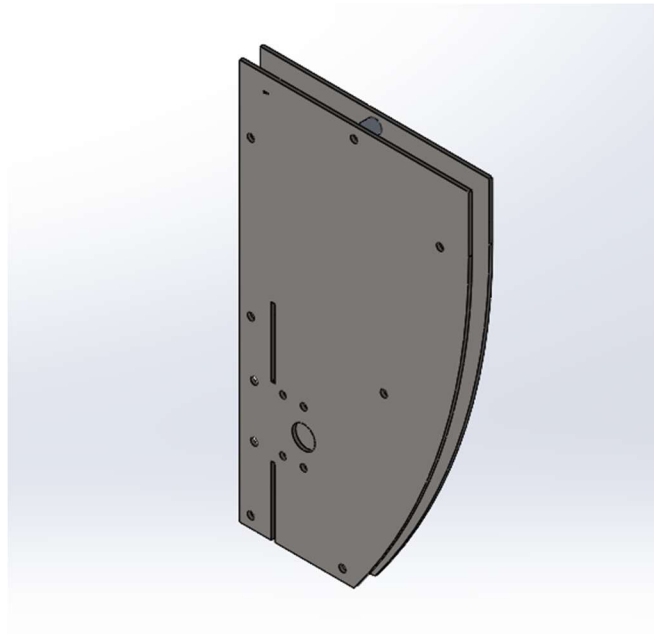


Figure 3.16 Driven side, assembled

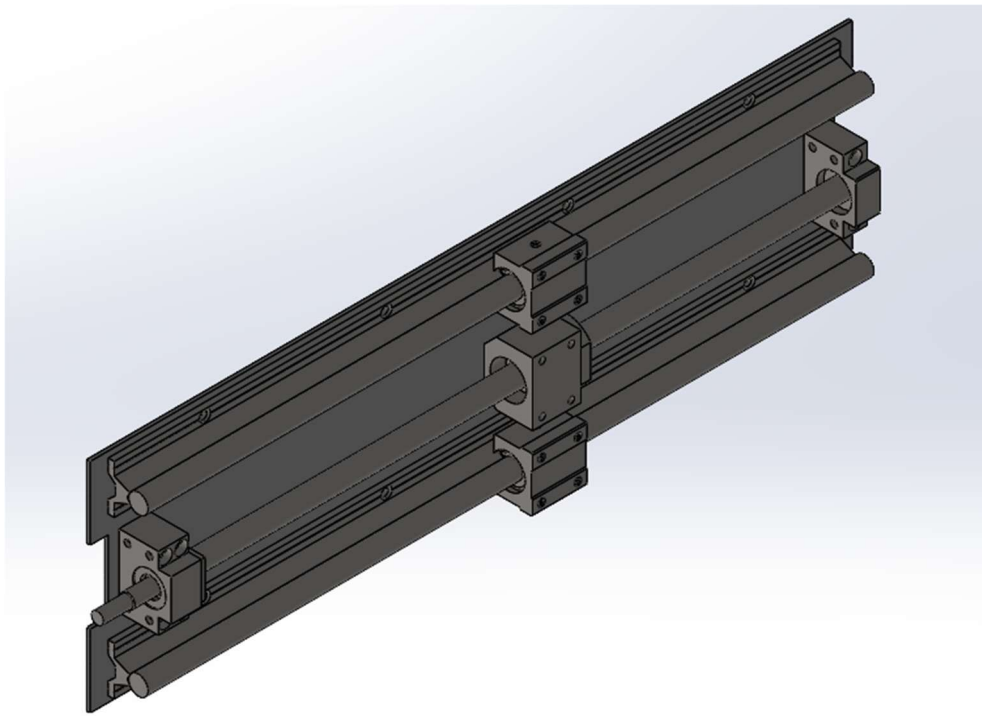


Figure 3.17 X-Axis, assembled

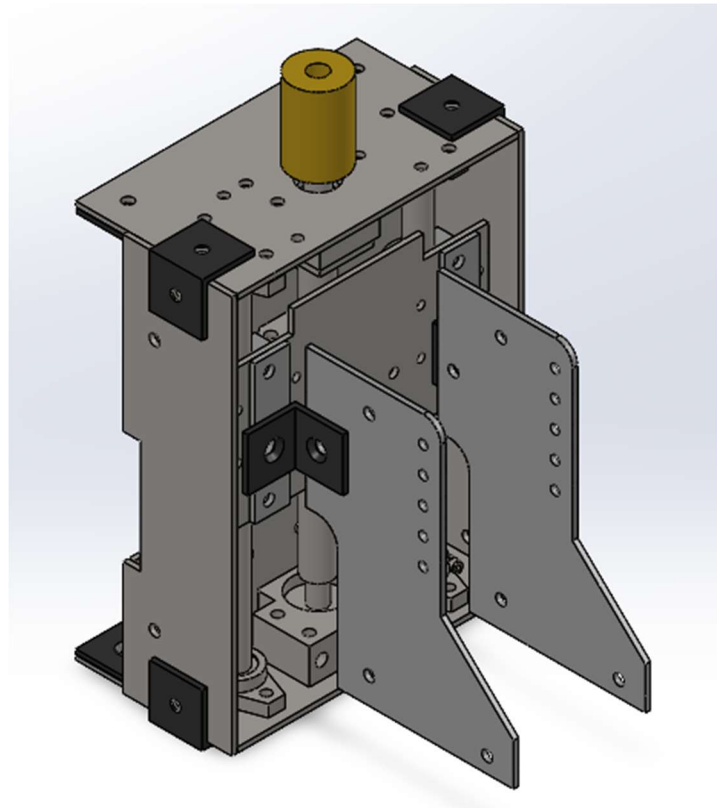


Figure 3.18 Z-Axis, assembled

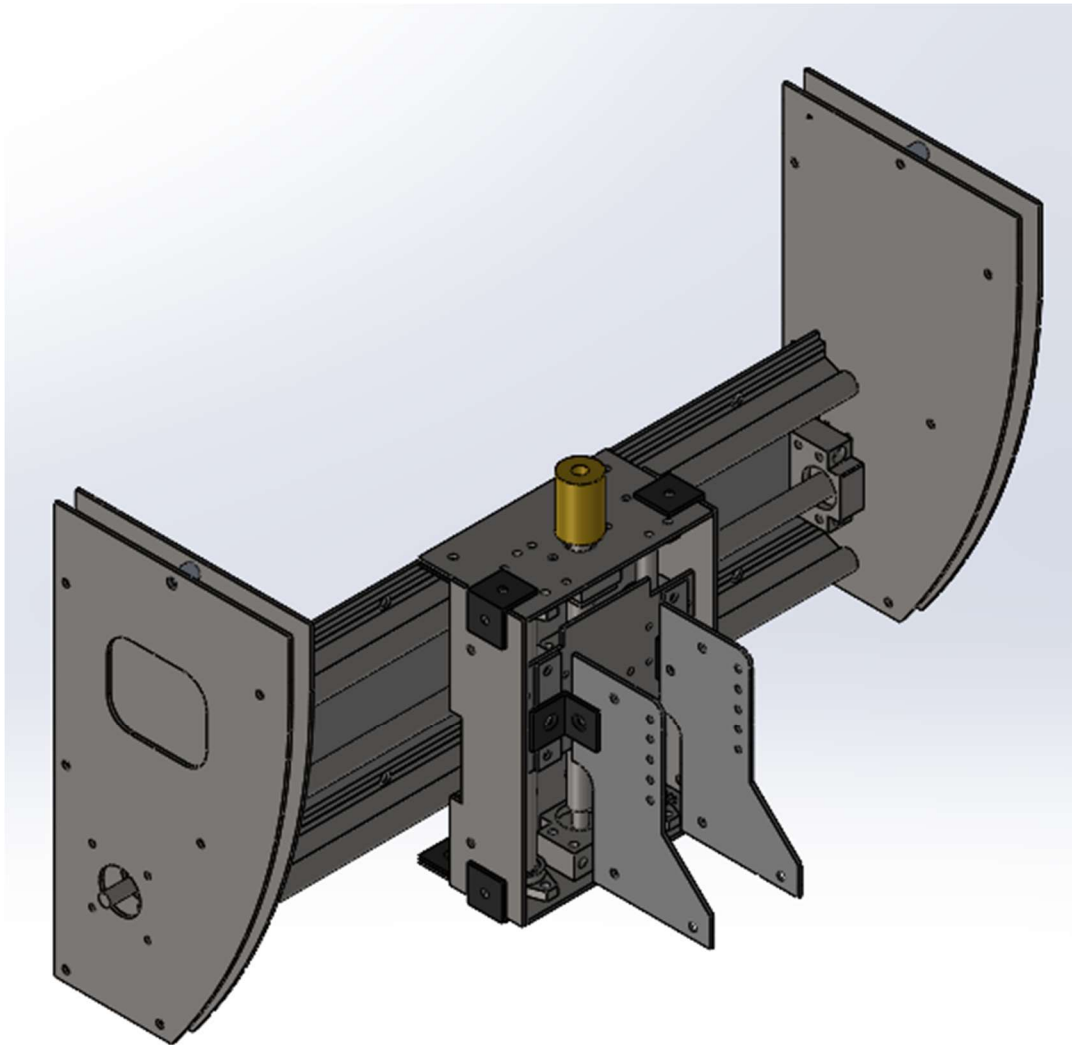


Figure 3.19 Welding mechanism, assembled

3.2.3 Conclusion

The welding mechanism is intentionally configured within a gantry-style framework, resulting in a reversed orientation that may appear unconventional at first glance. This arrangement is a deliberate engineering choice, ensuring structural simplicity, clear load paths, and unobstructed linear motion. By adopting this layout, the mechanism maintains stable torch alignment, consistent stand-off distance, and reliable tracking along the seam, all while supporting a compact and efficient overall system architecture.

3.3 Chassis

Cost efficiency and material availability played a central role in the design process. To ensure that the chassis can be manufactured using common workshop resources, we prioritized readily available structural elements. Square steel beams (Figure 3.3.1) with dimensions 40x40x4(mm) were selected as the primary framing members due to their favorable balance of rigidity, lightweight construction, and broad availability in small and medium dimensions. In contrast, alternative profiles such as I-beams (Figure 3.3.2) are not often available in smaller sizes.



Figure 3.20 Square section structural beam



Figure 3.21 I-section structural beam

3.3.1 Chassis structure

Through repeated design refinement and supporting structural analysis, the final chassis achieves the required stiffness while remaining easy to assemble and practical to transport. The result is a straightforward, reliable, and accessible design suited to the intended user environment.

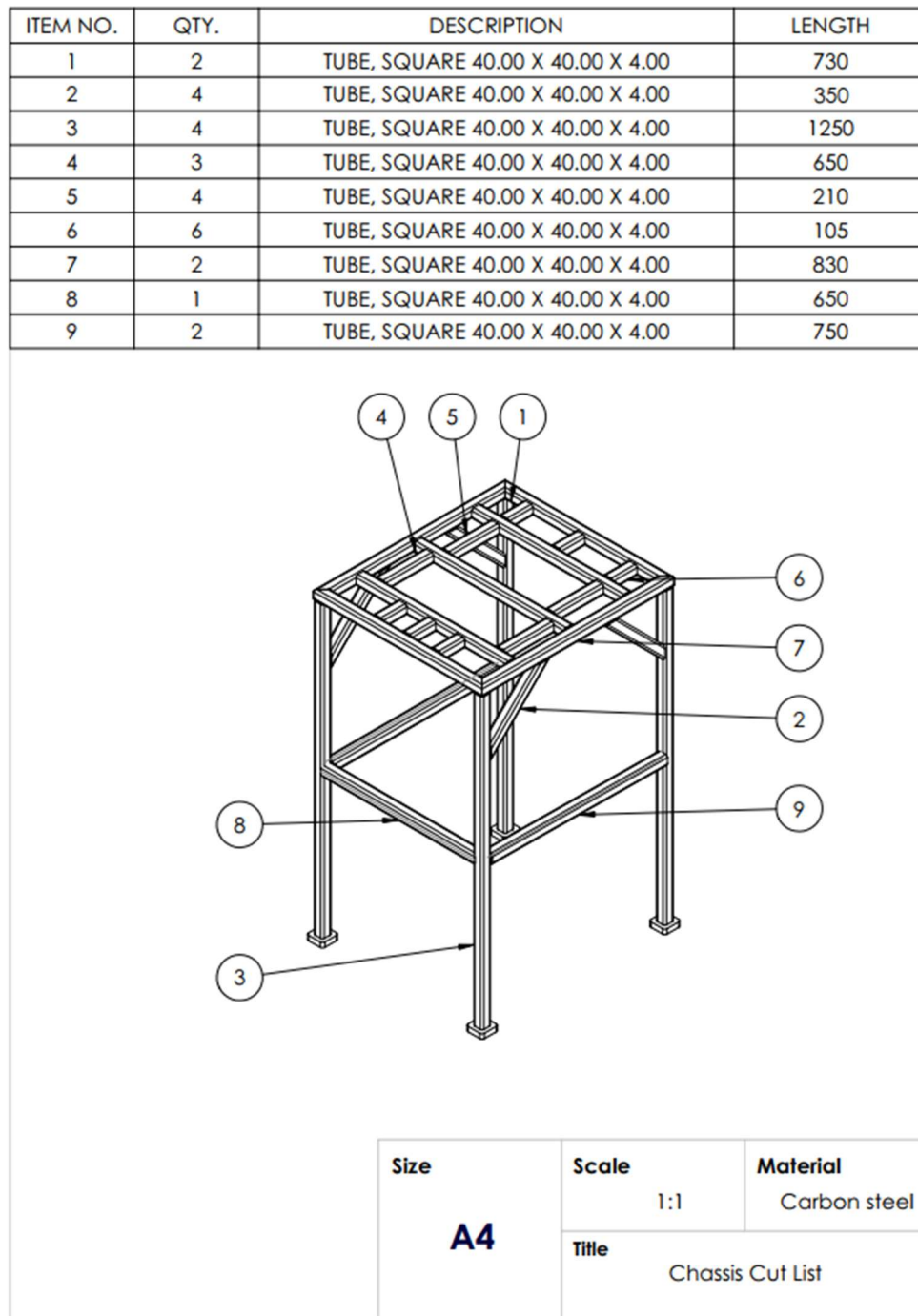


Figure 3.22 Chassis cut list

The following section presents an analysis of the assembled chassis. The Figure below (Figure 3.23) provides a 3D isometric view of the structure along with the integrated welding mechanism.

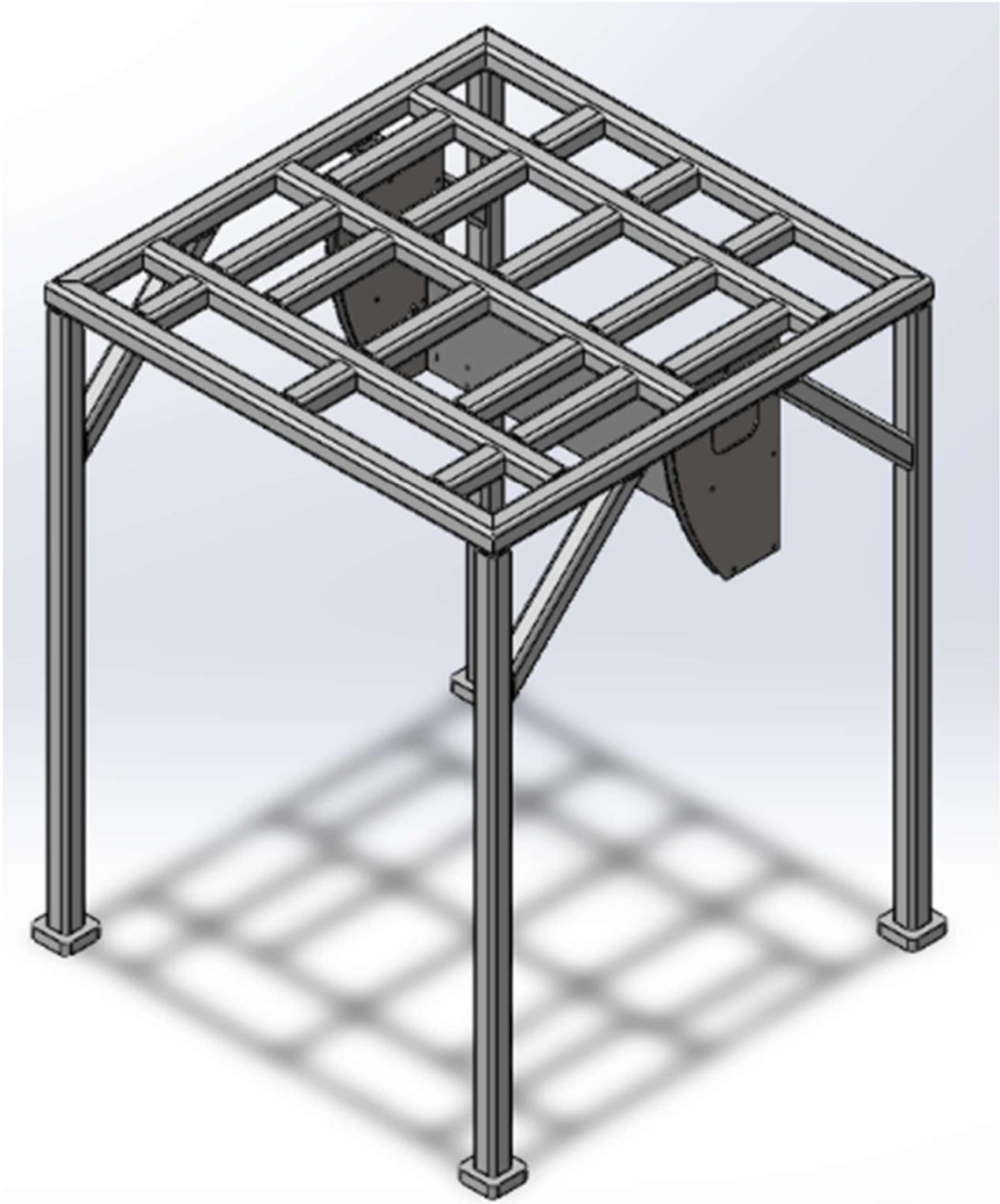


Figure 3.23 3D visualization of chassis

3.3.2 Chassis design

Running the design through SolidWorks Stress Study program to see the stress points and flaws in the design showed the following Figures:

Material Properties:

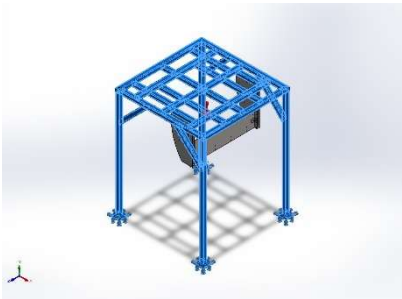
Model Reference	Properties
	Name: Plain Carbon Steel
	Model type: Linear Elastic Isotropic
	Yield strength: 2.20594e+08 N/m²
	Tensile strength: 3.99826e+08 N/m²
	Elastic modulus: 2.1e+11 N/m²
	Poisson's ratio: 0.28
	Mass density: 7,800 kg/m³
	Shear modulus: 7.9e+10 N/m²
	Thermal expansion coefficient: 1.3e-05 /Kelvin

Table 3.1 Properties of material used

The welding mechanism has an actual mass of 21 kg (Figure3.24), but a worst-case design load of **40 kg** (Figure3.25a) was used to ensure the chassis achieves the required stiffness. This load was applied as a **distributed mass** (Figure3.25b). The mechanism is welded to the upper frame of the chassis (Figure 3.26 and 3.27), allowing the load to be transferred securely and uniformly into the structure.

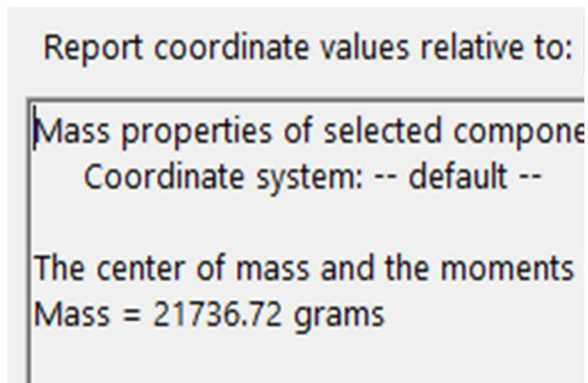


Figure 3.24 Mass of welding mechanism using mass properties feature on SOLIDWORKS

Remote Loads/Mass Details	
Study name	Static 1* (-Default-)
Load name	Distributed Mass-1
Entities	18 face(s)
Type	Displacement (Direct transfer)
Coordinate System	Global cartesian coordinates
Translation Values	—, —, — mm
Rotation Values	—, —, — deg
Reference coordinates	0 0 0 mm
Remote Mass	40 kg

Figure 3.25a Distributed mass properties

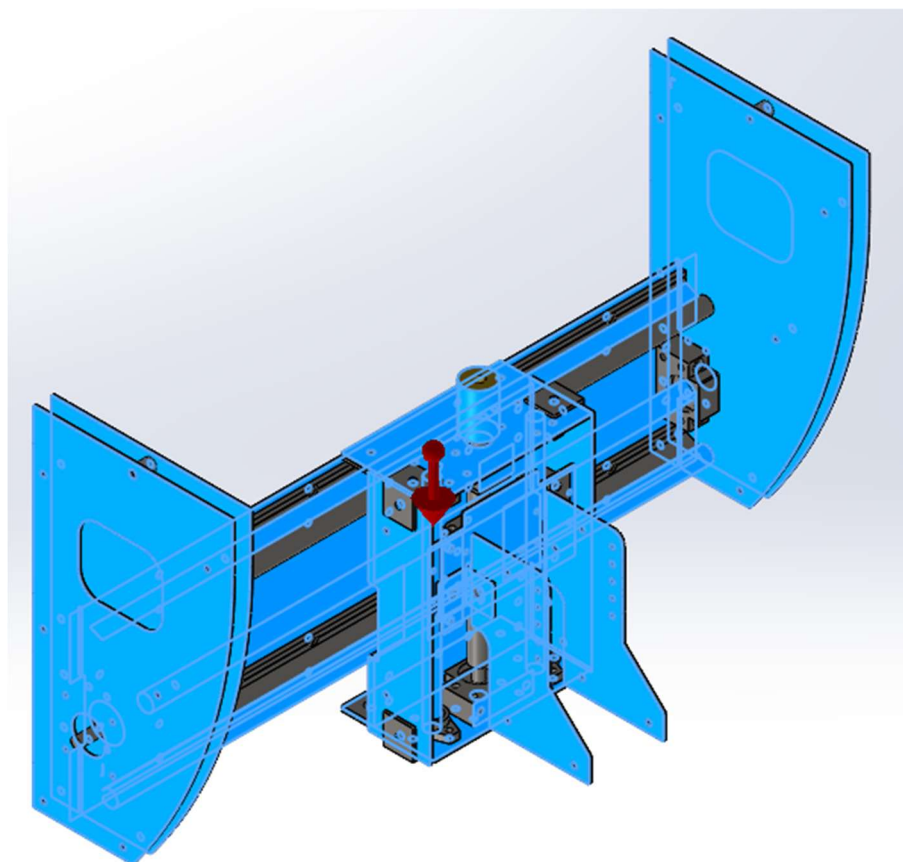


Figure 3.25b Visualized distributed mass, Shown by the red arrow

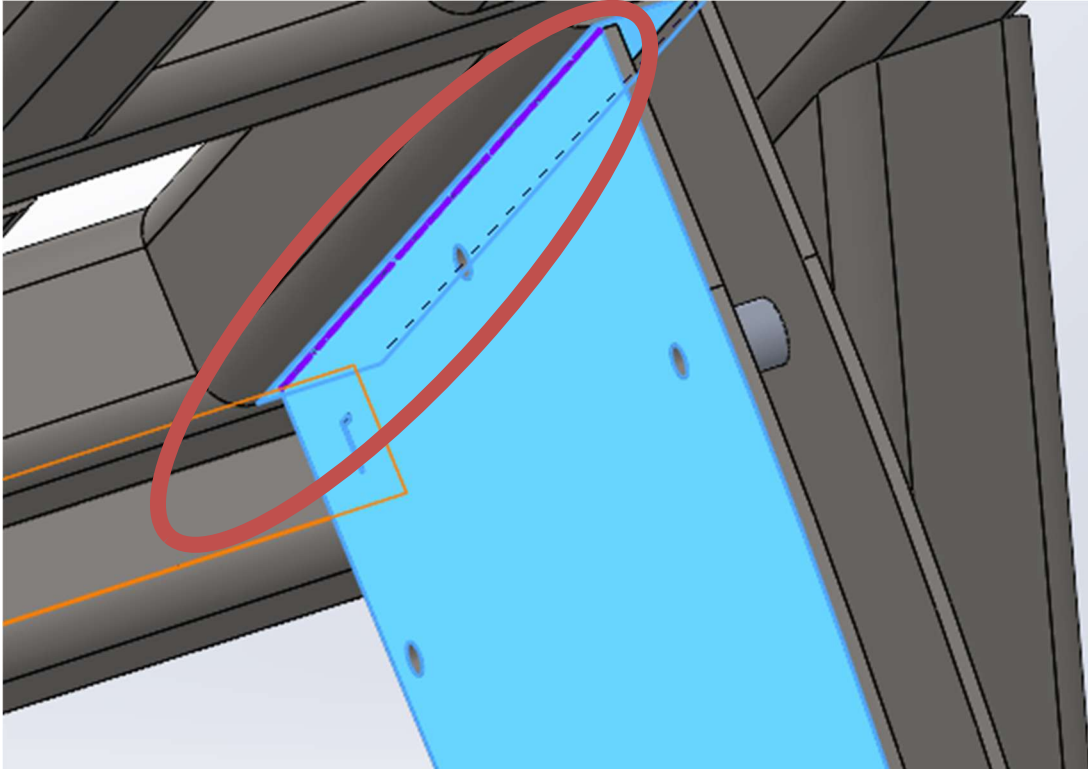


Figure 3.26 Welded joint on driver side

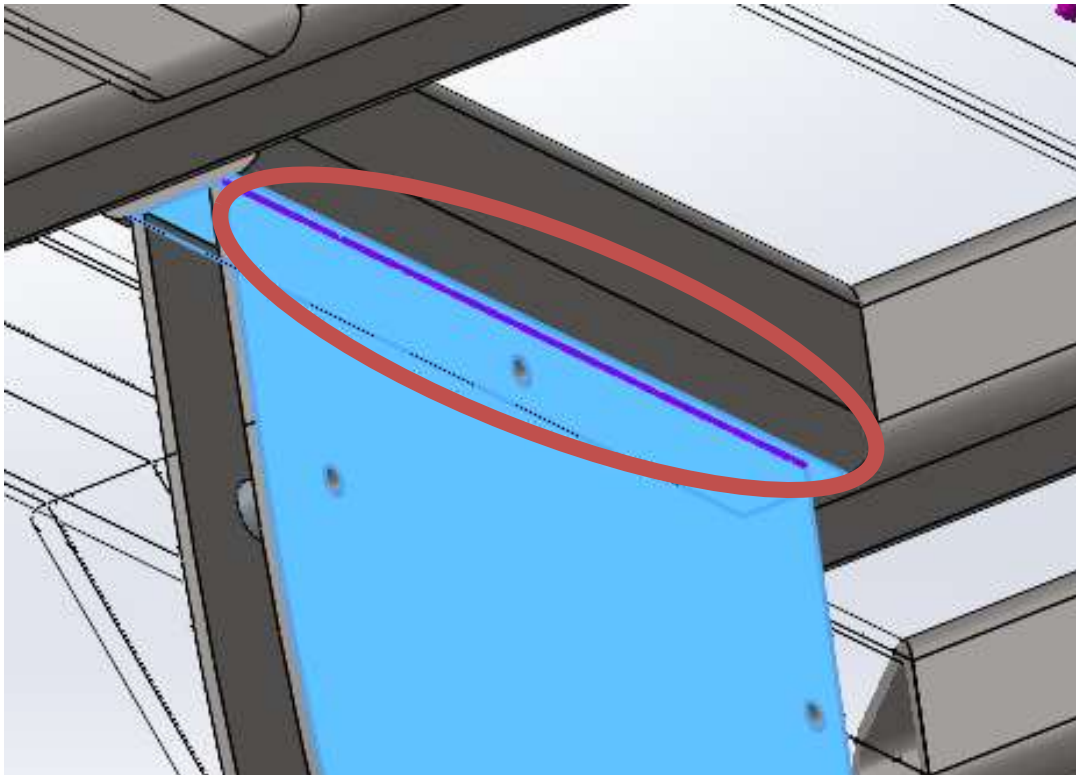


Figure 3.27 Welded joint on driven side

The chassis was meshed with a medium element size (Figure3.28), which provided optimal results while minimizing computational demands. The final mesh is shown in Figure 3.29.

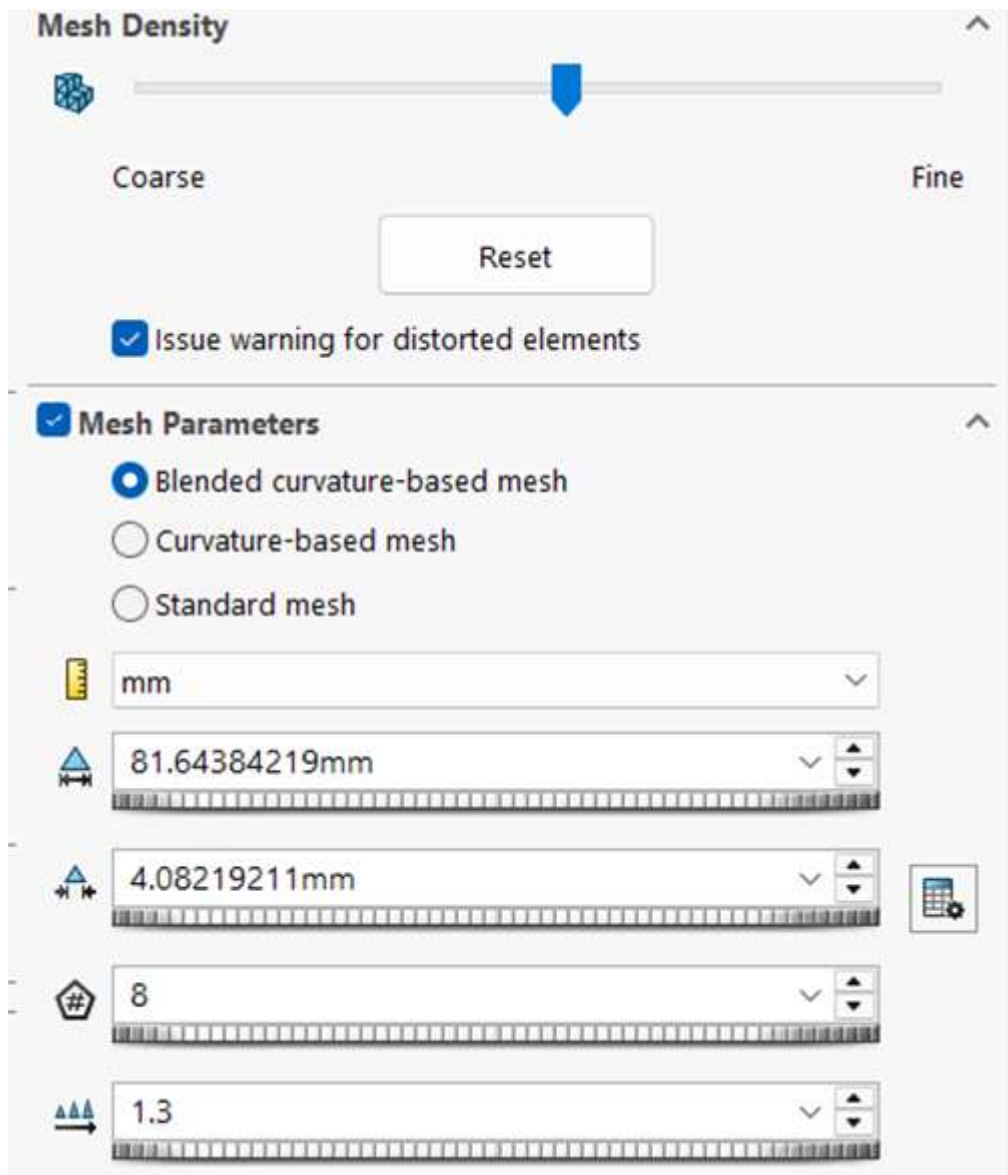


Figure 3.28 Mesh properties

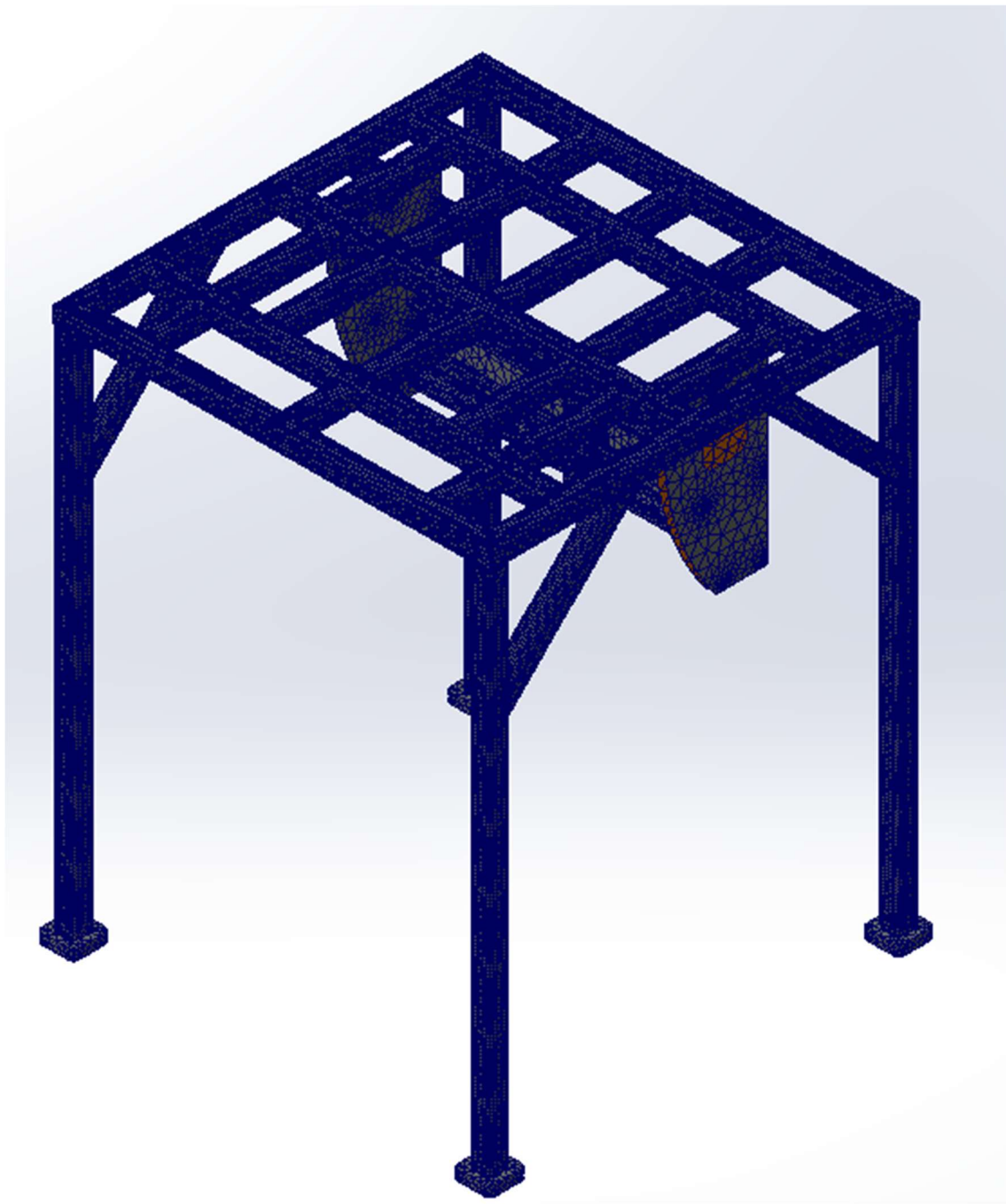


Figure 3.29 Mesh of assembled body

Boundary Conditions:

The end caps are shown in Figure 3.30. These places are assumed to be fully constrained

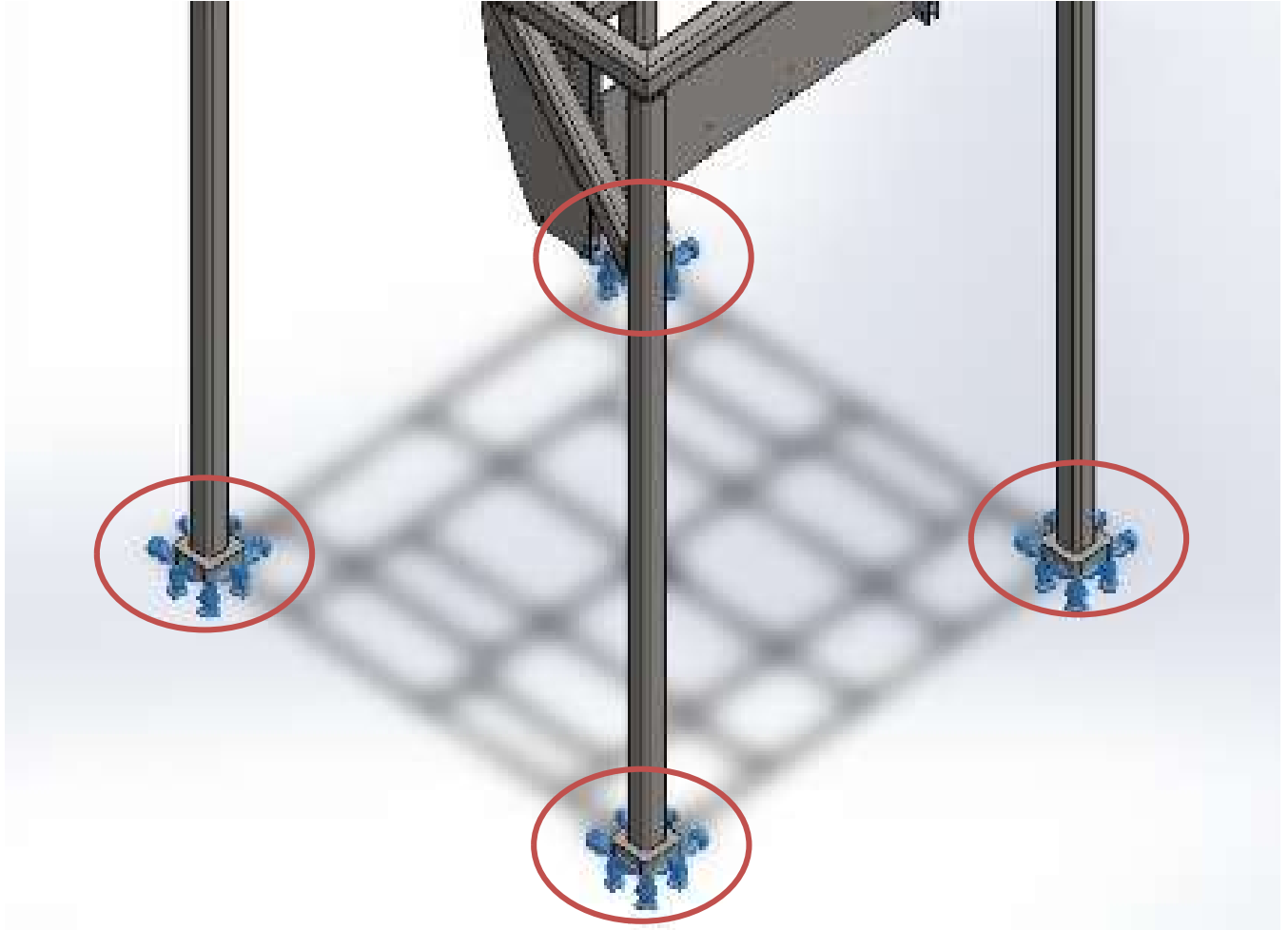


Figure 3.30 Fixtures of assembly

Figure 3.31 shows the stresses on the chassis using the Von-Mises failure theory. As it is shown, the most stress is in the center beam of the chassis. However, it is not anywhere near yield strength of plain carbon steel (table 3.1).

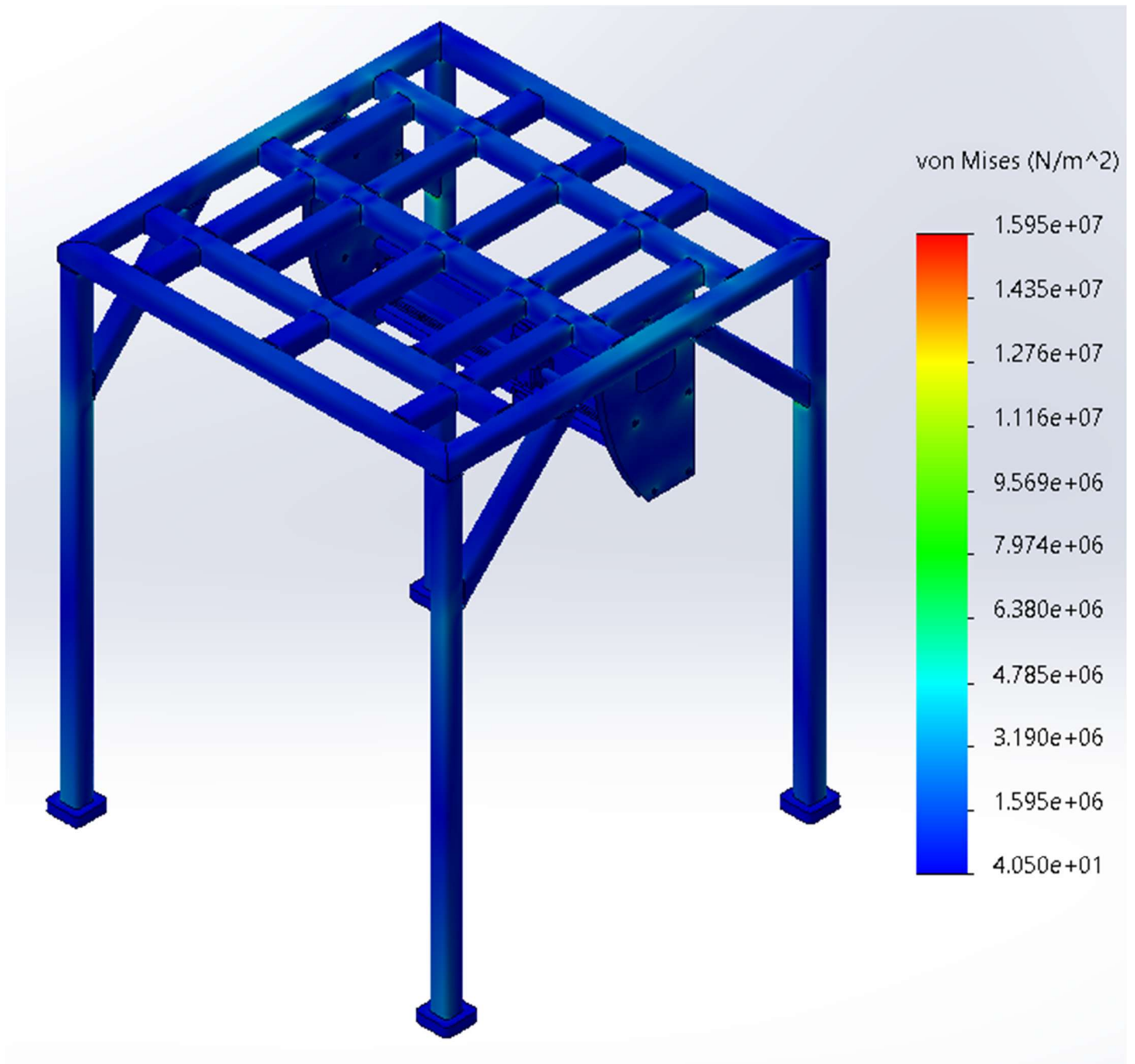


Figure 3.31 Stresses on Chassis

Figure 3.32 Shows the equivalent displacement on each part of the chassis. The maximum displacement is the center beams. Though it is the highest in this area, it is very low and will take a long time to reach this displacement of approximately 0.06mm.

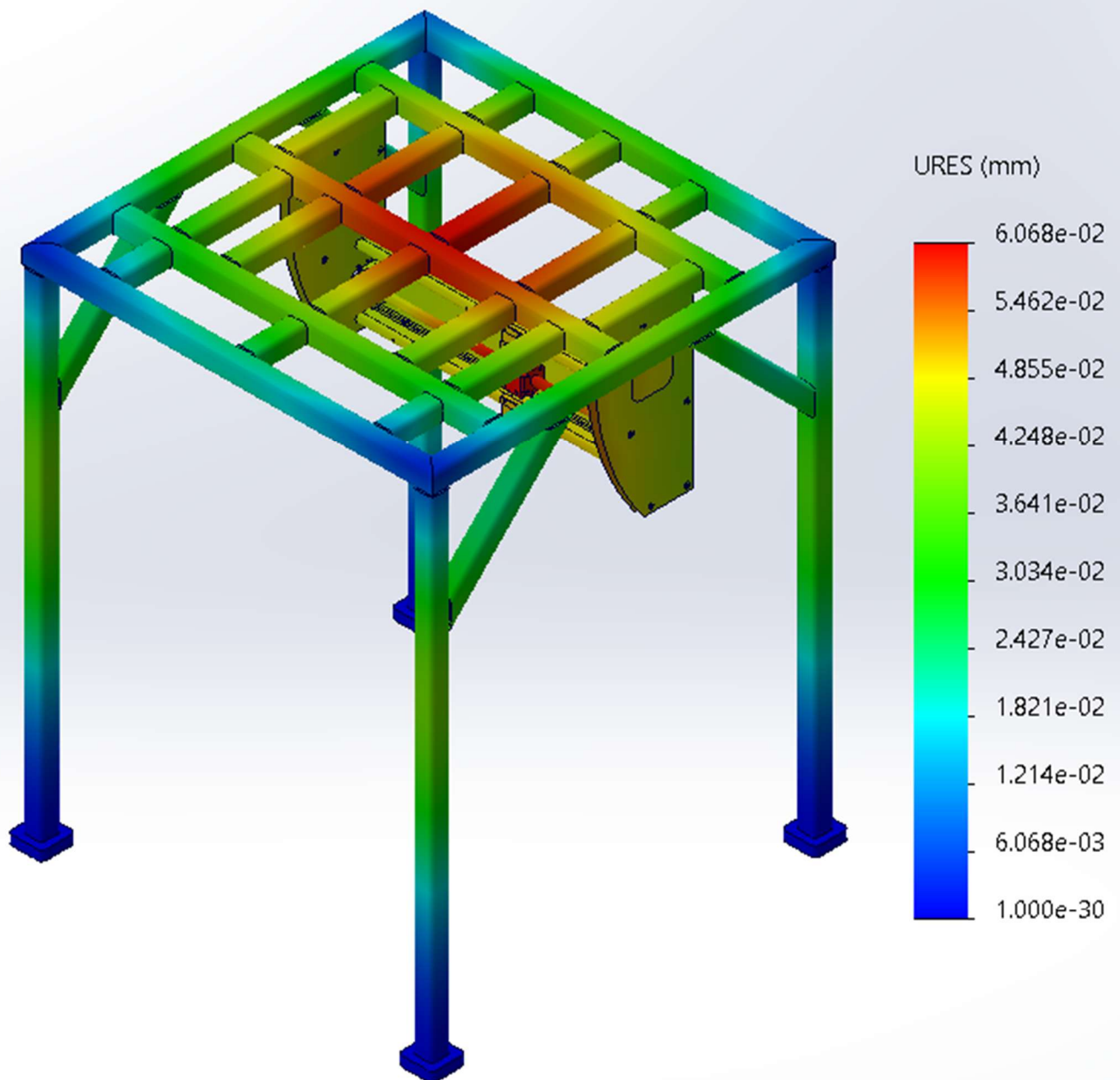


Figure 3.32 Displacement on chassis

Figure 3.33 shows equivalent strain on the chassis. We notice that it is minimal, due to the displacement (Figure 3.32) being minimal even at the critical point.

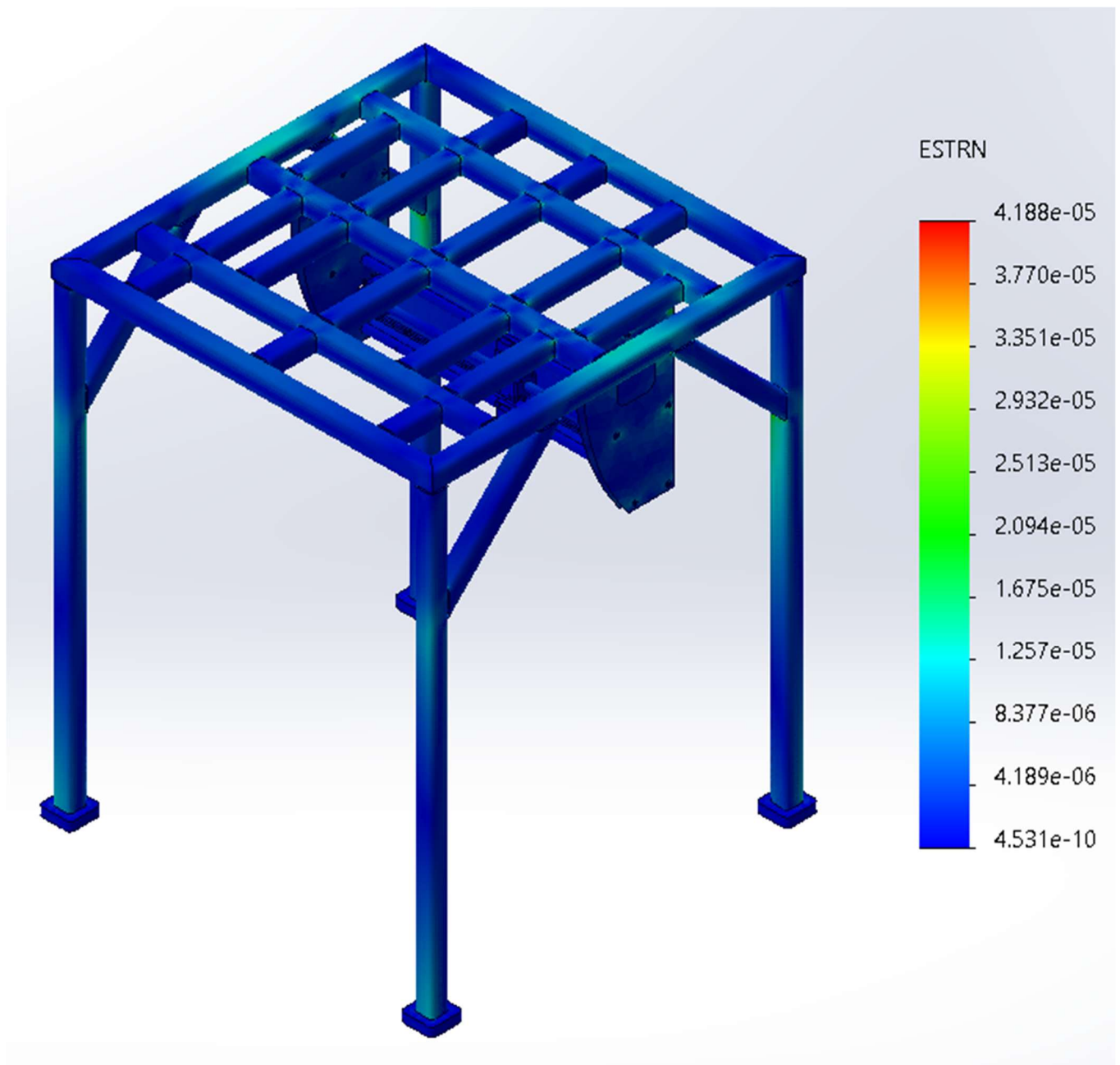


Figure 3.33 Strain on Chassis

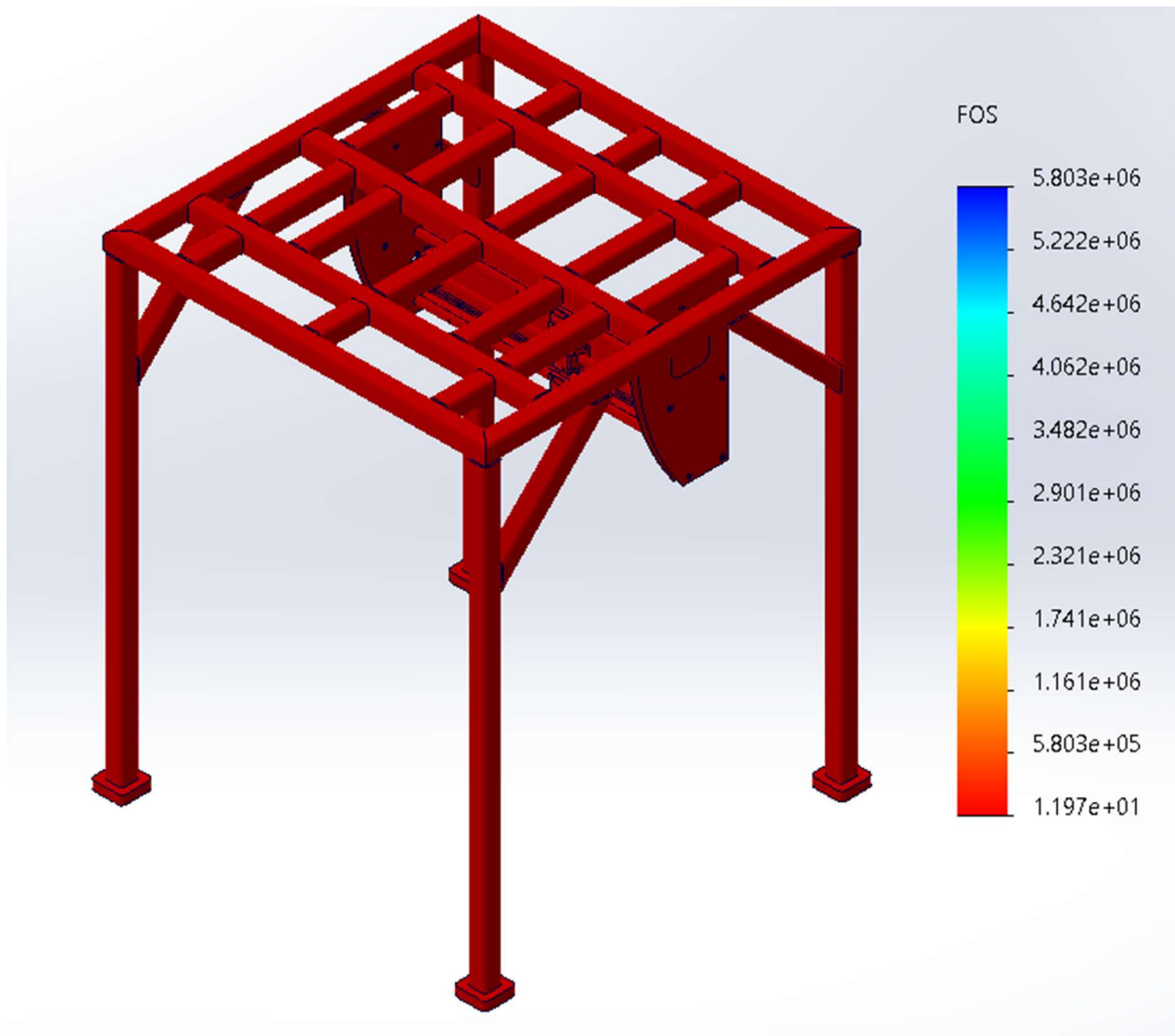


Figure 3.34 Minimum Factor of safety on Chassis, the system showed rigidity and robustness to static loads

3.3.3 Conclusion

The design demonstrated strong rigidity under the worst-case static load of 40 kg—higher than the expected operating load. Structural analysis produced a minimum factor of safety of **12** (Figure3.34), confirming a wide performance margin. Under dynamic loading, minor vibration may appear at higher speeds; however, the machine operates at relatively low rotational speeds (100–110 rpm), where such effects remain negligible. Fatigue on both axes is expected to be minimal, and the bearing systems are well suited for the light-duty operation typical of university labs and small workshops. Overall, the chassis delivers reliable stiffness and stability within its intended operating range, with reduced suitability only for higher-speed or heavier applications outside the project scope.

3.4 Workstation

The workstation serves as the primary operational hub, integrating the clamping systems and robotic assembly into a single unit. To optimize cost-effectiveness, the chassis utilizes standardized weldments consistent with the machine's primary framework. The design incorporates commonly available structural steel profiles to ensure material accessibility and ease of fabrication. Figure 3.35 details the cut list. Figure 3.36 shows the final assembly of the workstation.

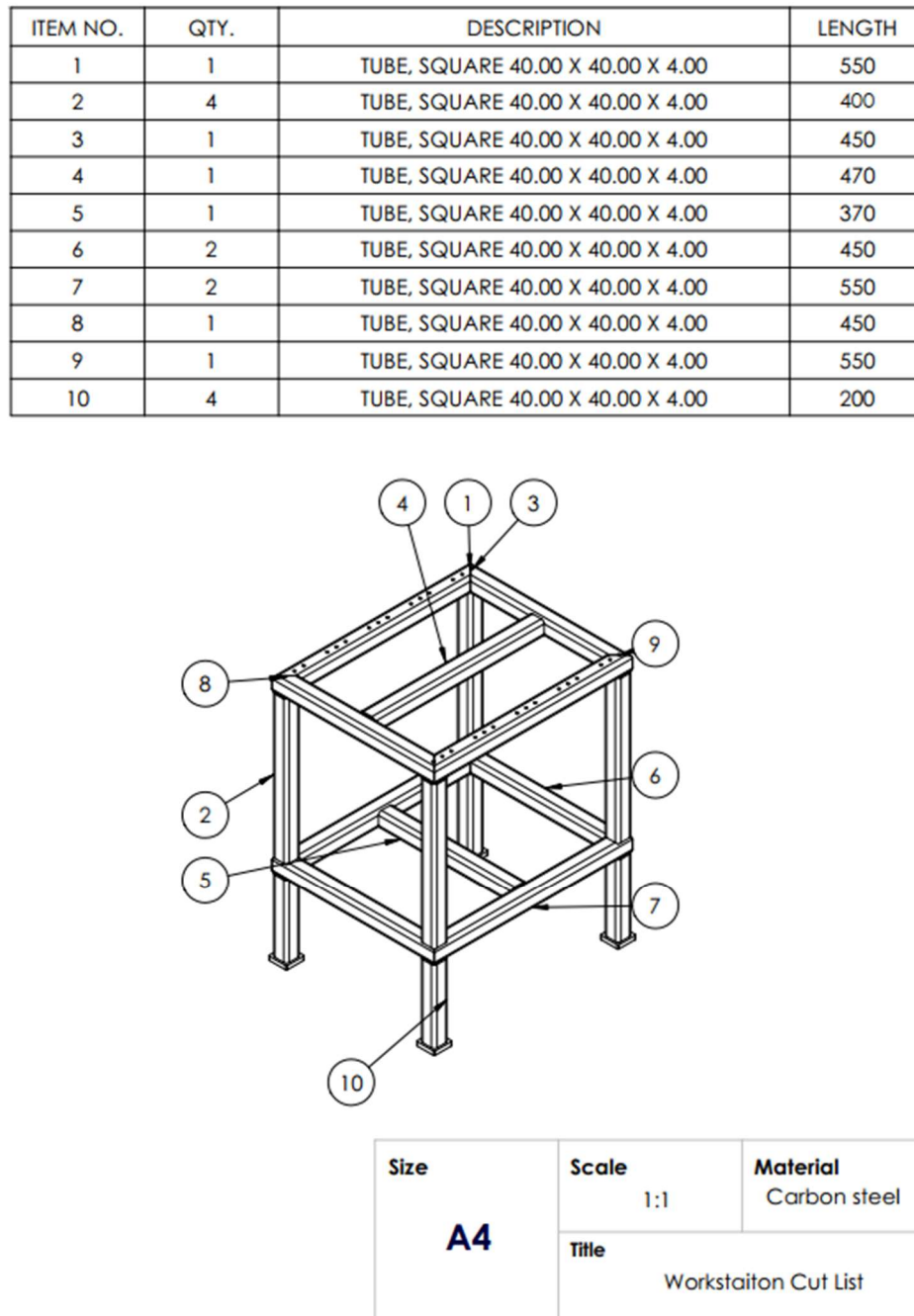


Figure 3.35a Workstation cut list

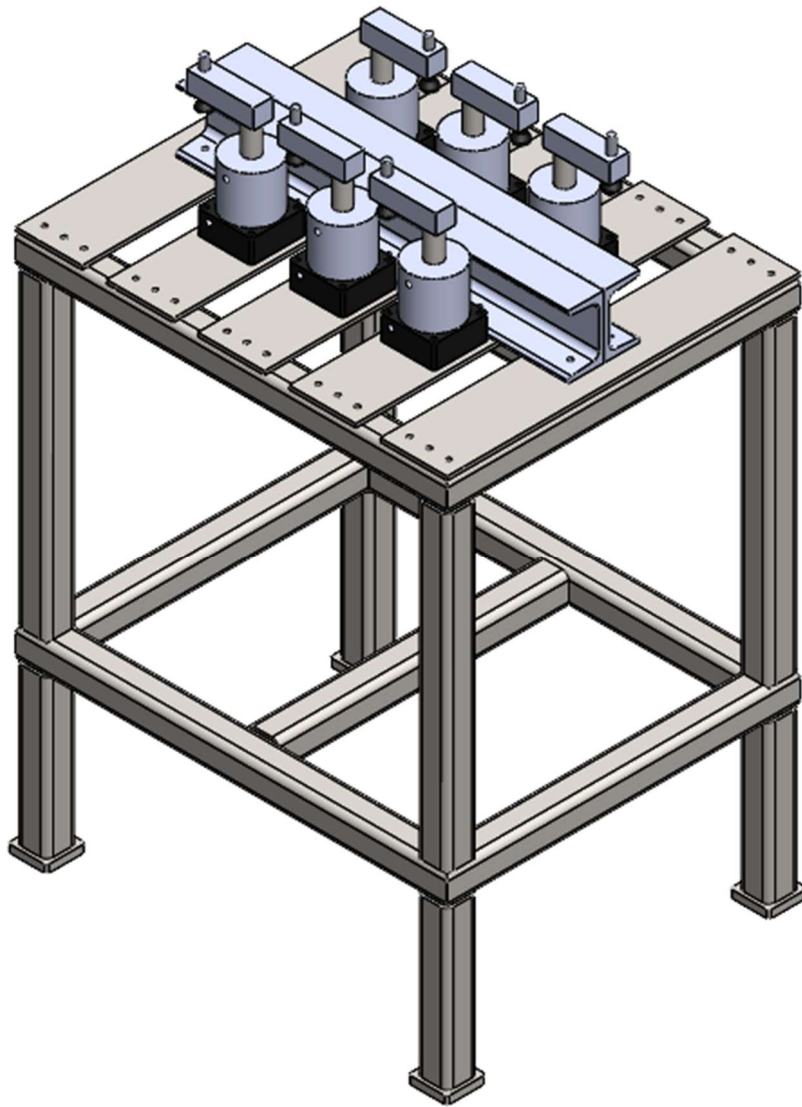


Figure 3.35b Assembled workstation

3.5 Mechanical components selected

Due to current market conditions, the component selection process was approached as validation rather than full custom design. Since simplicity, reliability, and availability are central to the project's philosophy, components were chosen based on what is readily accessible in the market while still meeting the required performance specifications. This approach ensures that the final system remains practical to assemble, cost-effective, and easy to maintain for the intended users.

3.5.1 Selection parameters

Travel speed: 100 mm/min

Maximum Load, X-axis: 20Kg

Maximum Load, Z-axis: 7Kg

Motor speed: 20 rpm

Maximum required speed for both axes: 100 rpm

Welding Span: 350mm

Z-axis stroke: 300mm



Figure 3.36 Main components of a ball screw system

3.5.2 X-Axis

Ball screw:

- Model: SFU2005, C7, Carbon Steel
- *Nominal Diameter*: 20 mm
- *Overall Length*: 500 mm (*effective stroke* $\approx 400 - 420$ mm)
- *Maximum Load*: 20 kg (≈ 196 N)
- $C_a = 11081.5$ N
- $C_o = 23339.8$ N
- $K = 245.17$ N/ μ m
- *Lead* = 5mm
- *Efficiency* $\eta = 0.90$
- *Dynamic Load Factor* $SF = 2$

– Dynamic Load:

$$P = SF \times F = 392.26 \text{ N} \quad (3.1)$$

– Life L_{10} :

$$L_{10} = (C_a / P)^3 \times 10^6 = 2.257 \times 10^{10} \text{ rev} \quad (3.2)$$

$$\text{Hours @100 rpm} = 3.76 \times 10^6 \text{ h}$$

– Static Safety:

$$SF_{static} = C_o / F = 119 \quad (3.3)$$

– Drive Torque:

$$T = (F \times \text{lead}) / (2\pi \times \eta) = 0.173 \text{ N} \cdot \text{m} \quad (3.4)$$

– Buckling Check:

$$F_c \approx 6370 \text{ N}$$

$$SF_{buckling} = 32.5$$

Bearing:

- BK/BF 15 ID
- *BK15 axial capacity* = 4200 N
- *BF15 C₀* = 5000 N
- *BF15 C* = 9400 N
- Axial Safety:

$$SF_{axial} = 4200 / 196.13 = 21.41 \text{ (3.5)}$$

- Static Safety:

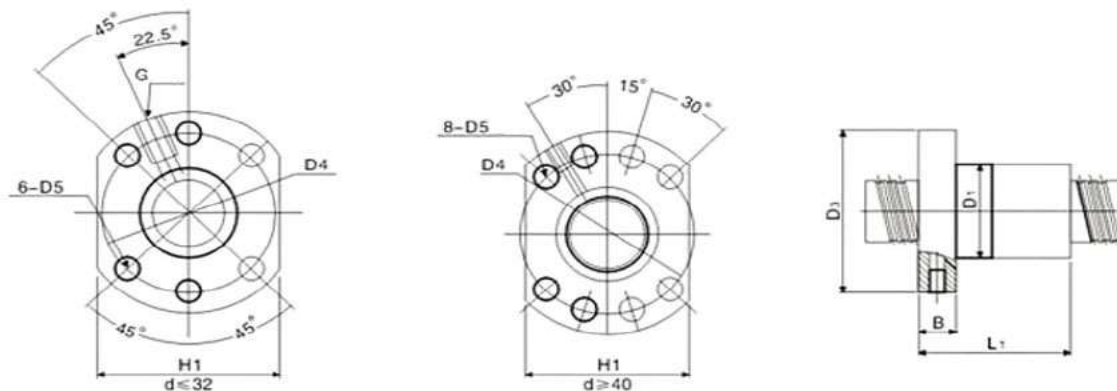
$$SF_{static} = 5000 / 196.13 = 25.49 \text{ (3.6)}$$

- Bearing Life L10:

$$L10 = (C / F)^3 \times 10^6 = 1.1009 \times 10^{11} \text{ rev (3.7)}$$

$$\text{Hours @20 rpm} = 9.1742 \times 10^7 \text{ h}$$

$$\text{Hours @100 rpm} = 1.8348 \times 10^7 \text{ h}$$



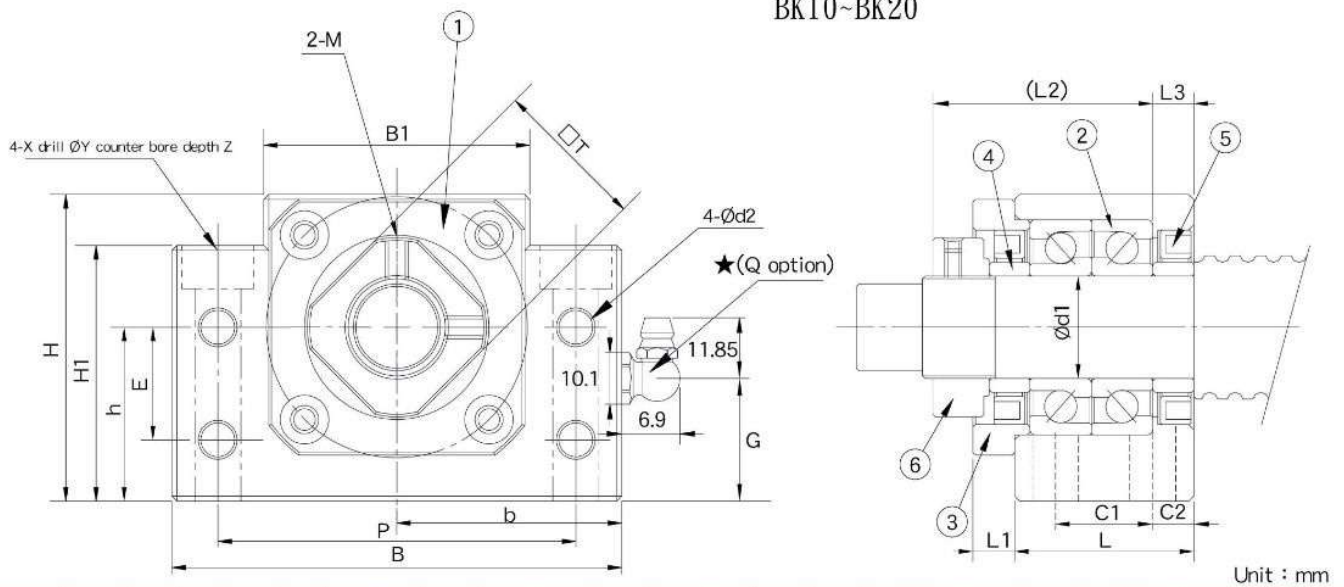
Model	Screw Size		Ball Diam.	Flange	Circuit	Ball Nut Mounting Dimension								Rate Load	
	Diam.	Lead				D1(g6)	D3	D4	D5	L1	B	G	H1	Dynamic Ca(KN)	Static Coa(kn)
RM1204	12	4	2,381	d≤32	3	22	42	32	4.8	35	8	M6	30	4	6.7
RM1605	16	5	3,175	d≤32	3	28	48	38	5.5	42	10	M6	40	7.65	13.2
RM1610	16	10	3,175	d≤32	2	28	48	38	5.5	42	10	M6	40	7.36	12.75
RM2005	20	5	3,175	d≤32	3	36	58	47	6.7	42	10	M6	44	8.6	17.1
RM2010	20	10	3,175	d≤32	2	36	58	47	6.7	42	10	M6	40	8.35	16.8
RM2505	25	5	3,175	d≤32	3	40	62	51	6.8	42	10	M6	48	9.8	23
RM2510	25	10	4,763	d≤32	3	40	62	51	6.8	85	15	M6	48	8.7	20.5

Figure 3.37 Ball screw datasheet

No.	Part name	Qty
1	Housing	1
2	Bearing	1 set
3	Holding lid	1
4	Collar	2
5	Seal	2
6	Lock nut	1set
7	Cushion Stopper (option)	1



BK10~BK20



Unit : mm

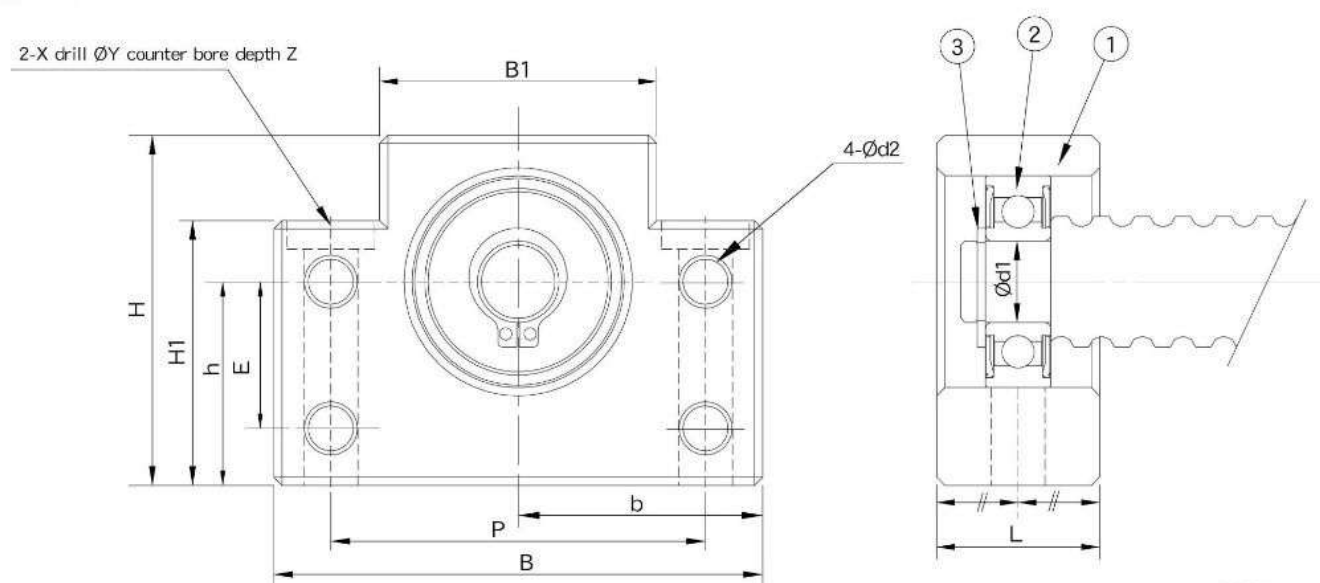
Model No.	Shaft diameter d1	L	L1	L2	L3	B	H	b ±0.02	h ±0.02	B1	H1	E	P	C1	C2	d2	X	Y	Z	M	T	G	Q	Weight (Kgs)
BK10	10	25	5	29.5	5	60	39	30	22	34	32.5	15	46	13	6	5.5	6.6	10.8	5	M3	16	15	M6	0.4
BK12	12	25	5	29.5	5	60	43	30	25	34	32.5	18	46	13	6	5.5	6.6	10.8	1.5	M4	19	18	M6	0.41
BK15	15	27	6	32	6	70	48	35	28	40	38	18	54	15	6	5.5	6.6	11	6.5	M4	22	18	M6	0.58
BK17	17	35	9	44	7	86	64	43	39	50	55	28	68	19	8	6.6	9	14	8.5	M4	24	30	M6	1.3
BK20	20	35	8	43	8	88	60	44	34	52	50	22	70	19	8	6.6	9	14	8.5	M4	30	24	M6	1.2
BK25	25	42	12	54	9	106	80	53	48	64	70	33	85	22	10	9	11	17	11	M6	35	37	M6	2.35
BK30	30	45	14	61	9	128	89	64	51	76	78	33	102	23	11	11	14	20	13	M6	40	37	M6	3.33
BK35	35	50	14	67	12	140	96	70	52	88	79	35	114	26	12	11	14	20	13	M6	50	37	M6	4.4
BK40	40	61	18	76	15	160	110	80	60	100	90	37	130	33	14	14	18	26	17.5	M6	50	43	M6	6.8

Figure 3.38 BK bearing datasheet

No.	Part name	Qty
1	Housing	1
2	Bearing	1 set
3	Snap ring	1
4	Cushion Stopper (option)	1



BF10~BF30



Unit : mm

Model No.	Shaft diameter d1	L	B	H	b ±0.02	h ±0.02	B1	H1	E	P	d2	X	Y	Z	Bearing	Snap ring	Weight (Kgs)
BF10	8	20	60	39	30	22	34	32.5	15	46	5.5	6.6	10.8	5	608ZZ	S 08	0.3
BF12	10	20	60	43	30	25	34	32.5	18	46	5.5	6.6	10.8	1.5	6000ZZ	S 10	0.3
BF15	15	20	70	48	35	28	40	38	18	54	5.5	6.6	11	6.5	6002ZZ	S 15	0.4
BF17	17	23	86	64	43	39	50	55	28	68	6.6	9	14	8.5	6203ZZ	S 17	0.75
BF20	20	26	88	60	44	34	52	50	22	70	6.6	9	14	8.5	6004ZZ	S 20	0.76
BF25	25	30	106	80	53	48	64	70	33	85	9	11	17	11	6205ZZ	S 25	1.43
BF30	30	32	128	89	64	51	76	78	33	102	11	14	20	13	6206ZZ	S 30	1.94
BF35	35	32	140	96	70	52	88	79	35	114	11	14	20	13	6207ZZ	S 35	2.25
BF40	40	37	160	110	80	60	100	90	37	130	14	18	26	17.5	6208ZZ	S 40	3.3

Figure 3.39 BF bearing datasheet

Coupling Validation After Machining Ball screw end to $\phi 17$ mm:

- *Original diameter: 20 mm*

- *Machined diameter: 12 mm*

- *Polar moment of inertia (J):*

$$\begin{aligned} J &= \pi * d^4 / 32 \quad (3.8) \\ &= 3.1416 * (0.012)^4 / 32 \\ &= 2.03 \times 10^{-9} m^4 \end{aligned}$$

Torque Demand of System

Max speed: 100 rpm

Estimated axial load (welding machine slide): 150 – 250 N

Screw torque (T):

$$T = (F \times \text{lead}) / (2\pi \times \eta) \quad (3.9)$$

Using $F = 250 \text{ N}, \eta = 0.9$:

$$T = 0.221 \text{ N} \cdot \text{m}$$

Apply safety factor 3 : $T_{req} = 0.66 \text{ N} \cdot \text{m}$

Allowable Shear Stress in $\phi 12$:

$$\tau = (T \times r) / J \quad (3.10)$$

$$\tau = (0.66 \times 0.006) / (2.03 \times 10^{-9})$$

$$\tau = 1.95 \text{ MPa}$$

Material allowable shear: $\approx 150 \text{ MPa}$

Result: Load is $75 \times$ lower than allowable.

Typical catalogue values:

Max torque: $10 - 25 \text{ N} \cdot \text{m}$

Max speed: $3,000 - 5,000 \text{ rpm}$

Demand:

$$T_{req} = 0.66 \text{ N} \cdot \text{m}$$

$$\text{Speed} = 100 \text{ rpm}$$

Clear margins:

$$\text{Torque margin} = 15 \times \text{ to } 40 \times$$

$$\text{Speed margin} = 30 \times \text{ to } 50 \times$$

The machined $\phi 17 \text{ mm}$ shaft easily withstands the transmitted torque with extremely low shear stress. The 14×17 flexible coupling provides far more torque and speed capacity than required. The configuration is mechanically safe with very high safety margin.



Figure 3.40 Flexible Coupler



Chapter 4: Pneumatic System

4.1 Introduction

Pneumatic pistons are used in the system to hold the workpiece securely during welding, and although the circuit is simple, it plays a critical role in ensuring accuracy and stability. During welding, the workpiece undergoes thermal expansion, which can cause shifting or warping if it is not properly restrained. The applied clamping force from the pistons prevents this movement and maintains correct alignment as the material expands and contracts. In addition, vibrations generated by the welding process and the machine's motion components can introduce small but significant displacements if the sheet is not firmly held. The pistons counter these effects by keeping the workpiece fixed throughout the cycle. From a safety standpoint, proper clamping prevents lifting or slipping that could damage the mechanism or create hazards for the operator. Most importantly, pneumatic clamping precision and repeatability; each welding cycle begins with the same stable reference, allowing the automated process to produce consistent and reliable results

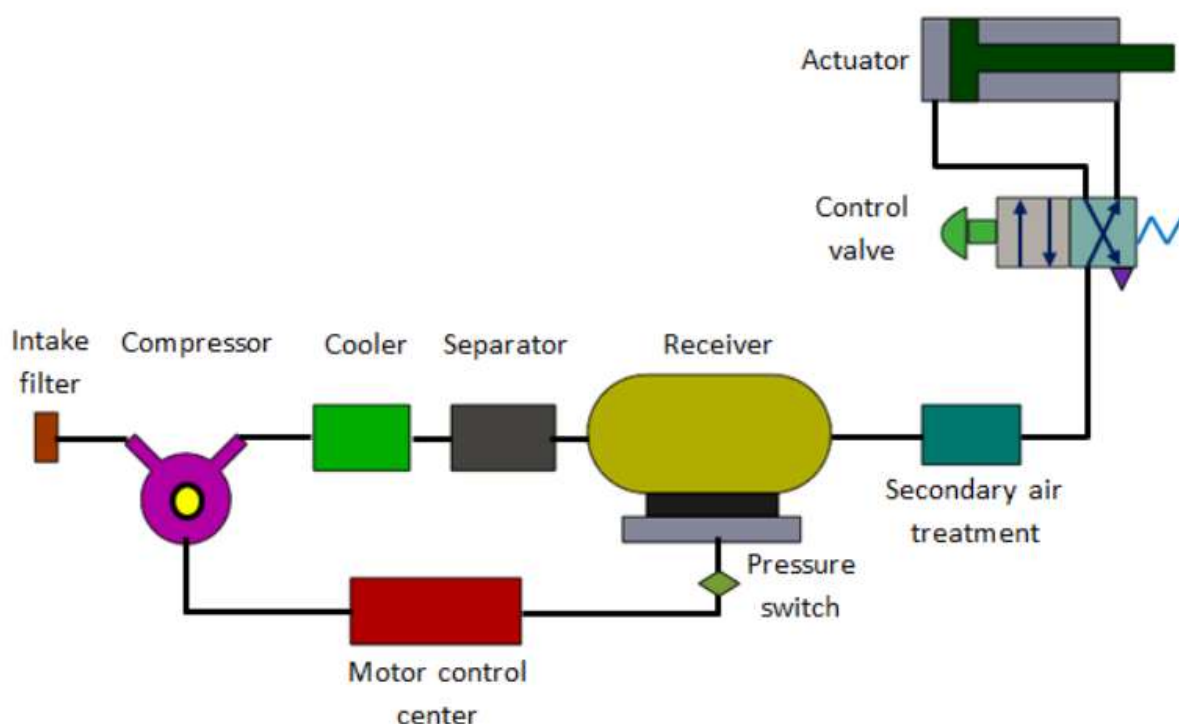


Figure 4.1 Main components of a pneumatic system

4.2 Clamping

4.2.1 Importance of clamping

This stage represents one of the most critical elements of any welding process, as proper clamping directly determines weld stability, thermal control, and overall joint quality. Effective clamping ensures that the workpiece remains fixed throughout heating, preventing shifts, distortion, or gaps caused by thermal expansion. A well-engineered clamp (Figure 4.2) supports a smooth, continuous weld path and allows the welding mechanism to maintain consistent stand-off distance, resulting in uniform penetration and a clean seam.

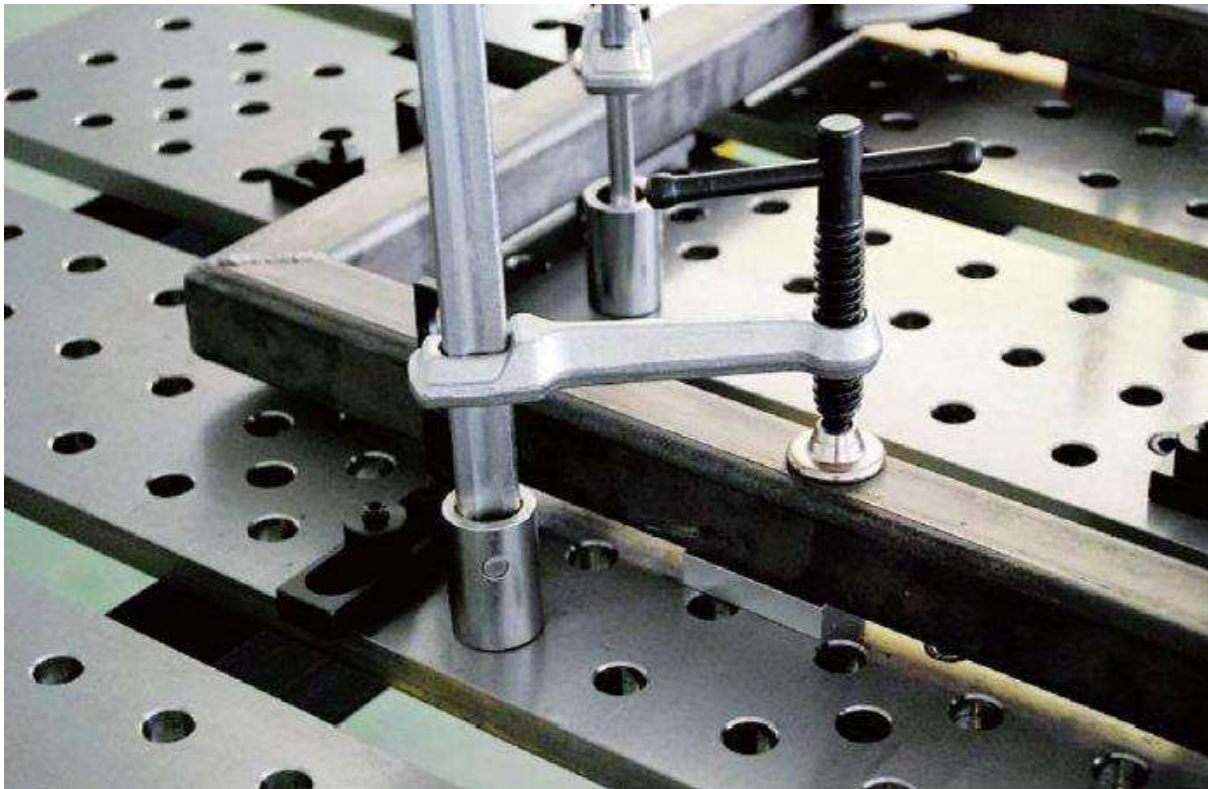


Figure 4.2 A properly fixated workpiece

4.2.2 Risks of improper clamping

Insufficient or uneven clamping can easily compromise the entire operation. Workpiece movement during welding leads to discontinuities, misalignment, excessive deformation, and in severe cases warping or burn-through of thin sheets. Such defects (Figure 4.3) not only weaken the joint but may also render the part unusable. Highlighting these risks underscores why the clamping system must be engineered with precision and why its reliability is central to achieving a stable and defect-free weld.

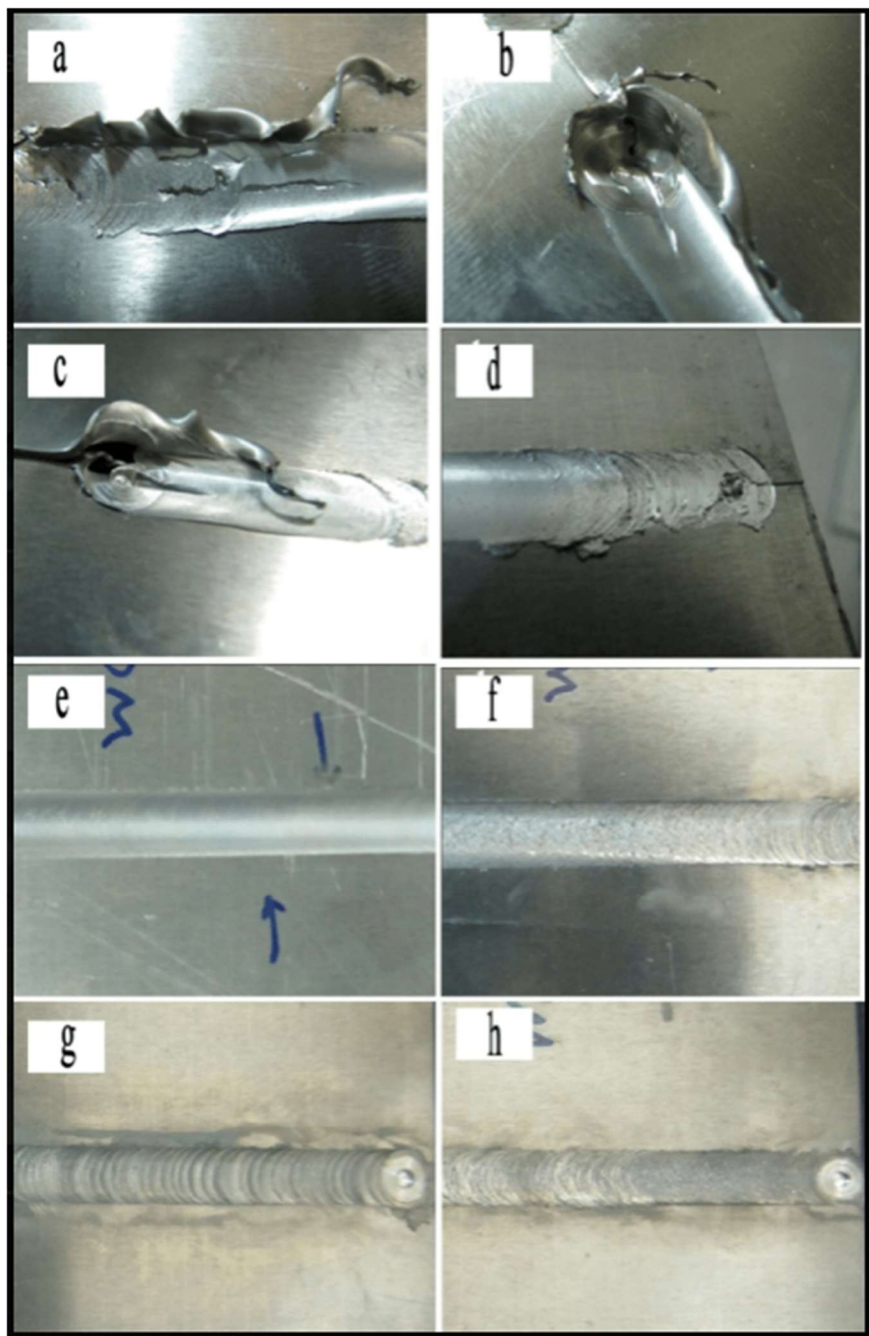


Figure 4.3 Various welding defects of thin sheet metal

4.3 Pneumatic circuit

The clamping system requires synchronized and reliable actuation to secure the workpiece before welding. In this design, all pneumatic pistons are connected to a single solenoid valve, ensuring that they engage simultaneously. The valve is triggered by a PLC signal once the workpiece is correctly positioned, providing consistent clamping force and stable alignment throughout the welding process (Figure 4.4).

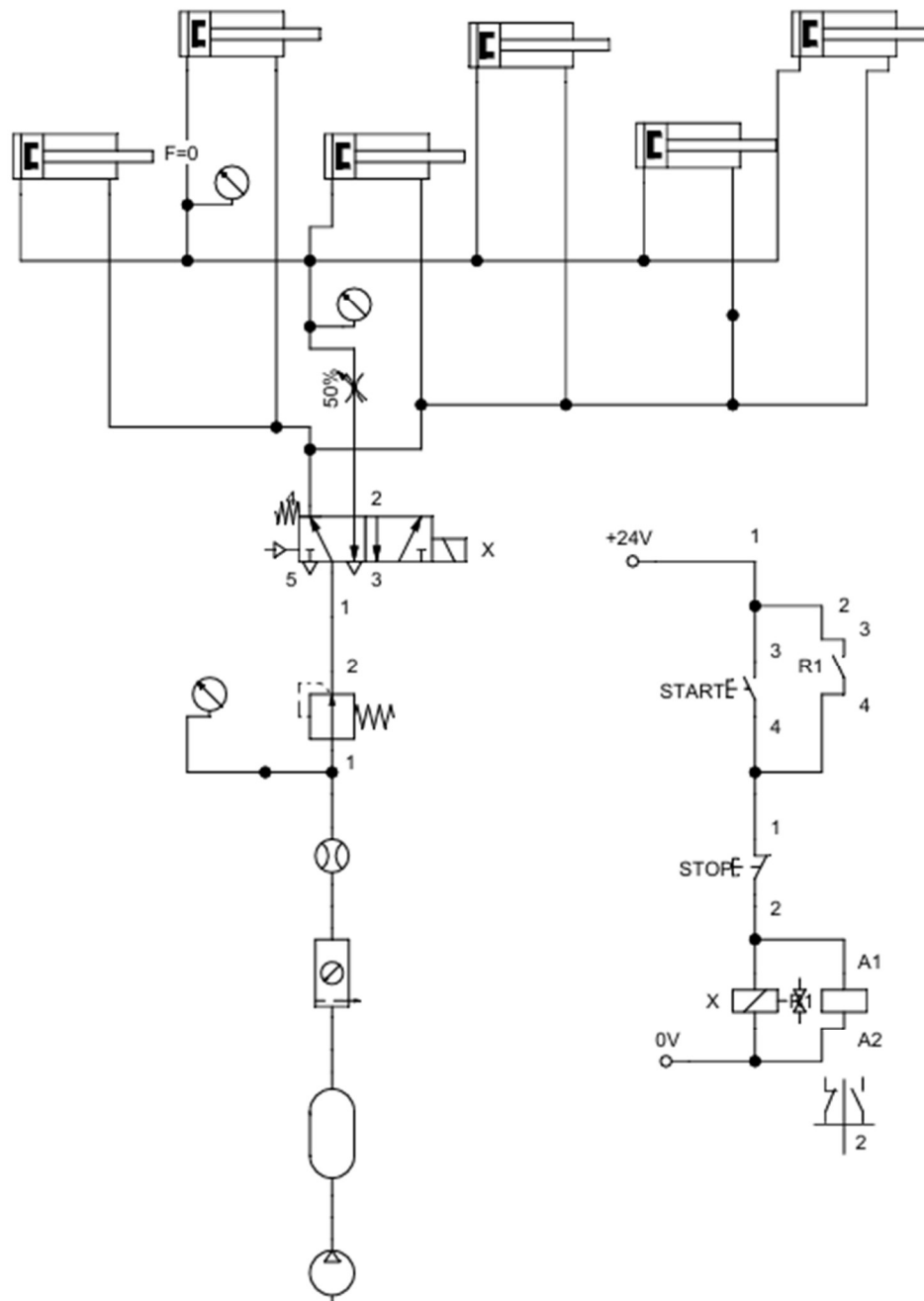


Figure 4.4 Pneumatic circuit for clamping system

4.4 Pneumatic Components

4.4.1 Selected Components

Compressor

- Copeland 50 Liters, 2 HP, Black
- Max Flow = 110 L/min
- Speed = 2500 rpm



Figure 4.5 Copeland compressor

Pistons

6 x Clamping Rotary Cylinder TJ 50x20



Figure 4.6 pneumatic cylinder (double acting)

Specifications

Bore size (mm)	12	16	20	25	32	40	50	63
Action	Double acting							
Rotation angle ^{Note 1)}	90° ±10°							
Rotary direction ^{Note 2)}	Clockwise, Counterclockwise							
Rotary stroke (mm)	7.5		9.5		15		19	
Clamp stroke (mm)	10, 20						20, 50	
Theoretical clamp force (N) ^{Note 3)}	40	75	100	185	300	525	825	1400
Fluid	Air							
Proof pressure	1.5 MPa							
Operating pressure range	0.1 to 1 MPa							
Ambient and fluid temperature	Without auto switch: -10 to 70°C (No freezing)							
	With auto switch: -10 to 60°C (No freezing)							
Lubrication	Non-lube							
Piping port size	M5 x 0.8				Rc1/8, NPT1/8, G1/8		Rc1/4, NPT1/4, G1/4	
Mounting	Through-hole/Both ends tapped common		Both ends tapped, Through-hole, Head end flange					
Cushion	Rubber bumper							
Stroke length tolerance	+0.6 -0.4							
Piston speed	50 to 200 mm/s							
Non-rotating accuracy (Clamp part) ^{Note 1)}	±1.4°	±1.2°			±0.9°		±0.7°	

Theoretical Output

Unit: N

Bore size (mm)	Rod size (mm)	Operating direction	Piston area (cm ²)	Operating pressure (MPa)			
				0.3	0.5	0.7	1.0
12	6	R	0.8	24	40	56	80
		H	1.1	33	55	77	110
16	8	R	1.5	45	75	105	150
		H	2	60	100	140	200
20	12	R	2	60.8	100	139	200
		H	3	90.2	149	208	298
25	12	R	3.7	112	185	258	370
		H	4.9	149	245	341	490
32	16	R	6	182	300	418	600
		H	8	243	400	557	800
40	16	R	10.5	319	525	731	1050
		H	12.5	380	625	870	1250
50	20	R	16.5	502	825	1149	1648
		H	19.6	596	980	1365	1961
63	20	R	28	851	1400	1950	2801
		H	31.2	948	1560	2172	3121

Note) Theoretical output (N) = Pressure (MPa) x Piston area (cm²) x 100

Operating direction
R: Rod end (Clamp)
H: Head end (Unclamp)

Weight/Through-hole Mounting

Unit: g

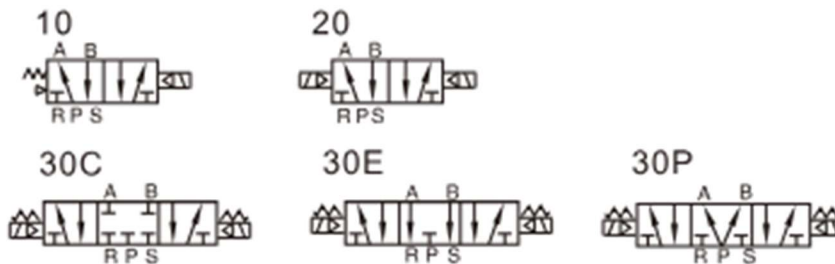
Clamp stroke (mm)	Bore size (mm)							
	12	16	20	25	32	40	50	63
10	70	100	250	280	500	595	—	—
20	87	123	290	320	525	640	1100	1520
50	—	—	—	—	—	—	1350	1805

Figure 4.7 Piston datasheet

4V300 Series



Symbol



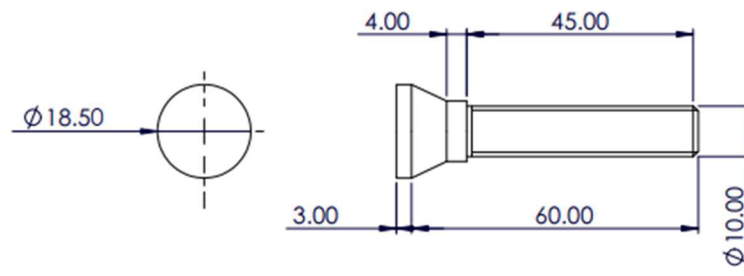
Specification

Model	4V310-08 4V320-08	4V330C-08 4V330E-08 4V330P-08	4V310-10 4V320-10	4V330C-10 4V330E-10 4V330P-10
Fluid	Air(to be filtered by 40μm filter element)			
Acting	Internal pilot or external pilot			
Port size [Note1]	In=Out=Exhaust=1/4"		In=Out=3/8" Exhaust=1/4"	
Orifice size(Cv) [Note4]	4V310-10,4V320-10:28.0mm ² (Cv=1.65) 4V330C-10:21.3mm ² (Cv=1.25)			
Valve type	5 port 2 position	5 port 3 position	5 port 2 position	5 port 3 position
Operating pressure	0.15~0.8MPa(21~114psi)			
Proof pressure	1.2MPa(175psi)			
Temperature	-20~70°C			
Material of body	Aluminum alloy			
Lubrication [Note2]	Not required			
Max.frequency [Note3]	4 cycle/sec	3 cycle/sec	4 cycle/sec	3 cycle/sec
Weight (g)	4V310-08:310 4V320-08:400	450	4V310-10:310 4V320-10:400	450

Figure 4.8 Solenoid DCV manufacturer specifications

4.4.2 Piston Clamp Hook

To optimize workpiece stability and structural integrity, a custom hook was designed for the pneumatic clamping mechanism. Initial conceptualization and geometry of the component are illustrated in the technical sketch in Figure 4.9, while the finalized brass end product after machining is shown in Figure 4.10. The component is engineered to be simple, compact, and lightweight, minimizing the added mass on the clamping assembly. Functionally, the primary impact of this brass hook is its ability to distribute concentrated mechanical forces across a wider surface area. By distributing the clamping pressure uniformly rather than focusing it on a single point of impact, the design effectively prevents localized stress concentrations, ensuring that the thin metal workpiece does not deform, crack, or break under structural load.



Size	Scale	Material
A4		Brass
	Title	Piston Hook

Figure 4.9 Sketch of the piston hook



Figure 4.10 Machined piston hook



Chapter 5: Electrical System

5.1 Introduction

The electrical system of the TIG welding machine is responsible for power conversion, arc generation, welding current control, and safe system operation. Because TIG welding requires a highly stable output and precise current regulation, the electrical architecture integrates high-frequency power electronics, a PLC-based control system, and safe isolation mechanisms. This chapter explains the electrical design, components, power stages, protection circuits, and the control logic implemented using the Delta PLC and HMI.

5.2 Electrical components selected

5.2.1 Programmable Logic Controller (PLC)

The core control unit selected for this TIG Welding Machine is the **Delta DVP-SV2 PLC** (Figure 5.1), a reliable DC/DC/DC model that fully meets the system's automation and coordination requirements. Its 24 V DC operation ensures safe and efficient control of all connected devices.

The DVP-SV2 was chosen due to its excellent performance-to-cost ratio, ease of programming, and high compatibility with other Delta components used in the system. Its fast scan time and stable communication make it suitable for precise motion control applications such as TIG welding.

The PLC provides:

- **16 digital inputs**, supporting both sink and source configurations
- **12 NPN digital outputs**, operating directly at 24 V DC
- **RS-232 and RS-485 communication ports**, allowing seamless integration with HMIs, drives, and external devices

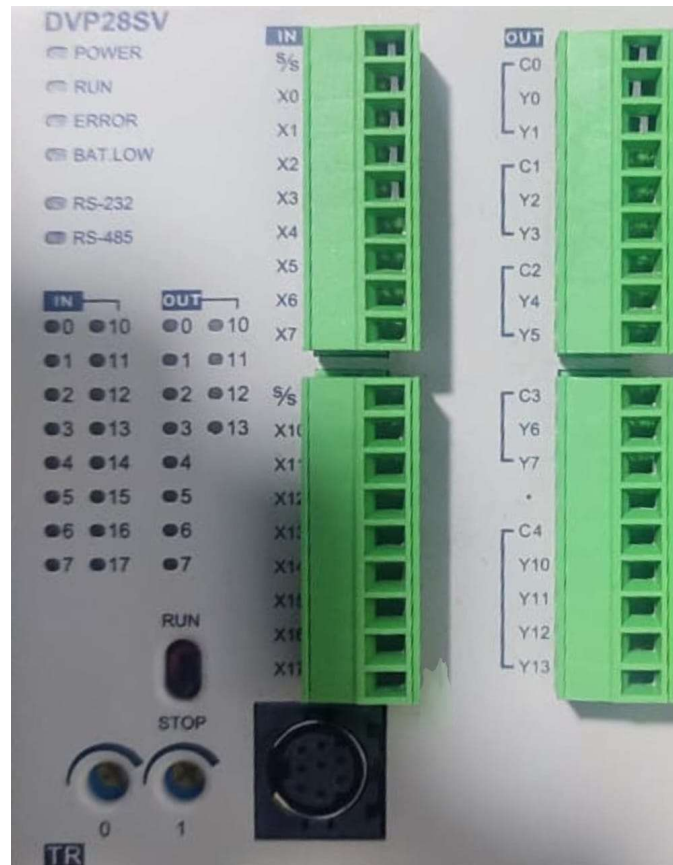


Figure 5.1 Delta DVP-SV2 PLC



Figure 5. 2 Delta DOP-107DV HMI

5.2.2 Human–Machine Interface (HMI)

To facilitate system supervision and user interaction, the **Delta DOP-107DV HMI** (Figure 5.2) was selected. This 7-inch touch panel provides an 800×480 resolution with a brightness of 400 nits, delivering a clear and responsive interface under various lighting conditions.

Key communication and hardware features include:

- **RS-232 port** for direct PLC communication
- **USB port** for configuration and data transfer
- **IP65 front-panel protection**, ensuring resistance to dust and water splashes in industrial environments

The DOP-107DV offers high reliability, intuitive navigation, and the flexibility required to adjust welding parameters and monitor system performance.

5.2.3 Power Supply

Since the entire control circuit operates at 24 V DC, a **Delta 24 V / 5 A power supply** (Figure 5.3) was selected. This power supply provides a stable, efficient, and cost-effective power source for the PLC, HMI, sensors, and drive control signals.



Figure 5.3 Delta 24 V / 5 A power supply

5.2.4 Motor and Drive System

A closed-loop hybrid stepper motor was needed to control the X-axis with high precision and repeatability. For this purpose, we selected the **YAKO YK286EC18A hybrid servo motor** (Figure 5.4), capable of delivering:

- 8 N·m holding torque
- Smooth operation at up to 100 rpm



Step angle:	1.8°	Length:	140mm
Holding torque:	8.2N.m	Rated current:	6.0A
Phase resistance:	0.53Ω	Phase inductance:	4.7mH
Rotor inertia:	3600g.cm ²	Motor weight:	3.75kg
With driver:	MS-S3		

Figure 5.4 YAKO YK286EC18A hybrid servo motor specifications

Voltage: 75Vac

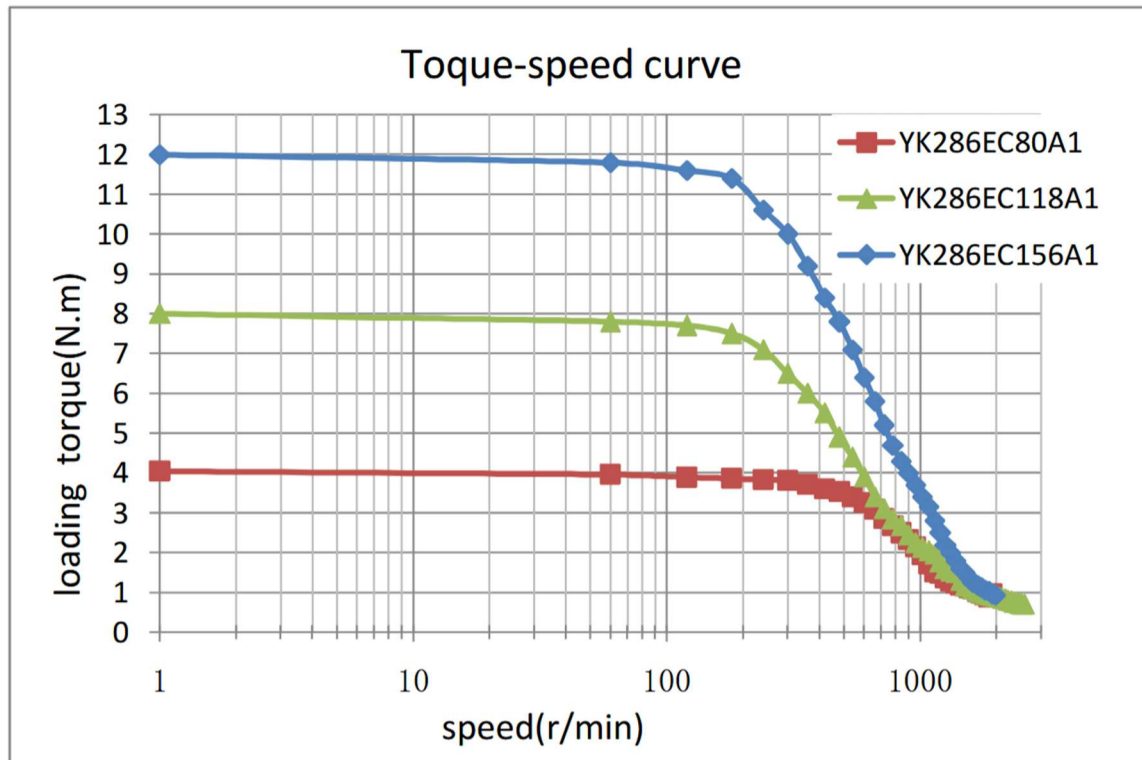


Figure 5.6 YAKO YK286EC18A hybrid servo motor characteristics curve



Figure 5.7 YAKO MS-S3 hybrid servo driver

This torque capacity is more than sufficient to handle the 20 kg carriage used in the machine. The motor is paired with the **YAKO MS-S3 hybrid servo driver** (Figure 5.7), which offers reliable pulse-based control and provides two essential feedback signals to the PLC:

- **Alarm output**, indicating any motor or driver-related fault
- **Position confirmation output**, confirming successful arrival at the commanded position.

5.2.5 Additional Control Components

To complete the control system wiring and ensure safety, several auxiliary components were included:

- **Two Normally Open push buttons**
- **One Normally Closed push button**
- **Two 10 A circuit breakers** used for overload and short-circuit protection

These elements allow safe operation, manual control during testing, and protection of both power and control circuits.



Figure 5.8 MCCB



Figure 5.9 Push Buttons

5.3 PLC programming

To control all the devices and components used in this project, the PLC will play a central and essential role. Every operation of the welding machine depends on a set of coordinated functions, and the PLC (Programmable Logic Controller) is responsible for managing all inputs, processing them according to the programmed logic, and generating the necessary outputs. In simple terms, a PLC receives signals from sensors, buttons, or the HMI, executes a programmed control algorithm, and then sends corresponding commands to motors, actuators, and other outputs. This makes the PLC the “brain” of the entire system.

Before building the full welding machine logic, we must first develop the functions that control the motors. The first motor we work with is the X-axis motor, which in our design is a stepper motor driven by pulses. The position command is sent from the HMI screen as a floating-point (REAL) value representing distance in millimeters. However, a stepper motor does not understand millimeters; it only understands pulse counts. Therefore, our first task is to convert this distance value into an integer representing the exact number of pulses the motor must receive to reach the desired position.

5.3.1 Conversion Between Millimeters and Pulses

The X-axis uses a **ball screw with 5 mm pitch**, meaning one rotation of the screw moves the carriage by 5 mm. The motor requires **200 pulses per revolution**, giving:

$$k = \frac{200}{5} = 40 \text{ pulses/mm}$$

Thus:

Distance → Pulses:

$$\text{pulses} = 40 \times \text{distance(mm)} \quad (5.1)$$

Pulses → Distance:

$$\text{distance (mm)} = \frac{\text{pulses}}{40} \quad (5.2)$$

Two reusable function blocks were created in ISPSOft These ensure accurate and consistent axis positioning:

- mm2pulse: converts HMI inputs from mm to pulses (Figure 5.10).
- pulse2mm; displays motor feedback in mm on the HMI (Figure 5.11).

pulse2mm

No.	Declaration Type	Identifiers	Address	Type	Initial Value	Identifier Comments
1	VAR_INPUT	input	N/A [Auto]	DWORD	N/A	
2	VAR_OUTPUT	output	N/A [Auto]	REAL	N/A	
3	VAR	temp	N/A [Auto]	REAL	N/A	

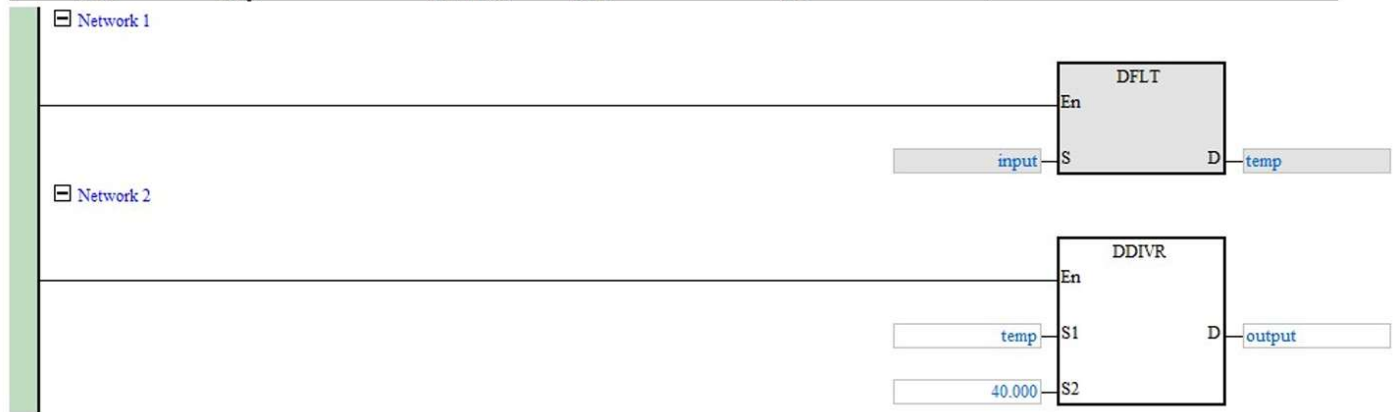


Figure 5.10 Pulse to mm ladder diagram

mm2pulse

No.	Declaration Type	Identifiers	Address	Type	Initial Value	Identifier Comments
1	VAR_INPUT	input	N/A [Auto]	REAL	N/A	
2	VAR_OUTPUT	output	N/A [Auto]	DWORD	N/A	
3	VAR	temp	N/A [Auto]	REAL	N/A	



Figure 5.11 mm to pulse ladder diagram

5.3.2 Homing Routine

A homing sequence is essential to establish a known reference position for the X-axis. A roller-type limit switch is used to detect the mechanical home point. Homing is essential because it allows the system to establish a reliable reference point—often called the absolute zero position—from which all future movements and measurements will be based. Without a precise homing procedure, the axis will not know where it physically is, which can lead to inaccurate positioning or unexpected motion.

In ISPSOft, the **DZRN (Return to Zero)** function block (Figure 5.12) was used to automate the procedure. DZRN:

- Move the axis slowly toward the limit switch
- Detects the switch activation
- Backs off slightly for accuracy
- Sets the absolute zero position

To ensure safe operation, a dedicated internal bit marks the completion of homing. Once activated, the DZRN function is disabled to prevent unintended re-homing during normal operation.

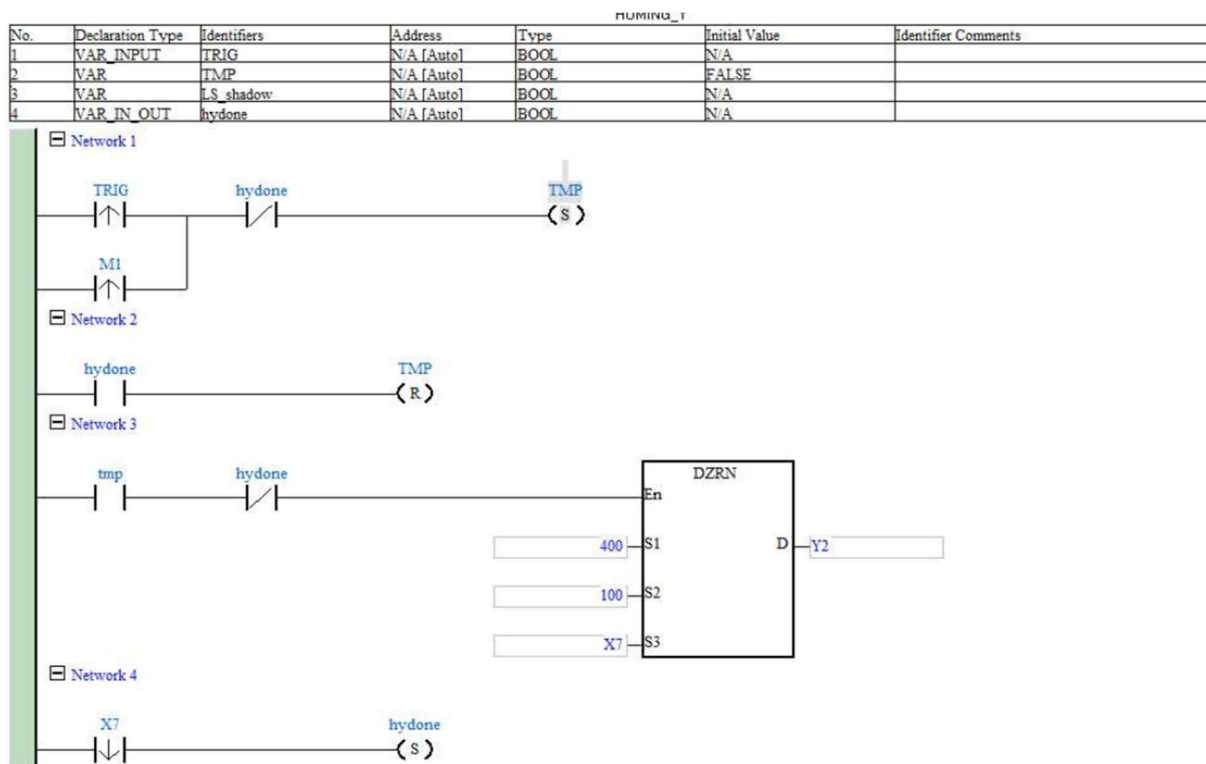
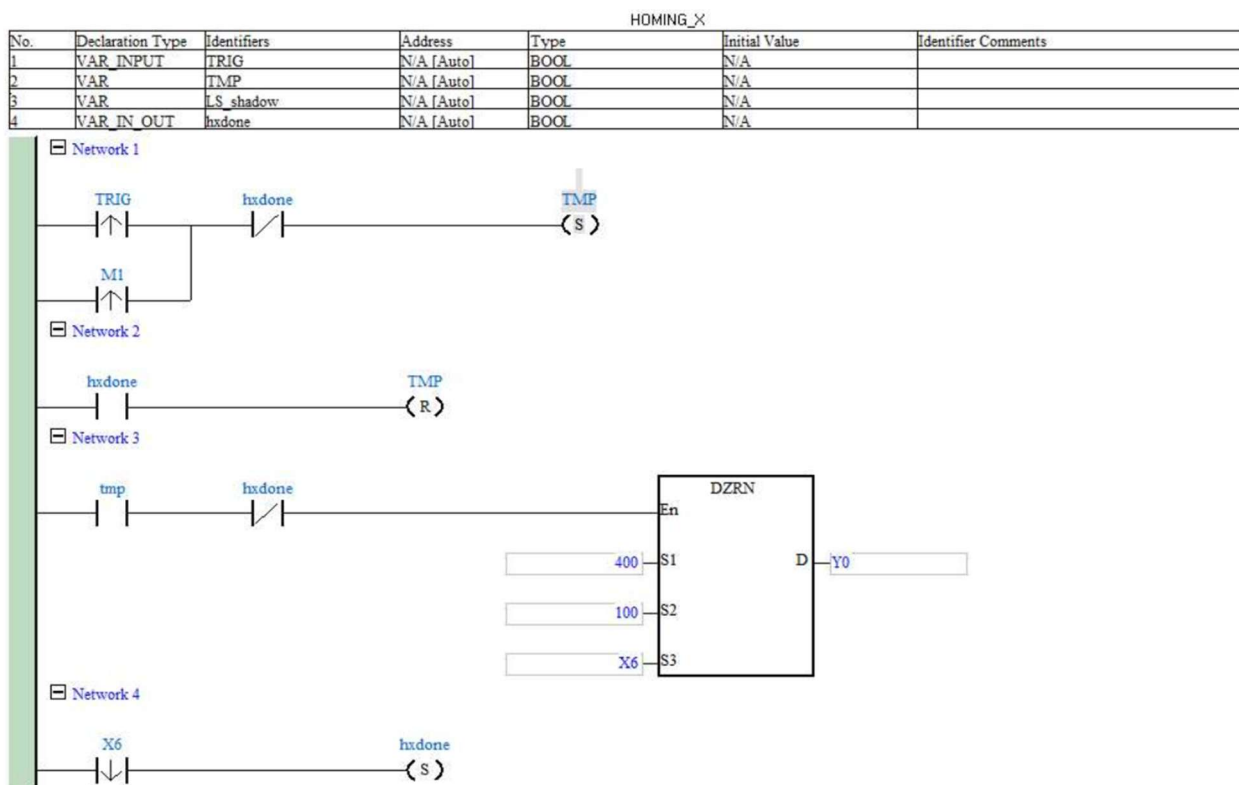


Figure 5.12 Homing function ladder diagram



At this stage, our input signals are fully prepared, and the homing process has been successfully completed. With the axis now knowing its absolute reference position, we can begin designing the logic that determines how the motor will interpret and handle the incoming motion commands.

The control process starts with two target positions sent from the HMI. These two positions represent the boundaries of the welding stroke—the motor must move back and forth between them during operation. However, when the system first starts, the motor could be standing at any arbitrary position along the axis. There is no guarantee that its initial location matches either of the two positions received from the HMI. Therefore, our first task is to determine which of the two target positions is closest to the current position.

To accomplish this, the PLC will compare the present motor location with both input positions. Since a PLC works purely with logic and numerical operations, we will subtract each target position from the current position and take the absolute value of both results. This removes any negative signs and ensures we compare distances correctly. Once we have these two distance values, we simply choose the smaller one. The motor will then move to the nearest target position using a predefined default speed (Figure 5.13).

No.	Declaration Type	Identifiers	Address	Type	Initial Value	Identifier Comments
1	VAR	fb00	N/A [Auto]	x_axis	N/A	
2	VAR	speed	D7030	DWORD	N/A	
3	VAR	starting_position	D7032	REAL	N/A	
4	VAR	ending_position	D7034	REAL	N/A	
5	VAR	next_position	D7036	DWORD	N/A	
6	VAR	latch	M3008	BOOL	FALSE	
7	VAR	unlatch	M3009	BOOL	N/A	
8	VAR	in_position	X3	BOOL	N/A	
9	VAR	fb01	N/A [Auto]	HOMING_x	N/A	
10	VAR	HNPProg0	M3016	BOOL	N/A	
11	VAR	jog1	M3017	BOOL	FALSE	
12	VAR	weld	M3018	BOOL	FALSE	
13	VAR	jog2	M3019	BOOL	FALSE	
14	VAR	latch2	M3024	BOOL	FALSE	
15	VAR	tmr0	M3025	BOOL	FALSE	
16	VAR	tmr1	M3026	BOOL	FALSE	
17	VAR	HNPProg1	M3031	BOOL	N/A	
18	VAR	v_distance	D7056	REAL	N/A	
19	VAR	y_axis0	N/A [Auto]	y_axis	N/A	
20	VAR	HOMING_Y0	N/A [Auto]	HOMING_Y	N/A	
21	VAR	yd	D7074	DWORD	N/A	
22	VAR	latchy	M3042	BOOL	N/A	
23	VAR	unlatchy	M3043	BOOL	N/A	
24	VAR	hxdone	M3039	BOOL	FALSE	

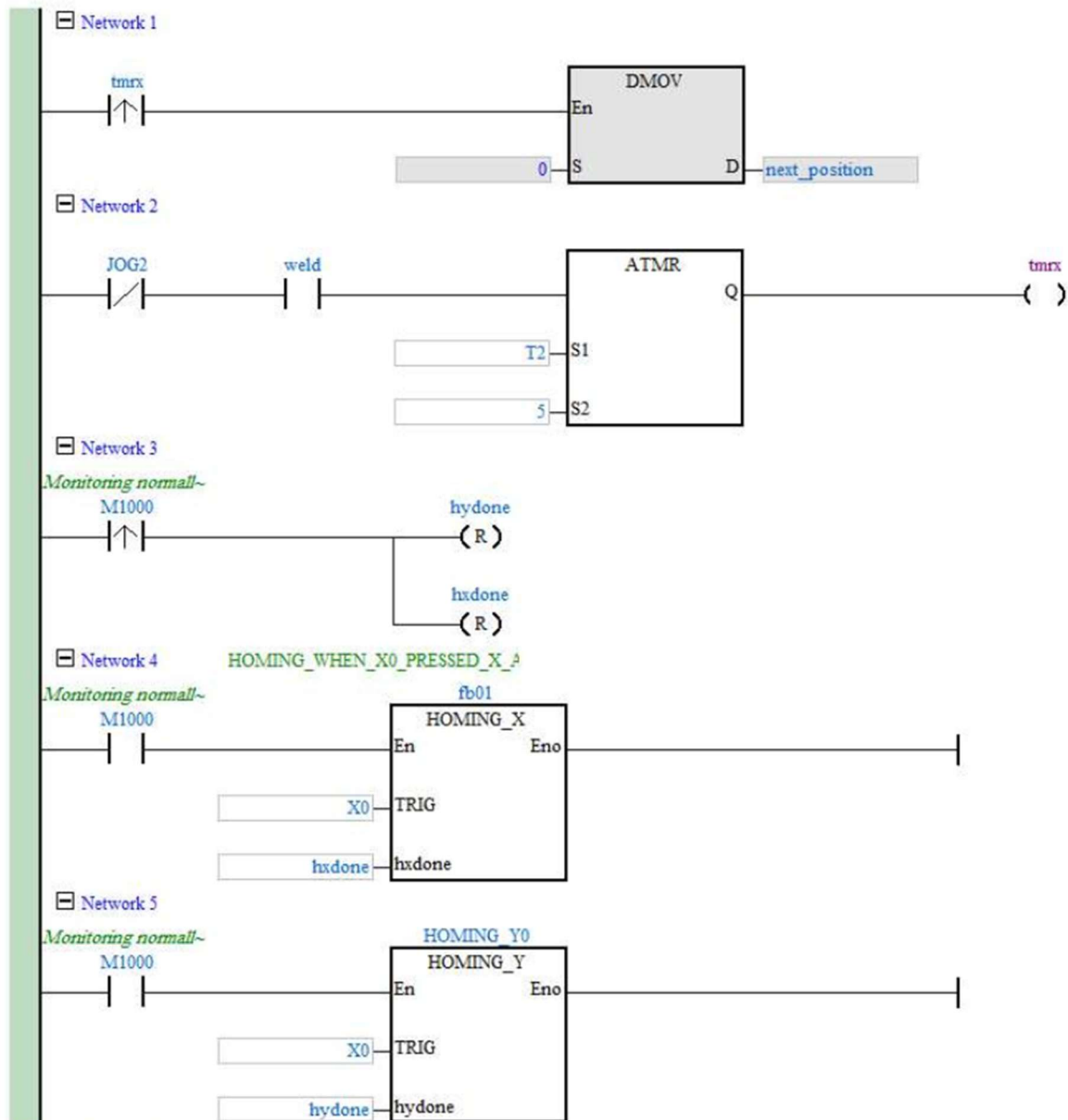


Figure 5.13 Homing routine ladder diagram

5.3.3 Operation Control

After reaching the first target position, the welding process should begin. The welding speed will be sent from the HMI into a dedicated register named **welding speed**. To manage the transition between default movement speed and welding speed, the PLC will rely on a set of internal status bits. These bits indicate which phase of motion the system is currently in— whether it is approaching the first position or performing the welding stroke between the two positions.

When the motor arrives at the nearest target position, the PLC will activate a special bit signaling that the first phase is complete. At this moment, we will introduce a 1-second delay timer. This pause allows the motor to settle mechanically, ensuring that the carriage is stable before welding begins.

Once the delay expires, the motor must travel to the second position provided by the HMI. To determine this next destination, we can reuse the same comparison logic used earlier. However, we must modify the function to include an additional condition: if the motor has already reached one of the two positions, it should automatically move to the other one next. This ensures continuous movement between the two welding boundaries and enables smooth back-and-forth welding motion (Figure 5.14a).

By building this structured decision process, evaluating the closest target, transitioning speeds using status bits, and alternating between positions, we establish a reliable foundation for the welding machine's automatic movement cycle.

Finally, we can describe the logic of the main program, which coordinates all the individual functions we created earlier. When the PLC first powers up, the program begins with an initialization rung that resets the homing-complete bit. This forces the machine to perform the homing routine every time the CPU starts, ensuring that the system always begins from a known and accurate reference point. Once this reset is executed, the homing function is called, returning the motor to its mechanical zero position.

After the homing sequence finishes, the CPU proceeds to read the two target positions and the welding speed sent from the HMI. These values are copied into internal registers where they can be processed and modified by the different functions. At this stage, the system is ready for operator input.

When the workers press the Start button, the logic responsible for deciding the next position becomes active. This function compares the current motor position with the two input positions, determines which one is closer, and outputs the appropriate next target. Once this decision is made, the DDRVA function block—responsible for controlling absolute positioning in Delta PLCs—is triggered. DDRVA then moves the motor to the selected position using the default movement speed (Figure 5.14b).

When the motor reaches the first target position, a one-second timer is activated. This short delay ensures the axis is mechanically stable before switching speeds or beginning the welding stroke. After the timer completes, the program recalls the next-position function, which now outputs the second position. The DDRVA block is executed again, this time with the speed parameter switched from the default speed to the welding speed.

Upon arrival at the second position, another one-second timer is triggered. Once the timer completes, the program prepares the final movement cycle. The next target is set to absolute zero, the speed is restored to the default value, and DDRVA is executed again to return the axis to its reference point (Figure 5.14c).

When this final movement is complete, all special bits, status flags, and internal registers must be reset to their initial values—except for the homing bit, which remains set to indicate that the axis is already homed. This full reinitialization is essential for allowing the machine to be reused repeatedly without retaining old values that could cause unexpected behavior or operational errors or bugs in future cycles (Figure 5.14d).

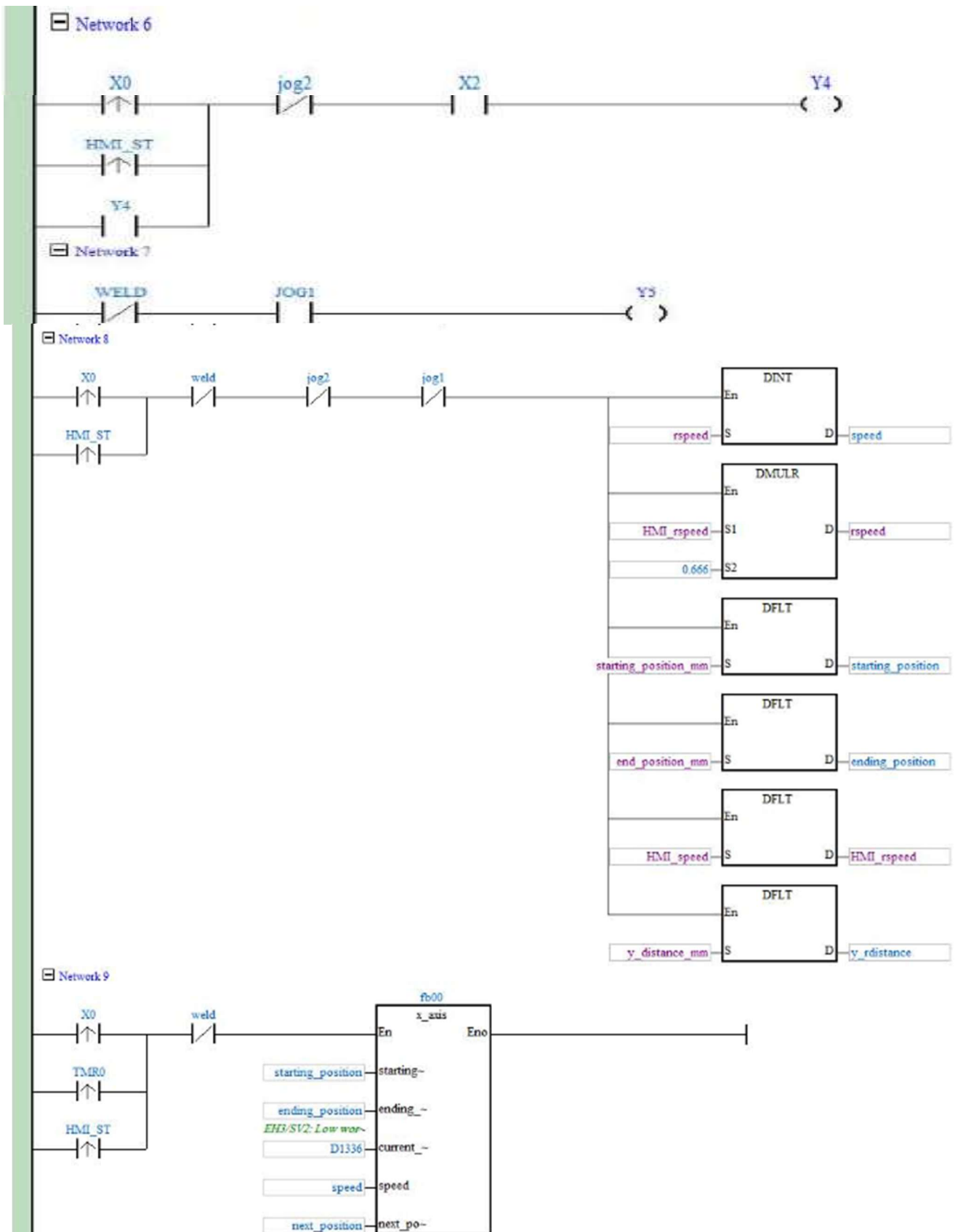


Figure 5.14a operation control ladder diagram

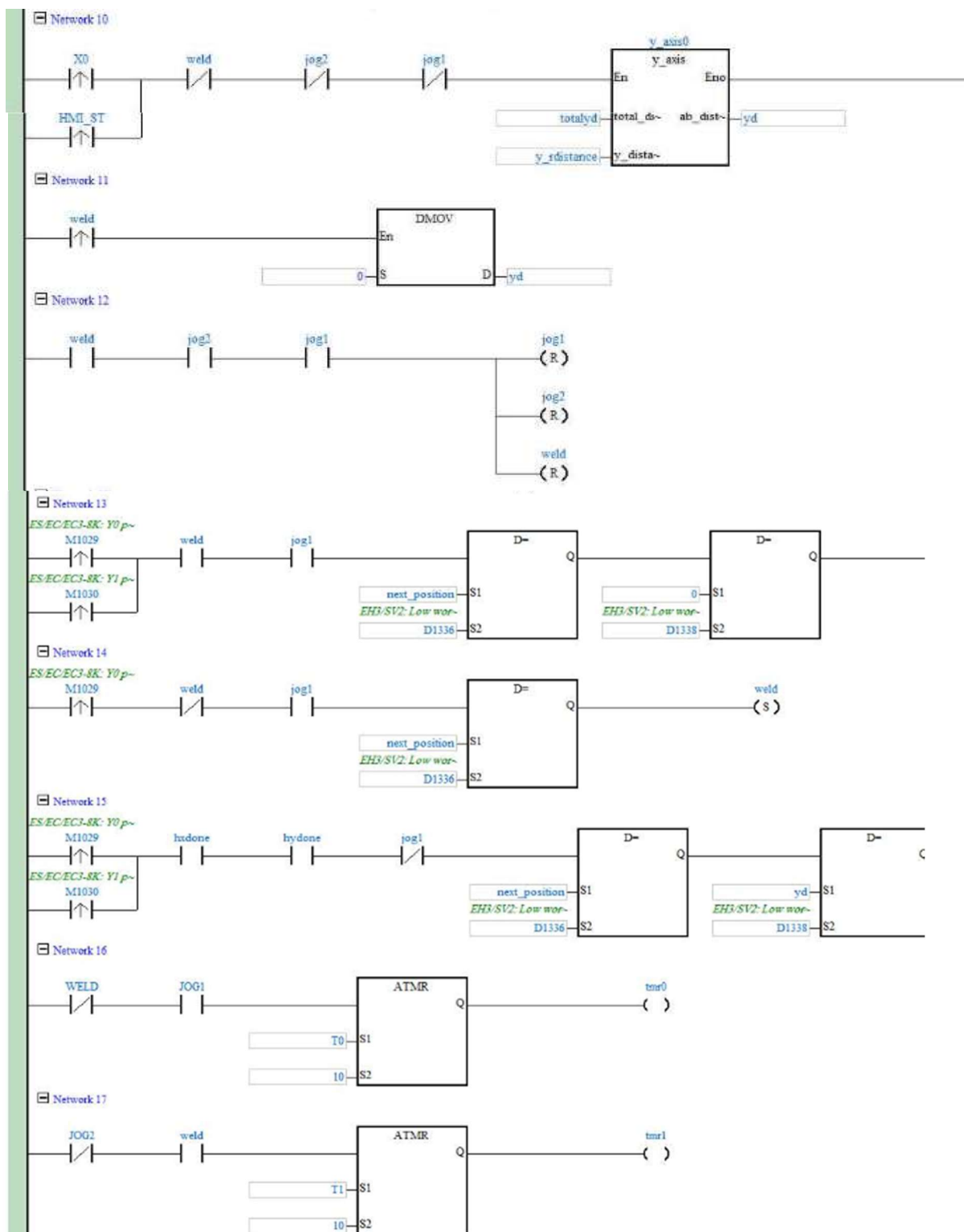


Figure 5.14b operation control ladder diagram

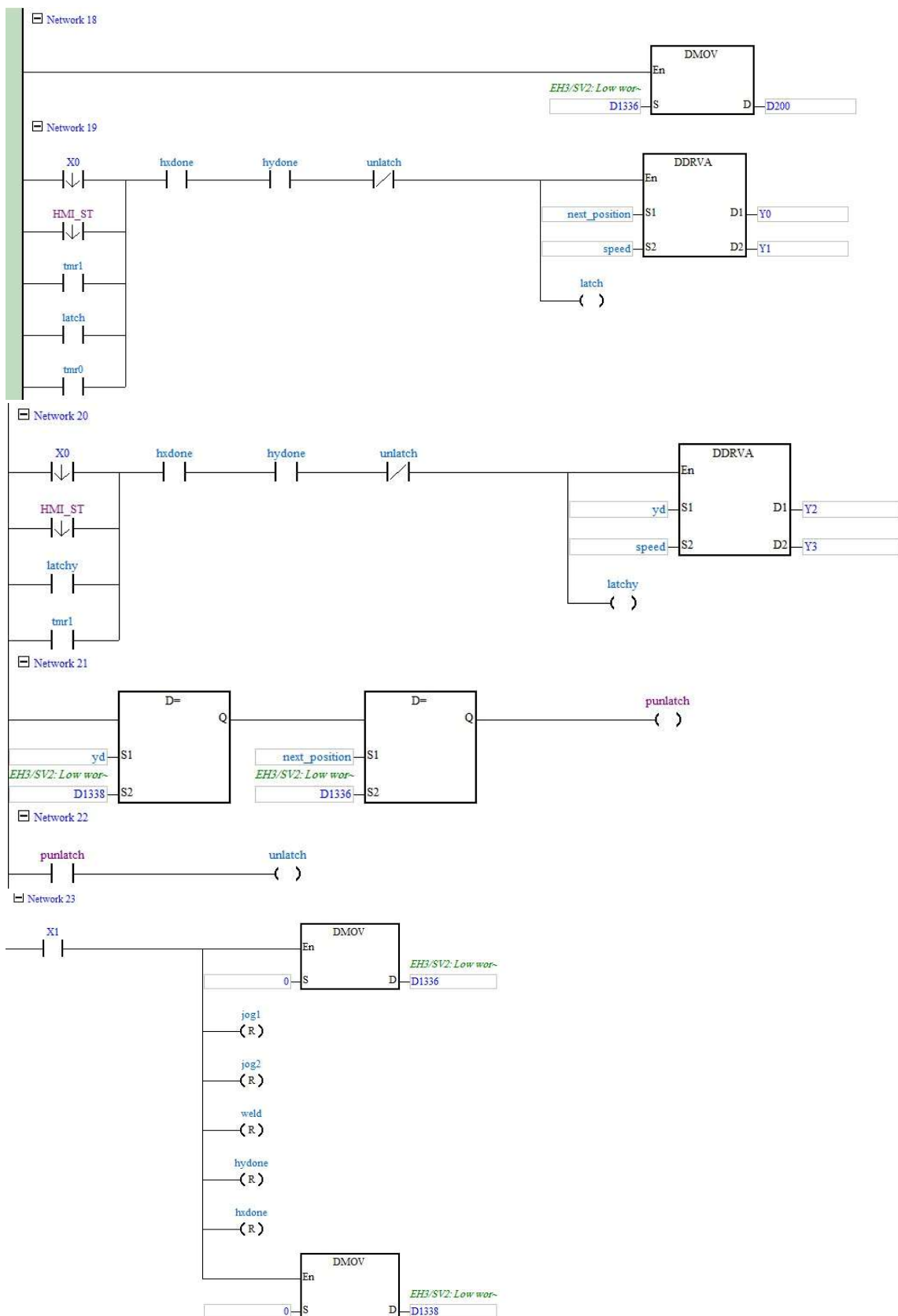


Figure 5.14c operation control ladder diagram

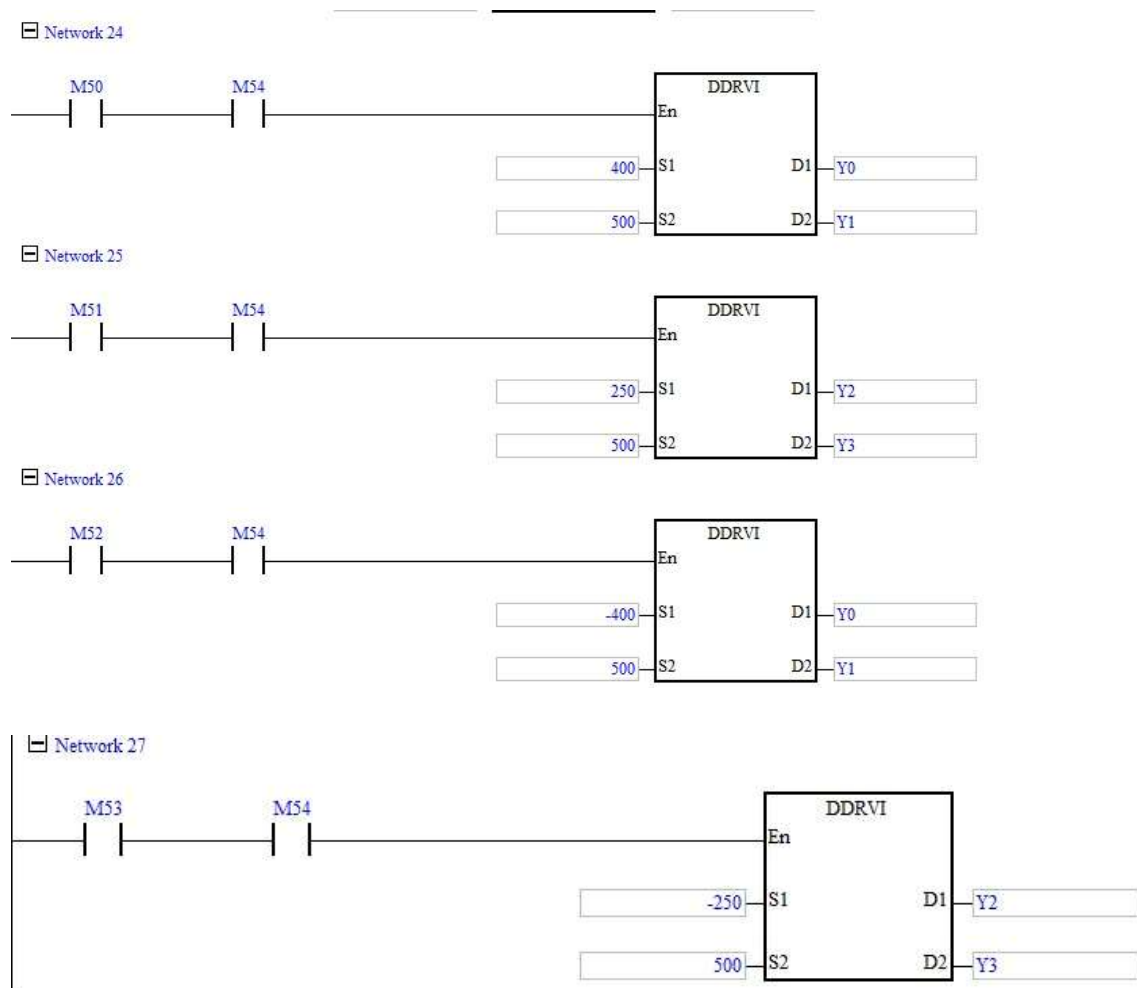


Figure 5.14d operation control ladder diagram

5.4 HMI screens

For both PLC and **HMI**, we have completed the foundational structure and interface framework that will serve as the basis for the remainder of the project.

Using this **HMI program**, our machine will be operated through **seven main screens**. The first of these is the **main Screen** (Figure 5.15), which appears immediately after the user logs in with their assigned **username and password**. This screen serves as the central navigation hub, allowing the user to access other screens and perform essential machine control functions.



Figure 5.15 Login screen

On the **Main Screen** (Figure 5.16), several buttons are available to navigate to other sections of the interface. For example, selecting the **Status** button directs the user to the **Status Screen** (Figure 5.17), where they can monitor the real-time condition of all **input and output signals** in the system.

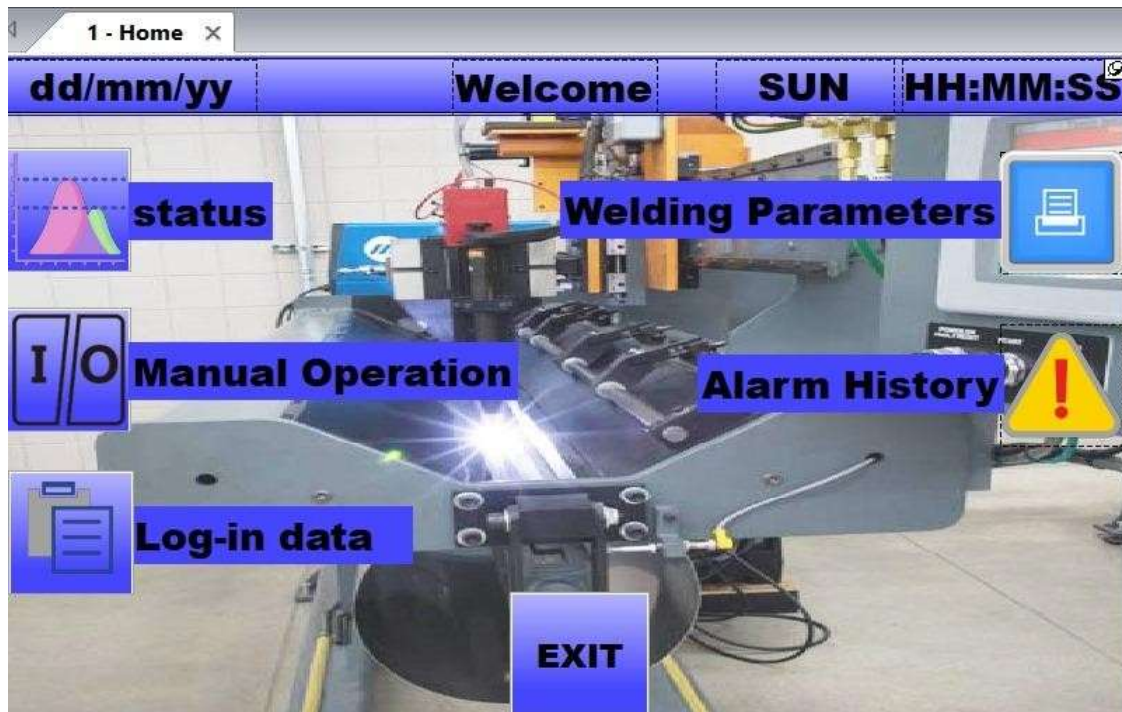


Figure 5.16 Main screen

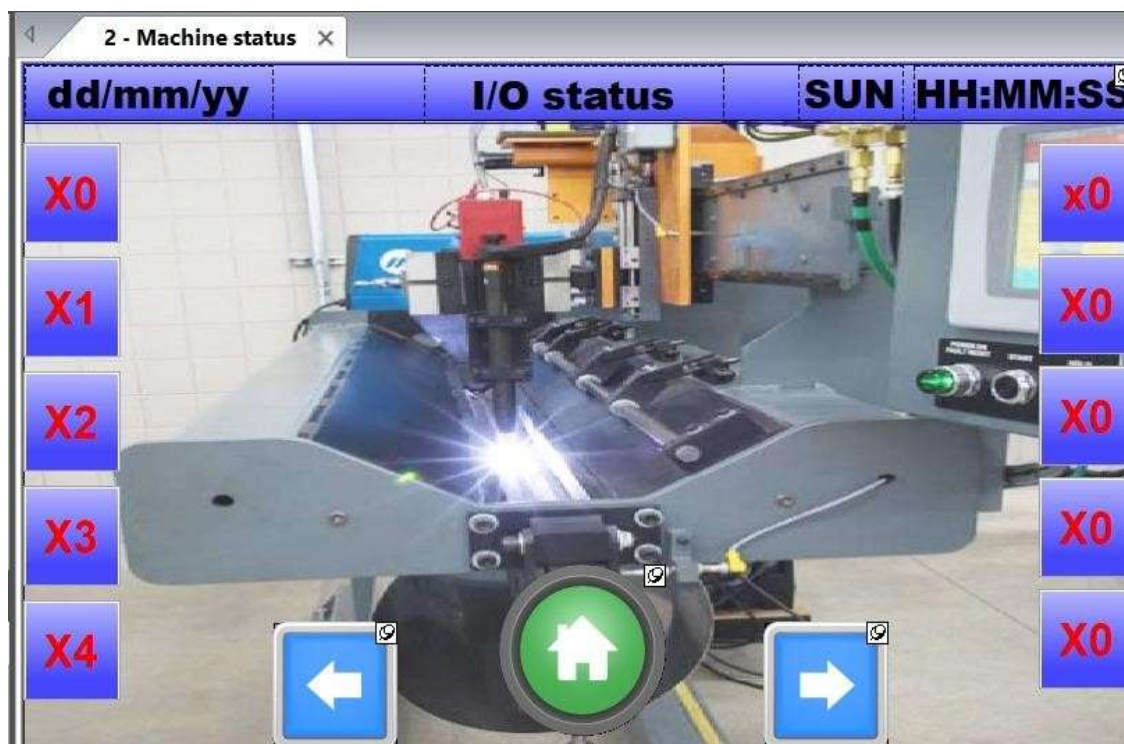


Figure 5.17 Machine status screen

Another screen is the **Manual Operation Screen** (Figure 5.18), which is currently under development. It can be accessed from the **Main Screen** by selecting the **Manual Operation** button.

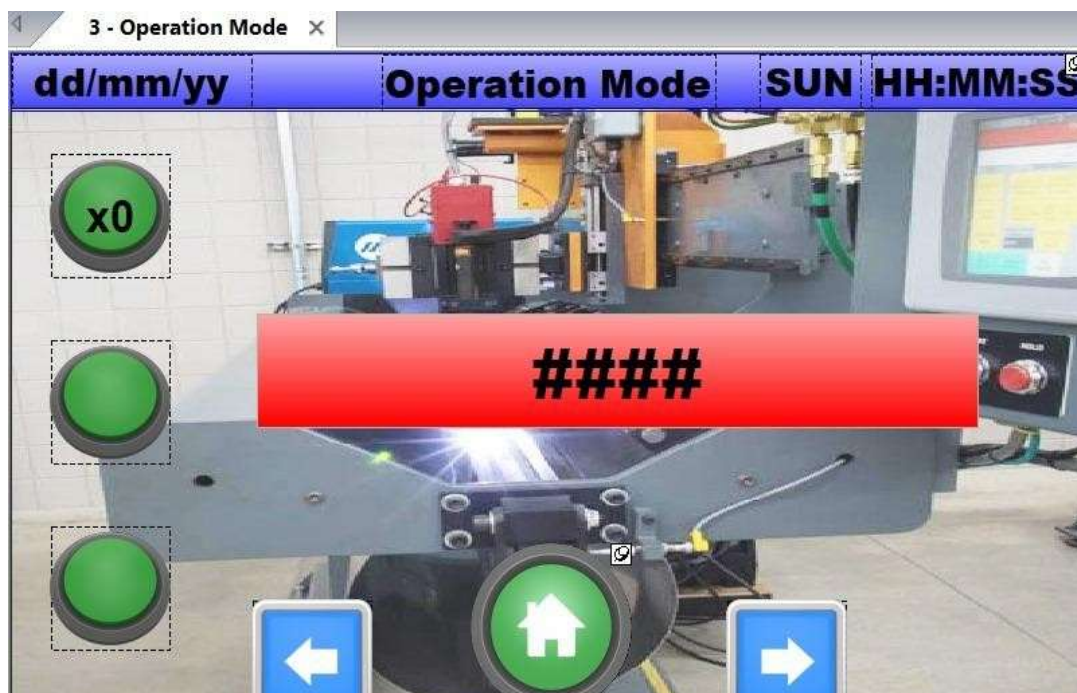


Figure 5.18 Manual mode

There is also a **Welding Parameters Screen** (Figure 5.19), accessible from the **Main Screen** via the “**Welding Parameters**” button. This screen allows the user to **adjust welding settings** and **control the welding process**, providing flexibility and precision in fine-tuning operational parameters.

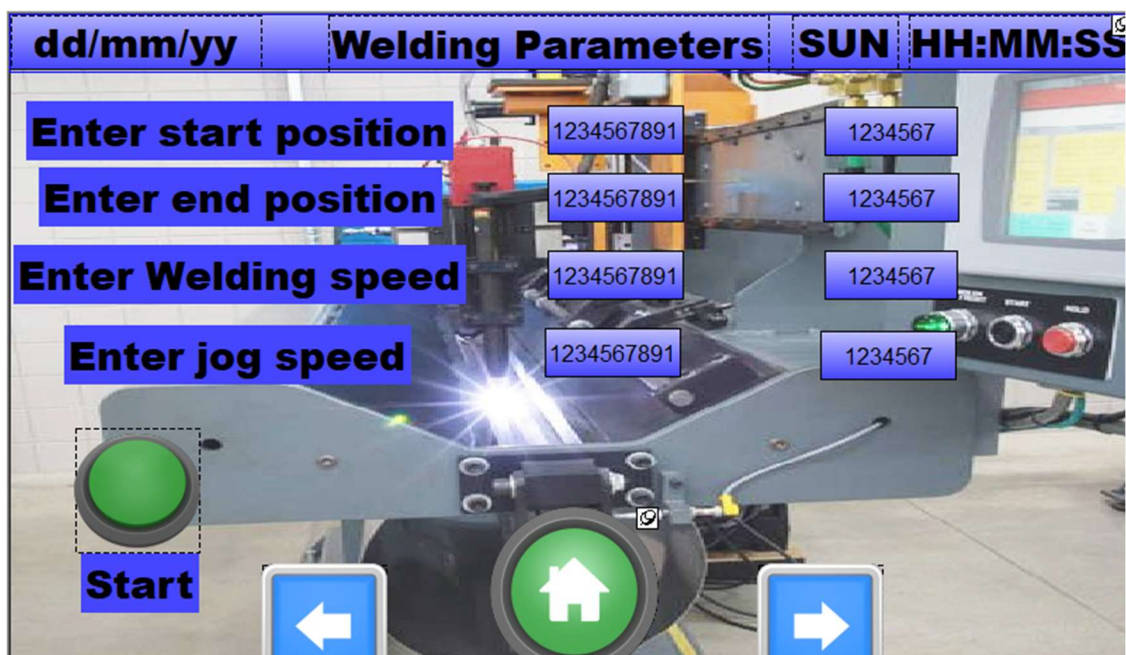


Figure 5.19 welding parameters screen

Two additional important screens are the **Alarm History Screen** (Figure 5.4.6) and the **Login Data Screen** (Figure 5.20).

The **Alarm History Screen** records all faults that occur during operation, providing detailed information such as the **fault name**, **date**, and **time** of each event, which helps with troubleshooting and maintenance.

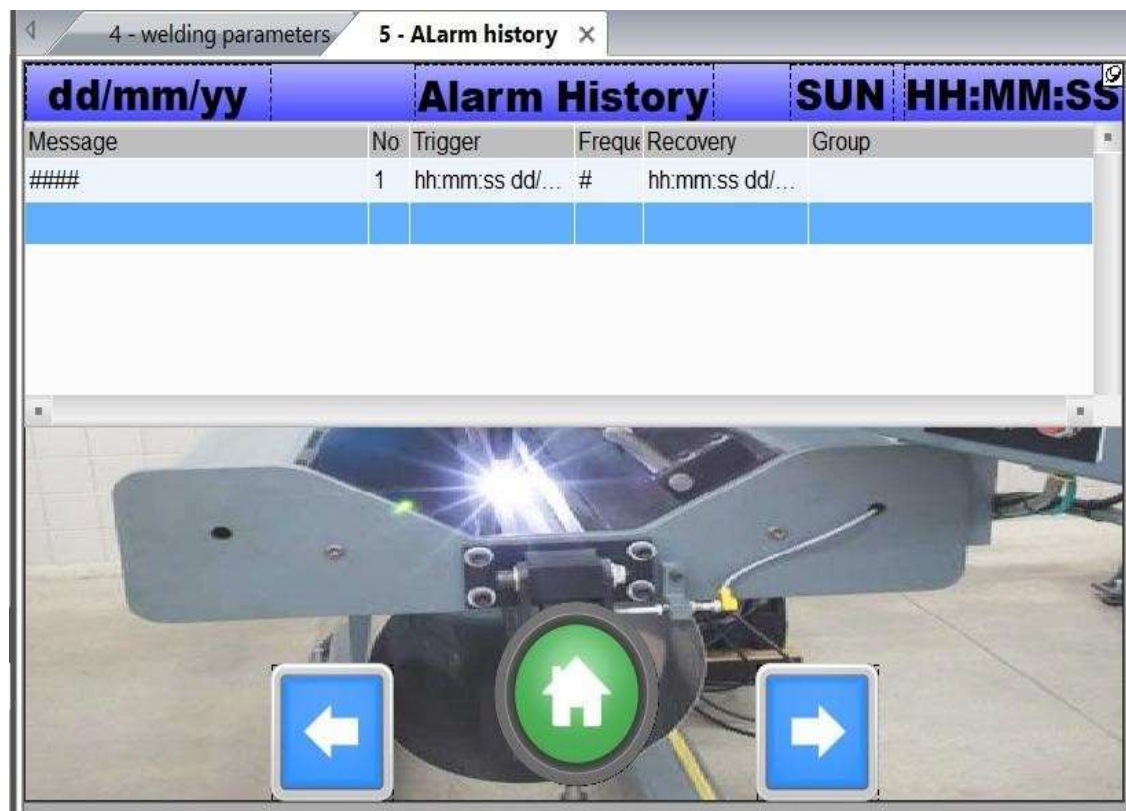


Figure 5.20 Alaram history screen

The **Login Data Screen** (Figure 5.21) keeps track of user access information. Both screens can be conveniently accessed from the **Main Screen** through their dedicated buttons.

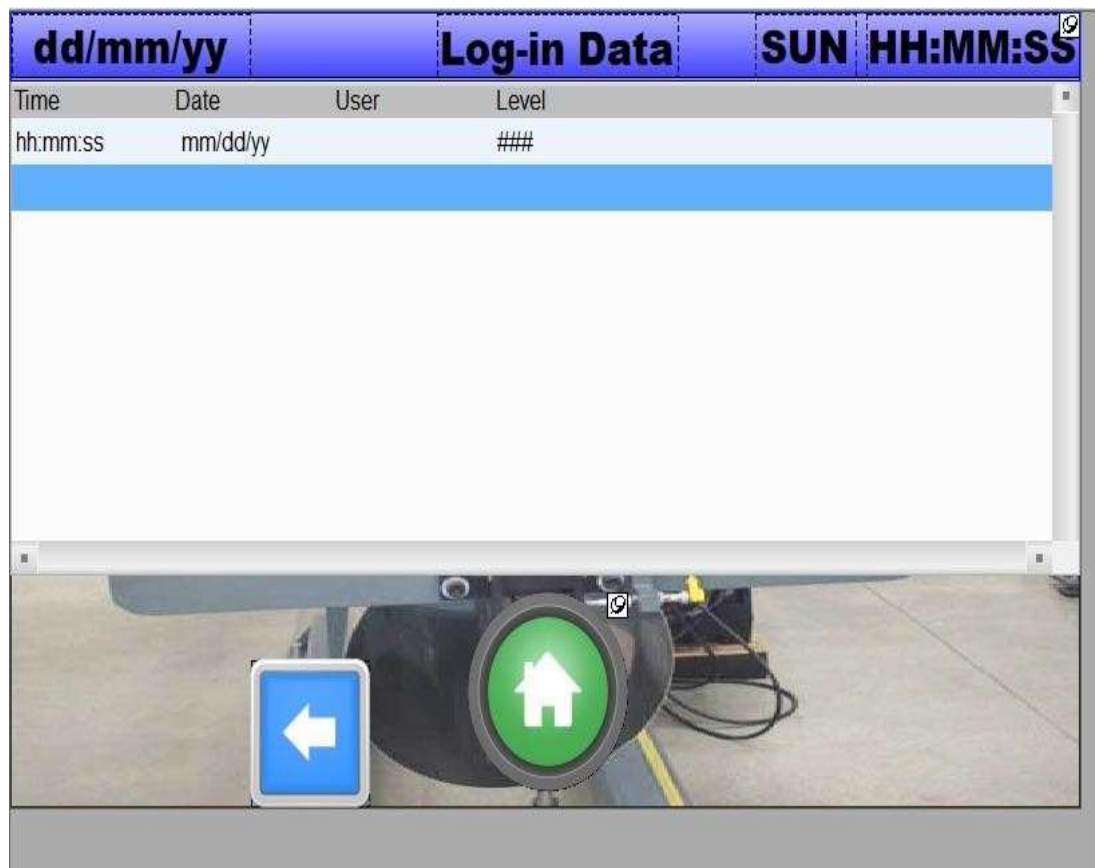


Figure 5.21 Log in data screen



Chapter 6: Industrial Internet of Things (IIoT)

6.1 Introduction

The rapid evolution of the Industrial Internet of Things (IIoT) has transformed traditional industrial systems by enabling interconnected machines, real-time monitoring, and intelligent control mechanisms. In alignment with these advancements, this graduation project aimed to design and simulate an integrated IIoT environment demonstrating both conceptual understanding and practical implementation.

The project consisted of several phases: learning core IIoT concepts, developing logic flows through Node-RED, implementing auxiliary logic using Java, constructing a dashboard with an authentication system, and simulating HMI screens in two functional modes. This report details the methodologies, tools, and outcomes associated with each development stage.

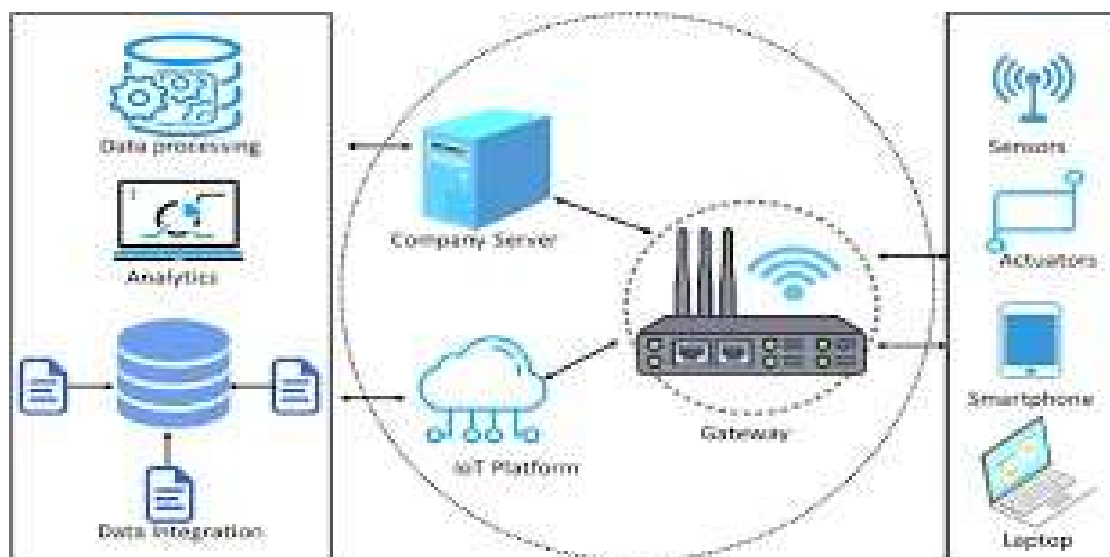


Figure 6.1 Main components of an IIoT system

6.2 Foundations and Theoretical Background

A foundational phase of this project was dedicated to studying the principles underlying Industrial Internet of Things (IIoT) systems. This included investigating standard architectural frameworks—typically composed of the perception, network, and application layers—alongside foundational edge and cloud computing paradigms to minimize latency via localized real-time data processing. This architectural review guided the investigation of system reliability, fault tolerance, and crucial cybersecurity considerations, such as authentication and data integrity mechanisms essential for secure industrial networks.

A primary focus was placed on evaluating network communication protocols to establish standard IIoT practices, specifically comparing the following:

- Transmission Control Protocol / Internet Protocol (TCP/IP): Serves as the core foundation of network communication, ensuring reliable, connection-oriented data transfer.
- Hypertext Transfer Protocol (HTTP): Utilizes a synchronous Request/Response model, though it is traditionally optimized for web applications rather than restricted industrial hardware.
- Message Queuing Telemetry Transport (MQTT): Operates on an asynchronous Publisher/Subscriber architecture characterized by low bandwidth consumption and a highly lightweight overhead.

Based on this comparative analysis, MQTT was selected as the primary communication protocol. Its architectural efficiency, low bandwidth requirements, and minimal overhead make it uniquely suited for real-time machine monitoring applications, ensuring continuous, fault-tolerant device-to-cloud telemetry.

6.3 Development of Logical Flows Using Node-RED

Node-RED was employed as the primary tool for constructing the project's logic and data flows due to its event-driven environment and intuitive visual programming interface. During the development process, I acquired proficiency in:

- Utilizing built-in nodes for data acquisition, processing, and output responses.
- Creating customized function blocks using JavaScript to implement logic not available through default nodes.
- Structuring complete workflows that simulated sensor inputs, machine responses, and user interactions.
- Integrating MQTT nodes to emulate IIoT communication between system components.

The resulting flows established the backbone of the project's interactive and automated features, reflecting real-world industrial processes.

6.3.1 Java Integration for Advanced System Logic

To complement the workflow logic created in Node-RED, basic Java was incorporated to implement more specific operational tasks that required procedural programming. Through this, I developed:

- A fundamental understanding of Java syntax, including variables, loops, conditionals, and functions.
- Small program modules capable of executing calculations, decision-making, and state management.
- Mechanisms to interface Java logic with the overall simulated control system.

Although the integration involved introductory-level Java programming, it offered valuable insight into backend logic design and gave the system more flexibility in handling diverse operations.

6.4 Simulation Project: HMI Control Screen

The practical work centered on creating an HMI control screen simulation—an essential operator tool in industrial systems. The simulation was structured into a registration interface, a main menu, and control panels that offer both automatic and manual modes of operation. Each part was designed to reflect common industrial workflows, including authentication, clear navigation, and flexible operational control.

6.4.1 Registration Interface

The simulation begins with a registration interface (Figure 6.2) designed to ensure that only authorized users gain access. Input fields were created for user data, submitted information was validated, and successful registration triggered a transition to the main menu. This structure reflects the emphasis on security and user management in IIoT environments.

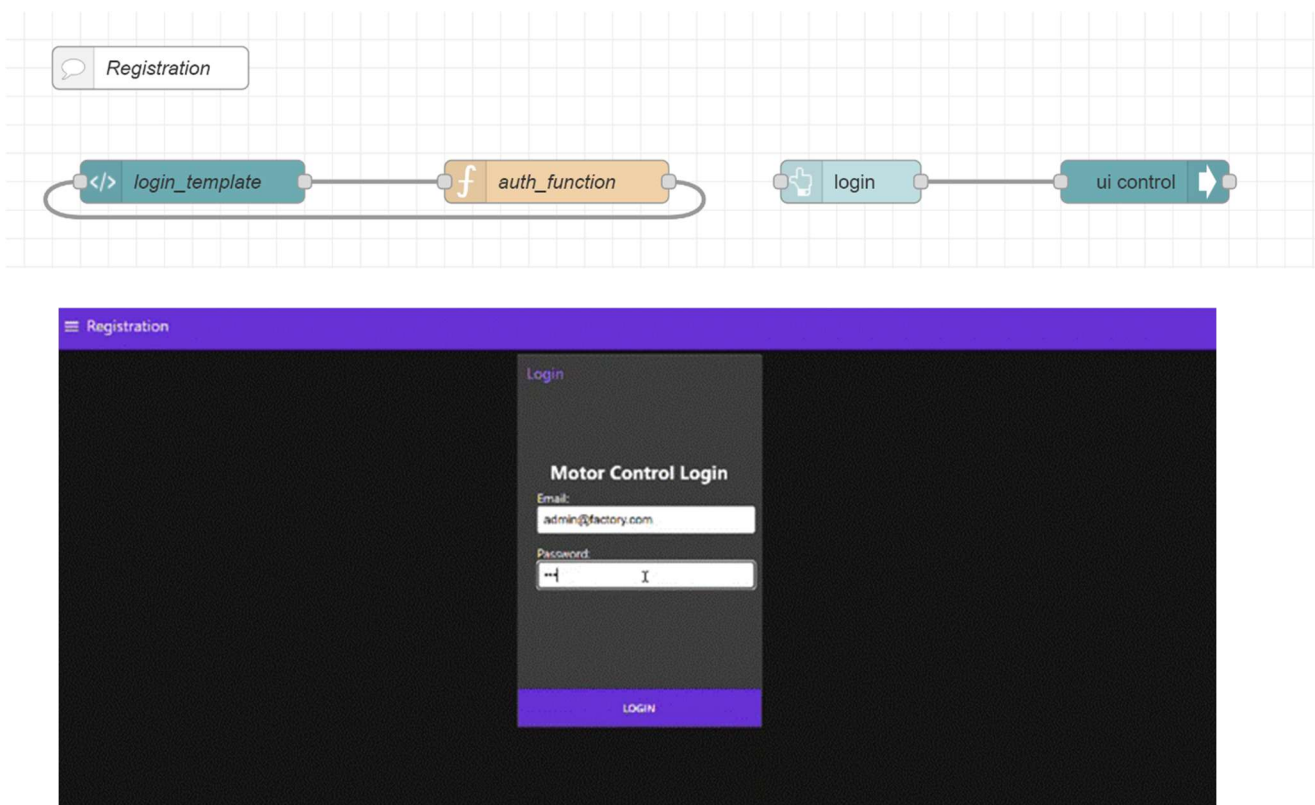


Figure 6.2 Registration screen

6.4.2 Main Menu Structure

After authentication, users enter a main menu (Figure 6.3) that provides access to two system functions. The navigation flow was kept simple and efficient to meet industrial standards for operator usability.

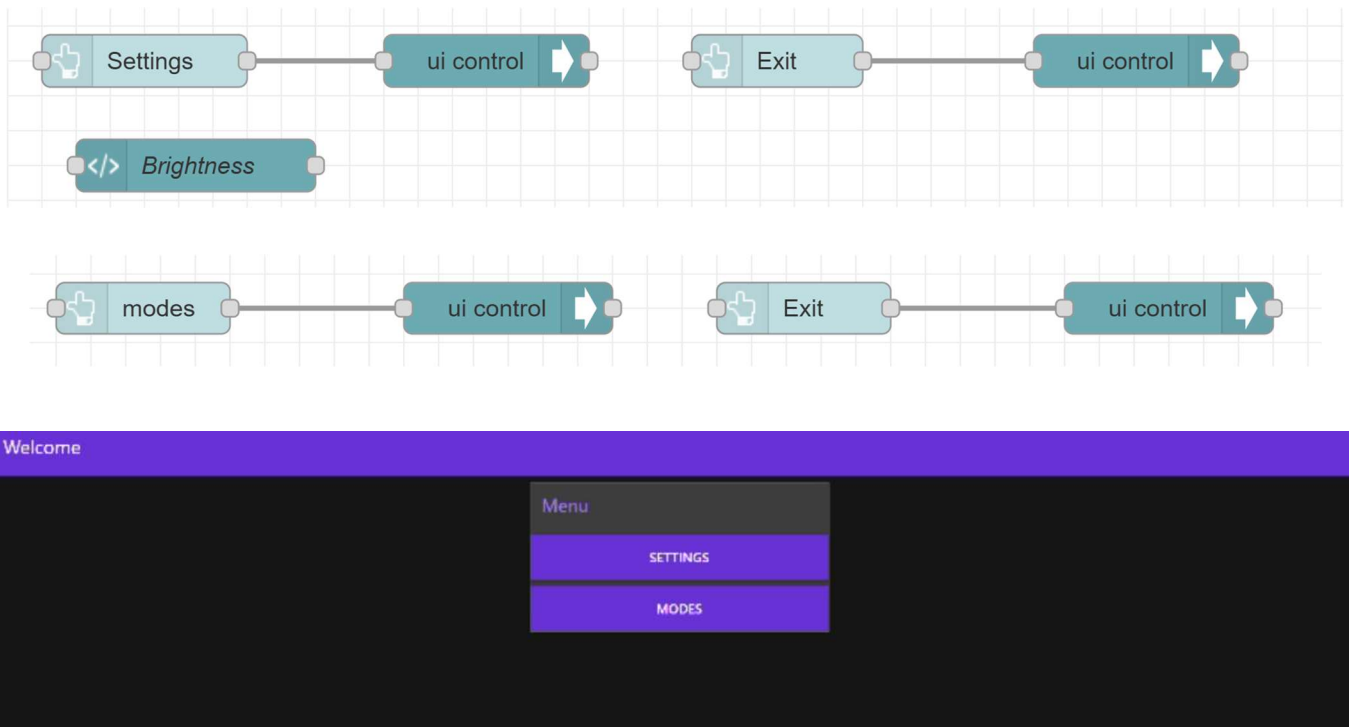


Figure 6.3 Main screen

6.4.3 Brightness Control

The first function is a brightness control (Figure 6.4) system offered in two modes. In automatic mode, brightness adjustments are carried out based on simulated sensor feedback, demonstrating an IIoT-style closed-loop response. In manual mode, the operator directly adjusts brightness levels using sliders or buttons, supporting calibration or on-demand changes.

```
1  <style>
2    .brightness-container {
3      background: #e0e0e0;
4      padding: 10px;
5      border-radius: 5px;
6      margin: 10px;
7    }
8  </style>
9
10 <div class="brightness-container">
11   <p style="margin: 0 0 10px 0; font-weight: bold;">Dashboard Brightness Control</p>
12   <div layout="row" layout-align="start center">
13     <p style="margin: 0 15px 0 0; min-width: 80px;">Brightness Level:</p>
14     <md-slider flex min="0.3" max="1.5" step="0.1"
15       ng-model="brightness"
16       ng-change="changeBrightness(brightness)"
17       aria-label="Brightness Control"></md-slider>
18     <span style="min-width: 30px; text-align: center; font-weight: bold;">{{brightness}}</span>
19   </div>
20 </div>
21
22 <script>
23   (function(scope) {
24     scope.changeBrightness = function(level) {
25       document.body.style.filter = 'brightness(' + level + ')';
26       document.documentElement.style.filter = 'brightness(' + level + ')';
27     };
28
29     scope.brightness = 1;
30
31     setTimeout(function() {
32       document.body.style.filter = 'brightness(1)';
33     }, 100);
34   })(scope);
35 </script>
```

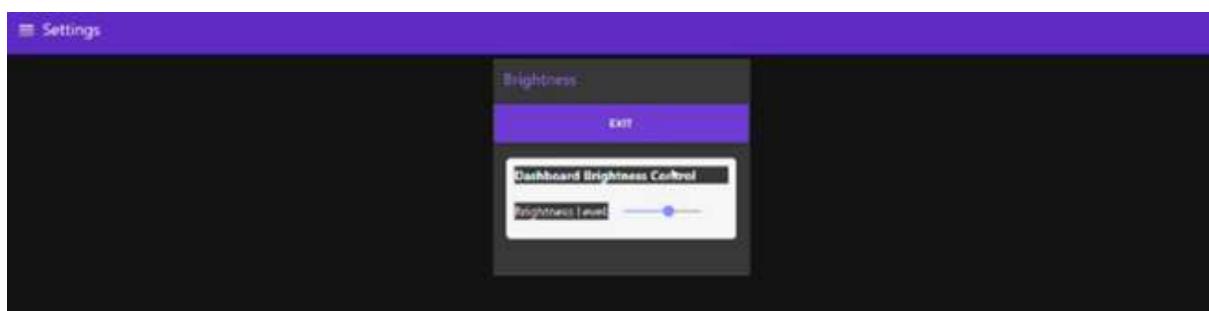


Figure 6.4 Brightness control screen

6.4.4 General Control Option

The second option provides another system function with the same two-mode structure. Automatic mode operates through predefined logic or sensor-based behavior, while manual mode grants the operator direct control such as switching the system on or off or adjusting values manually (Figure 6.5). This design reflects real industrial practice, where automated operation and human intervention must coexist reliably (Figure 6.6).

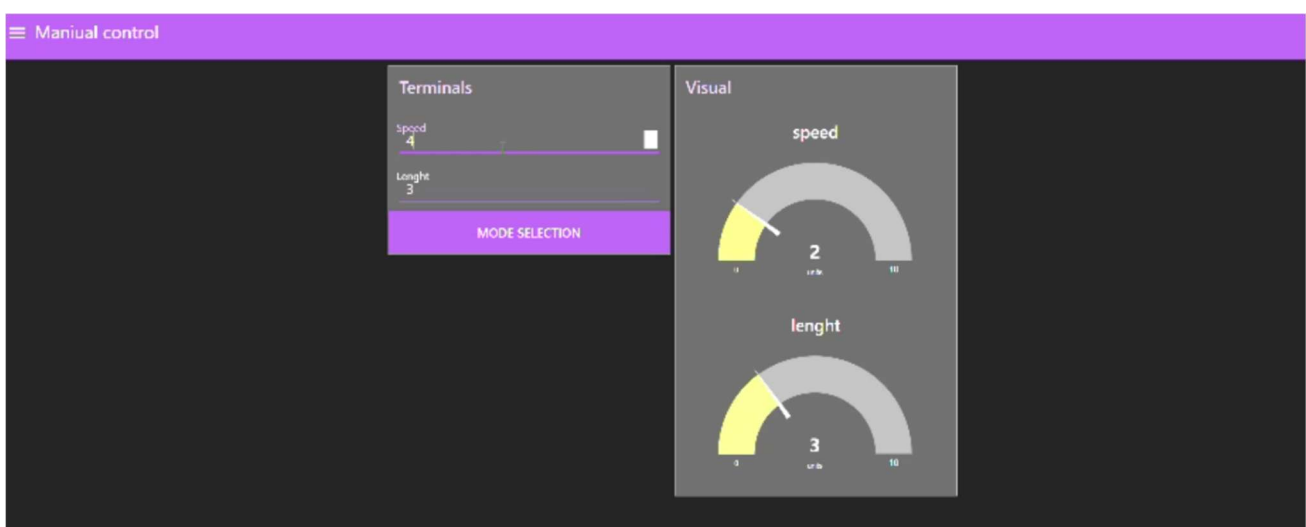
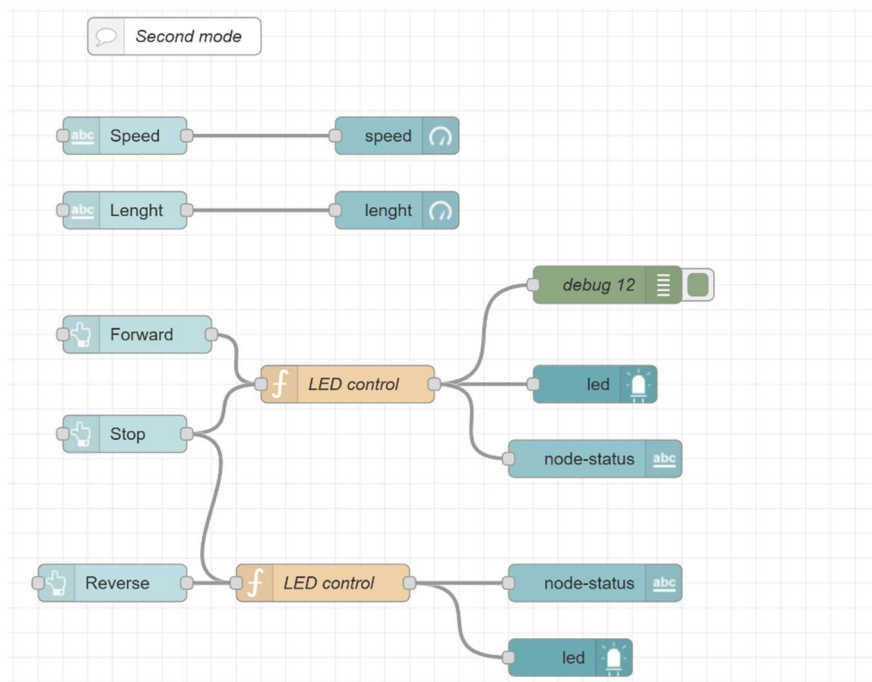


Figure 6.5 Manual control screen

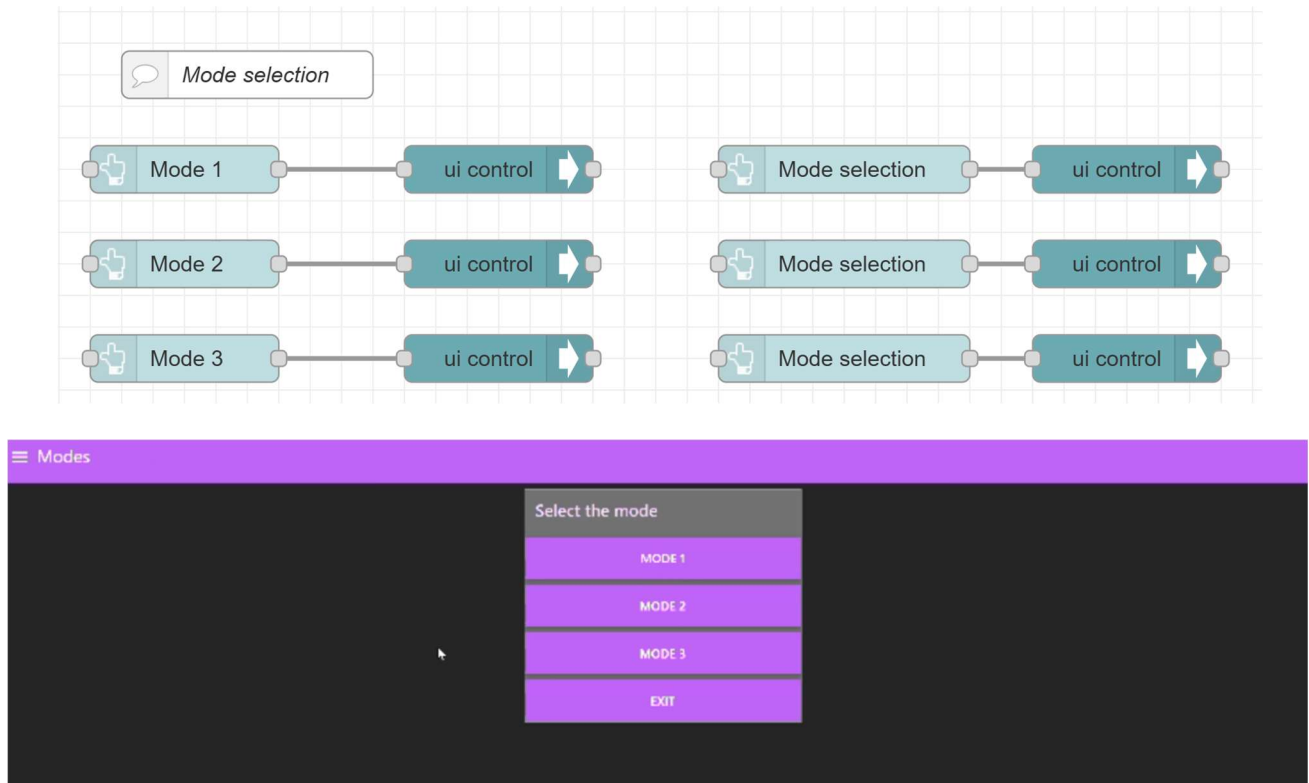


Figure 6.6 Preset control mode selection screen

6.5 Conclusion

This project successfully developed an IIoT monitoring and communication solution for the TIG welding machine, validating the practical integration of Industry 4.0 technologies in industrial automation. Key milestones achieved include mastering foundational IIoT concepts and protocols, building Node-RED communication flows, integrating HiveMQ brokers, and implementing dual manual/automatic HMI simulation workflows with robust read/write and authentication logic. While the completed system successfully replicates industrial telemetry and control, future development will focus on scaling this foundation. Planned enhancements include migrating to cloud data storage, developing mobile monitoring applications, building an advanced data analytics dashboard, and implementing remote machine control alongside predictive maintenance capabilities.



Chapter 7: Final Product

7.1 Introduction

This project presents the design and development of an automated, two-degree-of-freedom (2DOF) gantry-style autogenous GTAW machine, specifically engineered for joining thin metal sheets. Designed to bridge the gap between educational environments and industrial applications, the system is an ideal solution for technical laboratories, small-scale workshops, and independent fabricators. By strategically integrating commercial off-the-shelf (COTS) components with contemporary industrial control technologies, the architecture ensures accessibility, cost-effectiveness, and simplified maintenance without compromising on professional-grade performance.



Figure 7.1: Final assembled project

7.2 Subsystems

7.2.1 Chassis

To achieve an optimal balance between accessibility and structural performance, the gantry-style chassis is engineered using 40x40x3 mm square tubing. The architecture of the frame allows for processing and assembly via standard commercial cutting and welding equipment. Consequently, the design delivers exceptional rigidity, strength, and reliability while remaining highly accessible for small-scale production.



Figure 7.2: Manufactured chassis

7.2.2 Workstation

The structural framework, constructed from 40x40x3 mm square tubing, features a streamlined design that ensures high mechanical rigidity while facilitating straightforward maintenance. To secure the workpiece, industry-standard pneumatic cylinders are implemented as the primary clamping mechanism. These actuators were selected for their specific linear motion profile, which effectively secures the sheet metal while maintaining essential spatial clearance, ensuring zero interference with the welding path and the GTAW torch assembly.

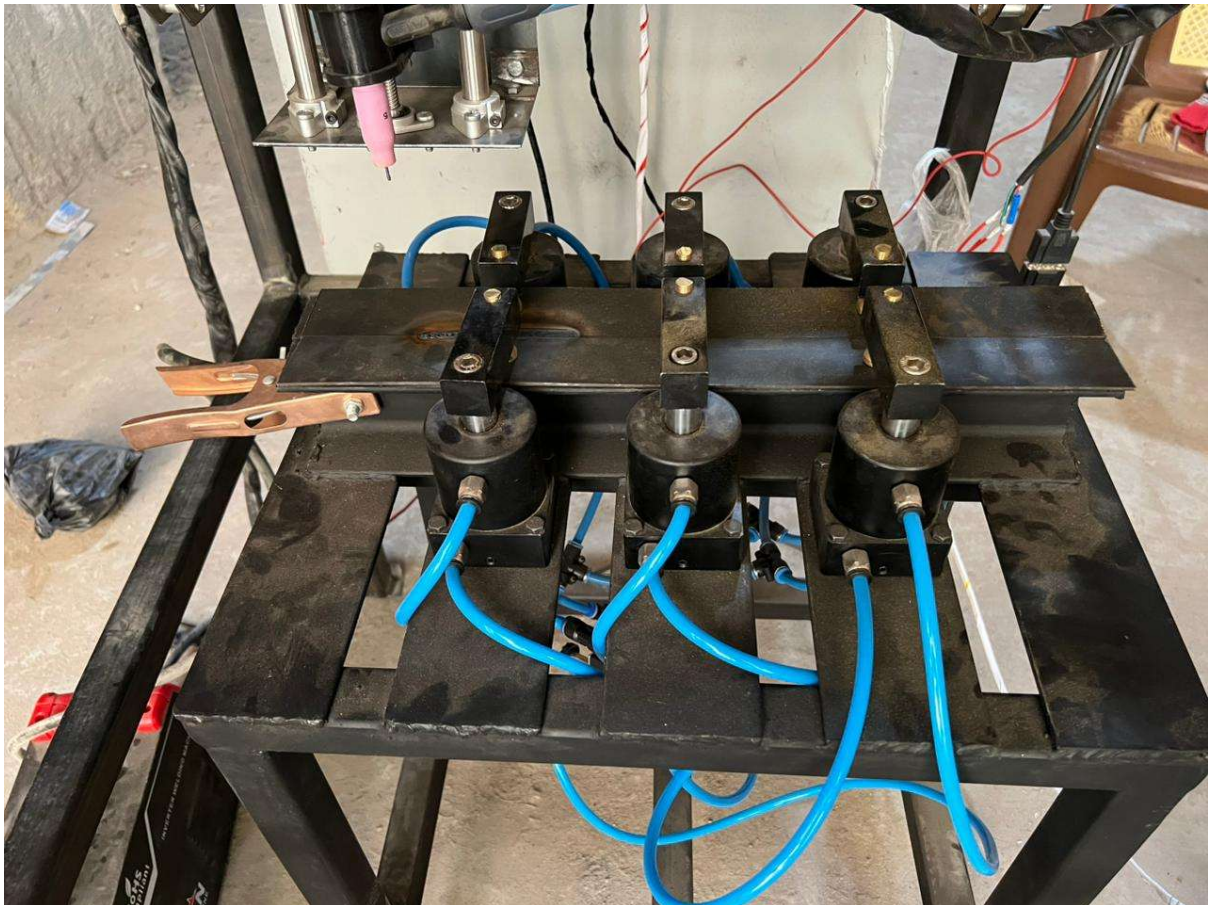


Figure 7.3: Manufactured and assembled clamping workstation, with pistons clamping a workpiece

7.2.3 Welding Mechanism

To ensure a robust yet responsive automated system, the auxiliary chassis and tool mounting assembly are constructed from 2 mm and 3 mm sheet metal. This configuration successfully delivers a rigid static body while ensuring a low-inertia carriage for the GTAW torch. Additionally, the parts are purpose-built for seamless fabrication via standard commercial CNC laser machines, emphasizing an efficient and highly reproducible manufacturing process.

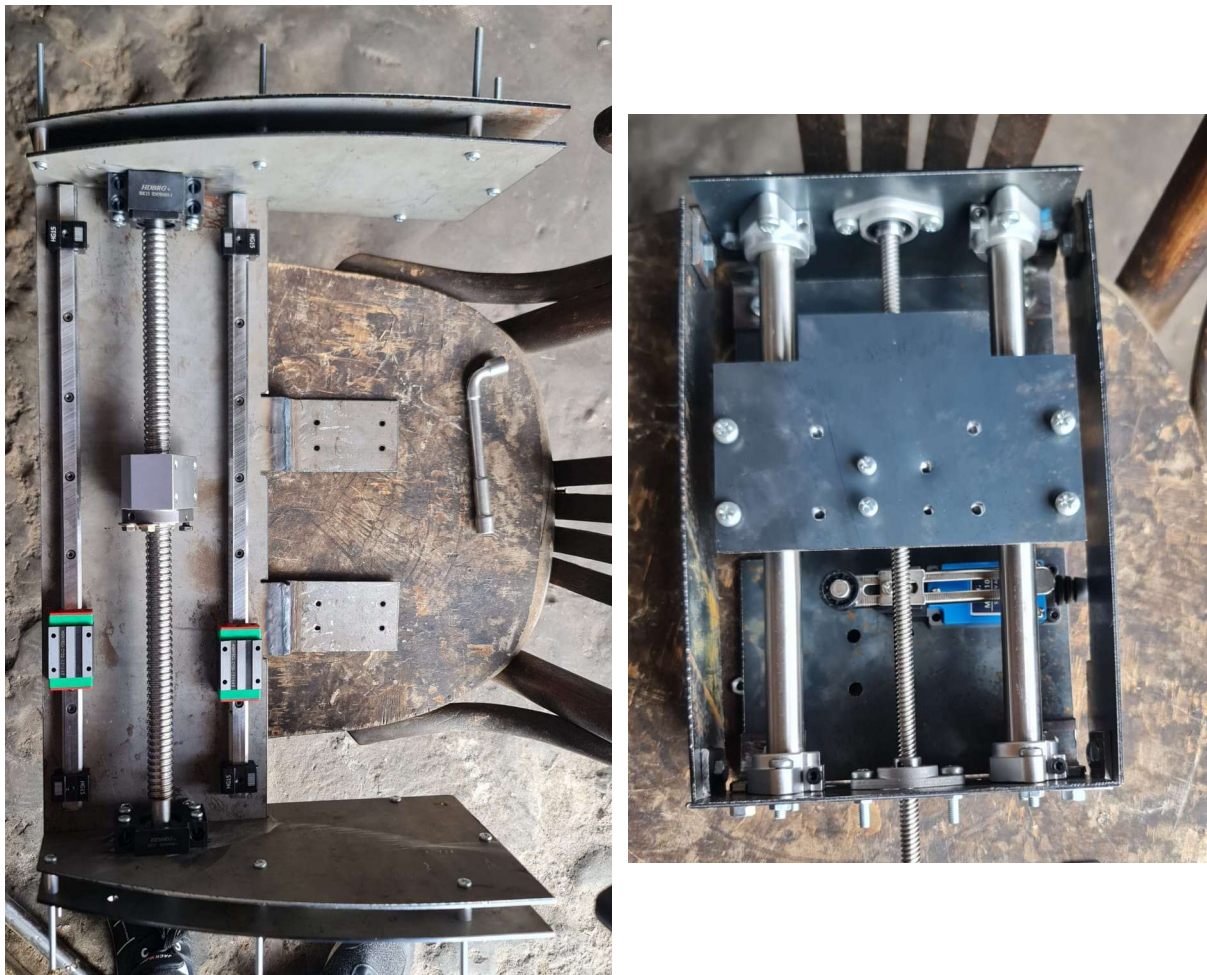


Figure 7.4: Manufactured subassemblies of welding mechanism, X-axis (left) and Z-axis (right).



Figure 7.5: Final assembled and joined welding mechanism, with motor and torch attached

7.2.4 Control Panel

The system's centralized control architecture is housed within a primary electrical enclosure. This central unit integrates all critical control and power hardware, including the Programmable Logic Controller (PLC), the Human-Machine Interface (HMI), motor drives, and auxiliary electrical circuitry. Consolidating these elements ensures a cohesive, organized, and protected environment for the machine's core electronic systems.

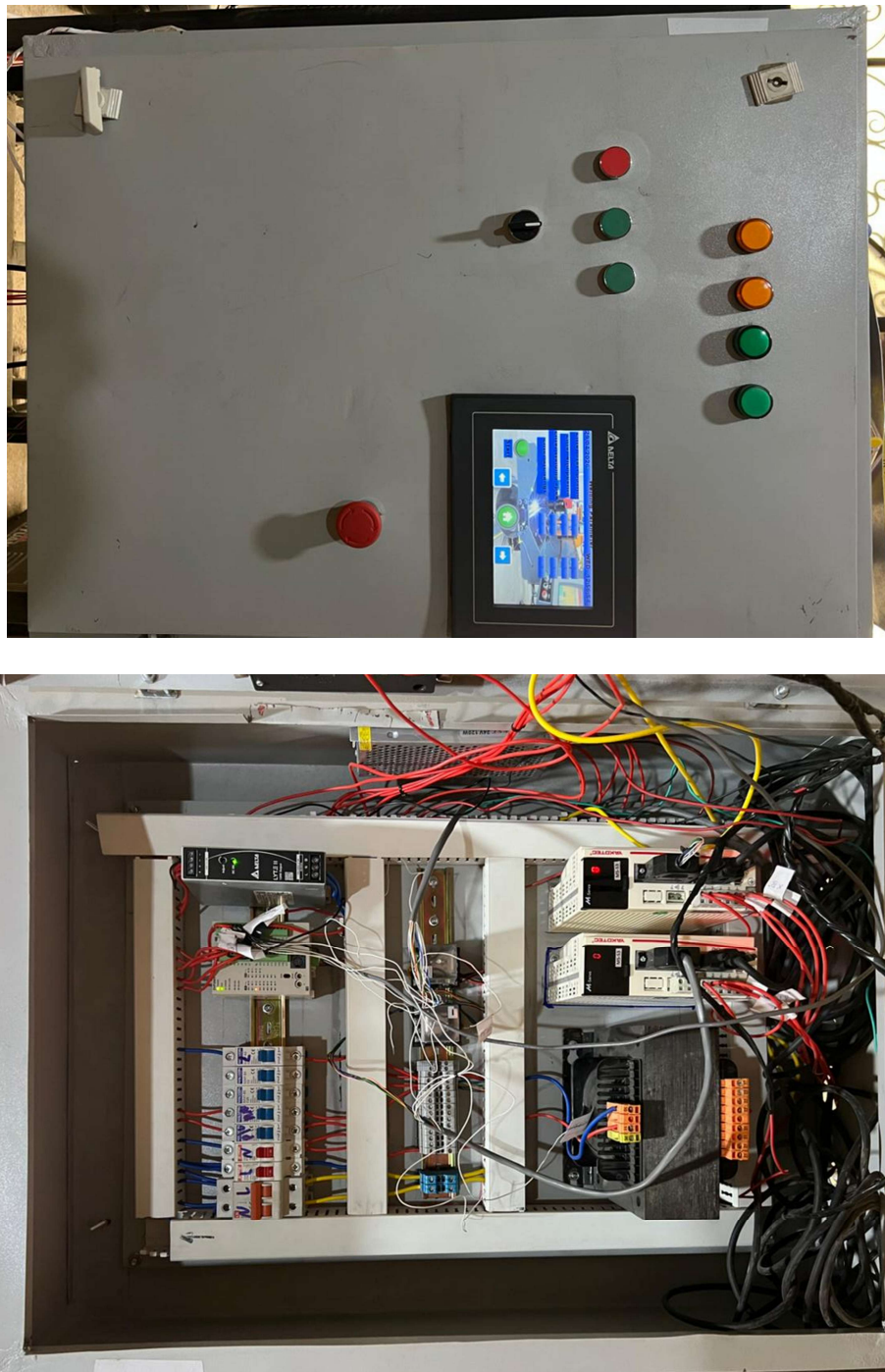


Figure 7.6: Control panel interior and exterior

7.3 Operation

The operational sequence begins with the operator inputting the specific welding parameters via the Human-Machine Interface (HMI). Concurrently, the sheet metal workpiece is positioned against the alignment gauge and securely clamped. Through the HMI, the operator selects the required operational mode, choosing between continuous seam welding or discrete spot welding. Upon cycle initiation, the system automatically executes a homing routine across both axes to establish a precise absolute coordinate reference before commencing the programmed welding path.



Figure 7.7: Showing homed position after finishing welding on highlighted workpiece



Chapter 8: Testing

8.1 Introduction

This chapter provides a comprehensive analysis of the welding iterations, documenting the experimental results, identified defects, and performance benchmarking. It details the operational progression of each trial and evaluates the system's evolution toward achieving optimized process parameters.

Advantages of autogenous GTAW are:

Significant cost and weight reduction, as eliminating filler metal consumables directly lowers material expenses and prevents adding unnecessary mass to the assembly. The process ensures identical chemistry and material matching, where the weld zone retains the exact chemical composition, corrosion resistance, and thermal expansion properties of the parent metal. It is also recognized for producing clean and low-profile beads, avoiding excessive reinforcement and yielding flush profiles that require minimal post-weld machining. The method is ideal for thin-gauge sheets, offering precise heat control for joining thin materials without the risks of deposit buildup or distortion. Finally, it facilitates simplified system automation, as mechanizing the process is significantly less complex without the need to integrate or synchronize an external wire-feed mechanism with torch travel.

Common defects in autogenous GTAW are:

Humping and periodic depressions (figure 8.1), which occur at elevated speeds as the unstable molten pool accumulates into isolated crests, leaving periodic voids in the bead. During cooling, the joint is susceptible to solidification and centerline cracking (figure 8.2) because it lacks filler-wire alloying elements to counteract tensile stresses and segregates. Furthermore, excessive heat input often results in undercutting and centerline sagging, where the joint volume is depleted or pulled downward by gravity. The process also remains highly vulnerable to porosity, as no deoxidizing agents are present to neutralize atmospheric gas entrainment or surface contaminants. Finally, the thermal demands of achieving full penetration frequently cause severe grain coarsening in the fusion zone and Heat-Affected Zone (HAZ), which significantly reduces the mechanical ductility and impact toughness of the weldment.

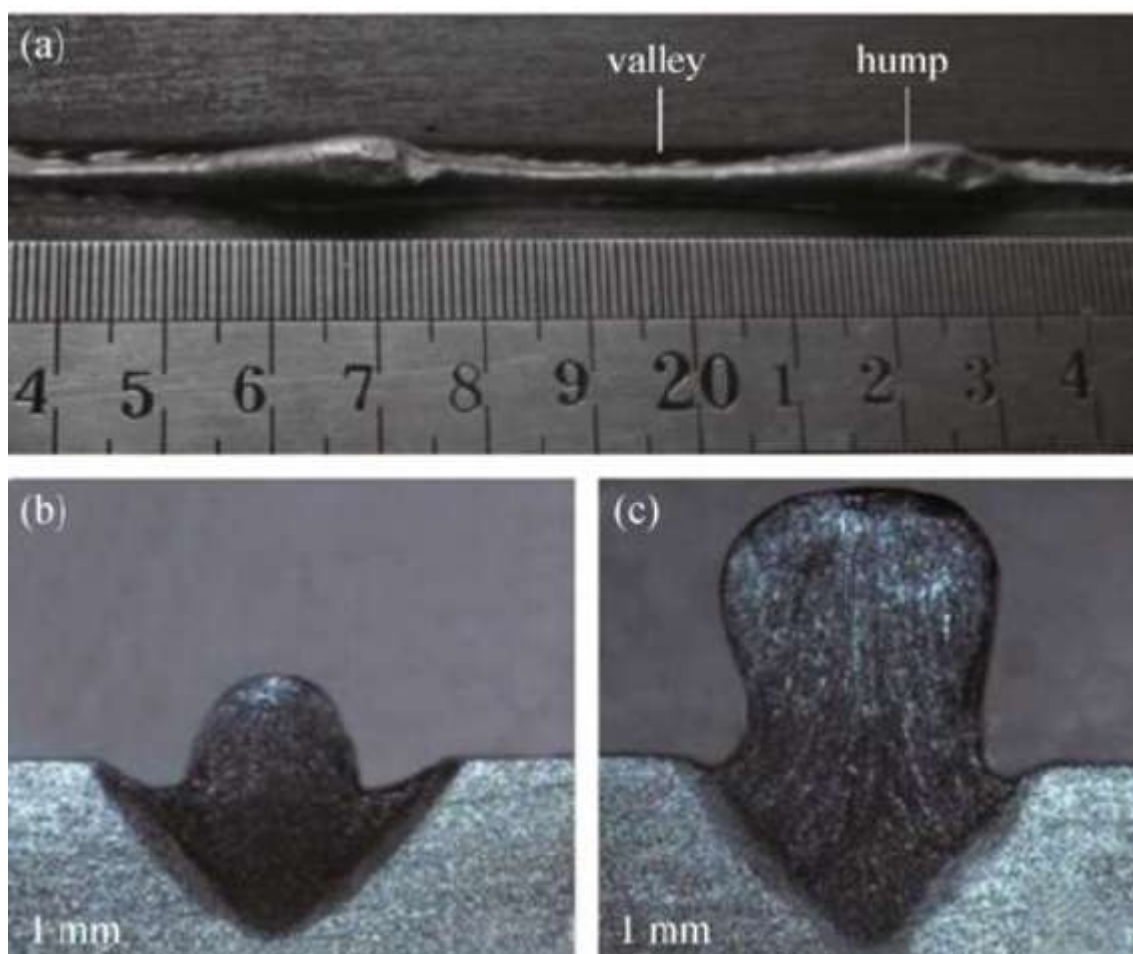


Figure 8.1 Humping bead in high speed GTAW: (a) weld bead appearance; (b) transverse cross-section at valley; (c) transverse

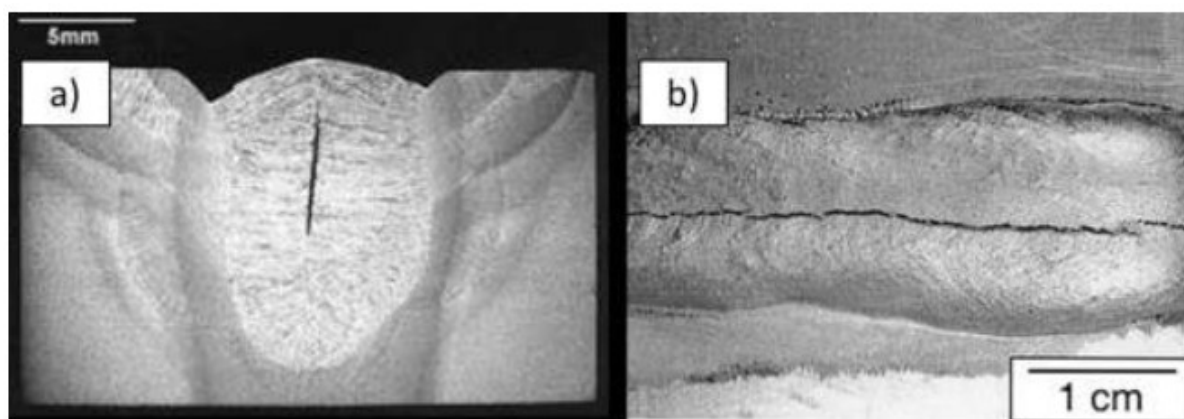


Figure 8.2 Centreline weld solidification cracks, a) buried sub-surface, b) surface penetrating

8.2 Welded Workpiece

Given the technical complexity of autogenous GTAW, the system is susceptible to various operational anomalies and parameter deviations. To mitigate these risks, extensive empirical testing was conducted across multiple workpieces. These trials specifically simulated common configuration errors to characterize their impact on weld bead morphology and to identify conditions that pose a significant risk of mechanical damage to the equipment.

Shown in figure 8.3, the first iteration was on galvanized 1mm sheet metal, with the following parameters:

- Travel speed = 140 mm/min
 - Current = 30 A
 - Torch gap = 1mm

Burn-through occurred in the workpiece due to excessive current density, a frequent challenge when joining thin-gauge sheet metal. While localized fusion was achieved, the thermal input exceeded the material's heat dissipation capacity. Although the industry standard suggests a current range of 25–35 A/mm of material thickness, empirical refinement via trial and error remains necessary to optimize the balance between penetration depth and structural integrity.



Figure 8.3: First iteration. 1mm galvanized steel sheet metal

Subsequent iterations utilized 3 mm carbon steel to standardize parameters and ensure process repeatability. This uniform substrate provided a controlled baseline for calibrating system variables and achieving consistent weld quality.

Figure 8.4a shows weld done with the following parameters:

- Travel speed = 120 mm/min
- Current = 100 A
- Torch gap = 1.5 mm

For the weld bead in figure 8.4b the parameters were the following:

- Travel speed = 135 mm/min
- Current = 110 A
- Torch gap = 1.5 mm

The specimen in figure 8.4a exhibits a porous weld bead, indicating a suboptimal travel speed and a current density below the required threshold. Conversely, the workpiece in figure 8.4b demonstrates improved bead morphology following the optimization of operational parameters.



Figure 8.4a: Weld bead with travel speed and current lower than optimum.



Figure 8.4b: Weld bead with optimized travel speed and current.

Additional welding defects encountered during testing included:

- Excessive Travel Speed: At 165 mm/min, fusion failed to occur as the thermal residence time was insufficient for the development of a stable weld pool (see Figure 8.5).



Figure 8.5: Weld bead resulted from too fast travel speed

- Insufficient Arc Gap: A torch proximity of < 1 mm resulted in tungsten electrode contamination and consumption. This condition is hazardous, potentially inducing electrical faults or system malfunctions in subsequent operations (Figure 8.6). The resulting electrode degradation and wear are documented in Figure 8.7.



Figure 8.6: Torch too close, causing a strong but undesirable weld.



Figure 8.7: Worn Tungsten electrode

After several iterations and parameter optimization, the final welding parameters were:

- Travel speed = 130 mm/min
 - Current = 115 A
 - Torch gap = 1.5 mm

Figure 8.8 illustrates the optimized weld bead, achieved through rigid clamping and precise axial alignment of the torch along the seam. The integration of these mechanical constraints with refined operational parameters resulted in the high-quality bead morphology depicted.



Figure 8.8: Final weld bead after many iterations and parameter optimization.

8.3 Testing

The specimens underwent Ultrasonic Testing (UT) to evaluate surface and near-surface integrity. The subsequent figures provide a comparative analysis of the non-destructive test results for each respective weld bead.

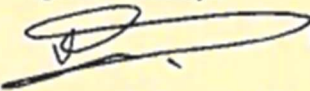
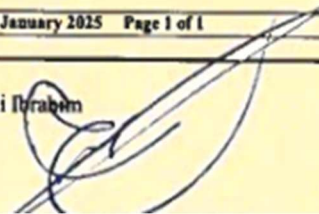
Scope of Work :																																																		
Part Description:		Two Welded Samples																																																
Material:	carbon steel		Thickness:	3 mm																																														
Item No.:	2-3		ITP:	100%																																														
Surface Conditions:	As Welded		Surface Temperature:	28 °C																																														
Procedure:	Ultrasonic Examination Procedure GOST-INSP.M-004		Acceptance Criteria:	ASME BPVC.VIII-2021																																														
			Drawings / Picture	See attached Drawings																																														
Technique Description:					Equipment and Materials:																																													
Couplant :	Grease				Transduce Angle:	0° ☐ 45° ☐ 60° ☐ 70° ☒																																												
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1	Weld 2	-	9.5	3	1	-----	-----	-----	UC - LOF - IP - P	-																																								
2	Weld 3	-	10	3	1	-----	-----	-----	LOFR - BURN THROUGH	-																																								
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An EGAC accredited (inspection body) No.032212 This inspection was carried out in compliance with ISO/IEC 17020 & ILAC P15. GOST confirms that all information obtained or created during its inspection activities is kept confidential by all personnel acting on its behalf. GOST-INSP.R-004 Issue No. 01 Issue date: 01 August 2023 Page No. 01 Last rev. date: 01 January 2025 Page 1 of 1																																																		
Examined By NDT Inspector Level II:		Eng. Mohamed Elsayed		 		Approved by General Manager Eng. Ali Ibrahim 		Final Examination Report																																										

Figure 8.9 Tested workpiece certifications

Figure 8.9 illustrates the comparative results of the final three test iterations. The specimens previously detailed in Figures 8.5a, 8.4b, and 8.8 are designated as Weld 2 and Weld 3, respectively. As noted in the initial analysis, Weld 2 exhibited significant porosity and undercutting—defects typical of suboptimal travel speeds and insufficient current density. Conversely, Weld 3 demonstrated superior bead morphology following parameter refinement. Despite these improvements, lack of root fusion remains a recurring challenge.

This defect is manifested as un-melted regions along the weld interface, which act as stress concentrators. These regions significantly compromise the mechanical properties of the joint, often leading to crack initiation and catastrophic failure under structural loading. Minimizing this defect requires rigorous control of operational parameters, precise joint preparation, and the maintenance of a stable arc.



Figure 8.10 A section of a welded joint shows lack of fusion



Chapter 9: Conclusion and Future Work

9.1 Conclusion

This chapter examines the systemic limitations inherent to the current prototype. Additionally, it highlights the essential developmental insights gained throughout the mechanical and electrical integration process. To guide subsequent research, this chapter establishes a framework for future work, proposing viable modifications to expand the machine's capabilities and industrial applicability.

9.1.1 Limitations of the project

While the developed system successfully achieves its primary objectives, several operational constraints must be acknowledged. Kinematically, the machine is restricted to two-degree-of-freedom (2-DOF) motion, limiting the end-effector to straight-line, planar welding trajectories rather than complex, multi-axis maneuvers. Process-wise, the system is exclusively designed for autogenous GTAW; the absence of a filler wire feeding mechanism intrinsically limits the maximum structural strength of the resulting welds. However, this constitutes a necessary design trade-off to prevent excessive heat input and burn-through when joining thin sheet metal. Spatially, the operational workspace is bound to accommodate workpieces with a maximum span of 300 mm. Consequently, the system is not intended for medium- or large-scale industrial manufacturing. Instead, its throughput and scale are optimized for low-volume production, fulfilling the cost-effective requirements of its primary target demographic: educational laboratories, small-scale workshops, and independent fabricators.

9.1.2 Lessons learned

The design and development of the 2-DOF autogenous GTAW machine provided several critical engineering insights. Primarily, the project demonstrated that high-quality, clean autogenous welds can be consistently achieved even when operating within strict mechanical and kinematic limitations. Furthermore, the integration of industrial-grade monitoring technology highlighted the immense value of data acquisition. By designing the system to continuously log operational metrics, the team gained practical experience in establishing databases for predictive maintenance, which significantly accelerates the identification of expected failures. Ultimately, the process of successfully bridging custom hardware with modern industrial automation standards was a highly educational milestone.

9.2 Future Work

Having successfully navigated the project's time and resource constraints, the developed system establishes a robust baseline for future mechatronic enhancements. Recommended avenues for continued development include:

- **Production Scalability:** Optimizing the structural design and manufacturing processes to allow the system to handle medium- to large-scale industrial production.
- **AI-Driven Diagnostics:** Integrating artificial intelligence into the control architecture to process operational data, enabling advanced predictive maintenance and autonomous fault detection.
- **Filler Metal Integration:** Designing an automated wire feeding mechanism to transition from purely autogenous welding to standard GTAW, thereby increasing the weld strength and overall utility.
- **Multi-Process Compatibility:** Expanding modularity to accommodate alternative arc welding processes, such as GMAW (MIG) and SMAW (Stick).

Ultimately, this project serves as a stepping stone into robotic welding, providing an accessible, cost-effective, and reliable automation solution for educational laboratories, small workshops, and fabrication enthusiasts.

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